Theory of Mathematics

Volume II: Real Analysis and Topological Foundations

Contents

Part I

Metric Foundations

Chapter 1

The Real Number System and Completeness

The real numbers are an ordered field completed by a least-upper-bound principle. Completeness converts approximation into existence and drives the convergence arguments used throughout analysis.

Learning goals

After this chapter, the reader should be able to:

- compute suprema and infima of explicitly described subsets of $\mathbb{R}$ and justify that the proposed bounds are sharp;
- use the least-upper-bound property to prove existence statements rather than merely estimate candidate values;
- apply the Archimedean property to produce integers or reciprocals satisfying a prescribed inequality;
- prove density of rational or irrational numbers between two given reals by constructing the required element;
- derive triangle and reverse-triangle inequalities in concrete absolute-value problems;
- construct examples that distinguish maxima from suprema and expose the failure of completeness in $\mathbb{Q}$;

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{structural theorem} \longrightarrow \text{application.}}$$

1.1 Ordered fields

Order is compatible with addition and multiplication. Positivity, trichotomy, and cancellation permit algebraic manipulation of inequalities.

1.2 Bounds and extrema

Upper and lower bounds describe sets globally; maxima and minima are bounds attained by elements of the set.

1.3 Suprema and infima

The supremum is the least upper bound and the infimum the greatest lower bound. Neither need belong to the set.

Example 1.1 (Proving a supremum rather than guessing it)**.** Let

$$A = \left\{ \frac{n}{n+1} : n \in \mathbb{N} \right\}.$$

Every element is less than 1, so 1 is an upper bound. To prove it is the least upper bound, let $\varepsilon > 0$. Choose $n > 1/\varepsilon$. Then

$$1 - \frac{n}{n+1} = \frac{1}{n+1} < \varepsilon,$$

so $n/(n+1) > 1 - \varepsilon$. Hence no number below 1 can remain an upper bound. Therefore $\sup A = 1$.

The two-part check—upper bound plus arbitrary approximation from below—is the reusable supremum method.

1.4 Completeness

Every nonempty real set bounded above has a supremum. This property fails in the rational numbers.

1.5 Archimedean property

No real number dominates all positive integers; equivalently, reciprocals of positive integers can be made arbitrarily small.

Example 1.2 (Archimedean choice with an explicit target)**.** Suppose $x < y$. We want an integer n for which $1/n < y - x$. By the Archimedean property choose

$$n > \frac{1}{y - x}.$$

Because both sides are positive, taking reciprocals reverses the inequality and gives $1/n < y - x$.

This apparently small move is the engine behind density proofs and many epsilon arguments: a positive geometric gap can always be resolved by a sufficiently fine reciprocal scale.

1.6 Density

Between distinct real numbers lie both rational and irrational numbers.

1.7 Absolute value and inequalities

Absolute value is the one-dimensional norm and satisfies triangle and reverse-triangle inequalities.

1.8 Nested intervals and roots

Completeness yields nested-interval principles and existence of positive square roots.

Example 1.3 (Nested intervals force a unique point). Let $I_n = [a_n, b_n]$ be nested nonempty closed intervals with $b_n - a_n \to 0$. Completeness yields at least one common point x. If y were another, then for every n

$$|x - y| \leq b_n - a_n.$$

Given $\varepsilon > 0$, choose N with $b_N - a_N < \varepsilon$. Then $|x - y| < \varepsilon$ for every $\varepsilon > 0$, hence $x = y$.

The length estimate is the verification that turns existence from completeness into uniqueness.

1.9 Core structural results

Theorem 1.4 (Archimedean property). *For every real x there is $n \in \mathbb{N}$ with $n > x$.*

Proof. Otherwise $\mathbb{N}$ would have a supremum s; then $s - 1$ would fail to be an upper bound, producing $n > s - 1$ and hence $n + 1 > s$, contradiction. □

Theorem 1.5 (Supremum approximation). *If $s = \sup A$ and $\varepsilon > 0$, some $a \in A$ satisfies $s - \varepsilon < a \leq s$.*

Proof. If not, $s - \varepsilon$ would itself be an upper bound, contradicting the leastness of s. □

Theorem 1.6 (Square-root existence). *For every $a > 0$ there is a unique $r > 0$ with $r^2 = a$.*

Proof. Take $r = \sup\{x \geq 0 : x^2 \leq a\}$. If $r^2 < a$, a small increase remains admissible; if $r^2 > a$, a small decrease remains an upper bound. Both contradict the definition of r. □

1.10 Worked examples

Example 1.7 (Supremum of a quadratic set). Let $A = \{x \geq 0 : x^2 < 5\}$. Identify $\sup A$. The set is nonempty and bounded. Its supremum s must satisfy $s^2 = 5$ by the square-root supremum argument, so $s = \sqrt{5}$.

Example 1.8 (Infimum by reflection). Prove $\inf A = -\sup(-A)$ for nonempty A bounded below. Lower bounds of A correspond under sign change to upper bounds of $-A$; reversing the order turns the least upper bound into the greatest lower bound.

Example 1.9 (No largest rational below an irrational). If α is irrational, show $\{q \in \mathbb{Q} : q < \alpha\}$ has no maximum. Given $q < \alpha$, density of $\mathbb{Q}$ gives a rational r with $q < r < \alpha$.

Example 1.10 (Reciprocals become small). Given $\varepsilon > 0$, prove some n satisfies $1/n < \varepsilon$. Choose $n > 1/\varepsilon$ by the Archimedean property and take reciprocals.

1.11 Solved dossiers

Each dossier is a substantial solved problem. Corpus mappings guide topic choice where explicit retained source blocks exist; otherwise the dossier is a fresh canonical completion.

Problem 1.11 (Supremum of a quadratic set)**.** Let $A = \{x \geq 0 : x^2 < 5\}$. Identify $\sup A$.

Solution. The set is nonempty and bounded. Its supremum s must satisfy $s^2 = 5$ by the square-root supremum argument, so $s = \sqrt{5}$. □

Problem 1.12 (Infimum by reflection)**.** Prove $\inf A = -\sup(-A)$ for nonempty A bounded below.

Solution. Lower bounds of A correspond under sign change to upper bounds of $-A$; reversing the order turns the least upper bound into the greatest lower bound. □

Problem 1.13 (No largest rational below an irrational)**.** If α is irrational, show $\{q \in \mathbb{Q} : q < \alpha\}$ has no maximum.

Solution. Given $q < \alpha$, density of $\mathbb{Q}$ gives a rational r with $q < r < \alpha$. □

Problem 1.14 (Reciprocals become small)**.** Given $\varepsilon > 0$, prove some n satisfies $1/n < \varepsilon$.

Solution. Choose $n > 1/\varepsilon$ by the Archimedean property and take reciprocals. □

Problem 1.15 (Integer bracketing)**.** Show every $x \in \mathbb{R}$ satisfies $m \leq x < m + 1$ for some $m \in \mathbb{Z}$.

Solution. Use the Archimedean property to restrict to finitely many candidate integers and choose the largest integer not exceeding x. □

Problem 1.16 (Rational density)**.** Prove that $x < y$ implies there is rational q with $x < q < y$.

Solution. Choose n with $1/n < y - x$ and then an integer m with $nx < m \leq nx + 1$; m/n lies between x and y. □

Problem 1.17 (Irrational density)**.** Prove there is an irrational number between any two reals.

Solution. Choose rational r in $(x - \sqrt{2}, y - \sqrt{2})$ and use $r + \sqrt{2}$. □

Problem 1.18 (Reverse triangle inequality)**.** Prove $||x| - |y|| \leq |x - y|$.

Solution. Apply the triangle inequality to $x = (x - y) + y$ and then interchange x, y. □

Problem 1.19 (Nested intervals are unique when lengths vanish)**.** If nested closed intervals have lengths tending to zero, show their intersection has at most one point.

Solution. Two common points would have distance bounded by every interval length, hence by arbitrarily small positive numbers, so they coincide. □

Problem 1.20 (Why $\mathbb{Q}$ is incomplete)**.** Explain why $\{q \in \mathbb{Q} : q^2 < 2, q \geq 0\}$ has no rational supremum.

Solution. Its real supremum is $\sqrt{2}$, and any rational candidate below or above can be improved using the order argument; $\sqrt{2} \notin \mathbb{Q}$. □

Problem 1.21 (Geometric partial sums)**.** Find the supremum of $1 + 1/2 + \cdots + 1/2^n$.

Solution. The sum equals $2 - 2^{-n}$, so 2 is an upper bound approached arbitrarily closely and hence is the supremum. ☐

Problem 1.22 (Completeness synthesis)**.** Explain how a supremum packages arbitrarily accurate lower approximations into one number.

Solution. Completeness supplies $s = \sup A$, and supremum approximation places elements of A within every positive distance below s. ☐

1.12 Exercises with complete solutions

The shorter exercise layer is kept separate from the dossiers.

Exercise 1.23. Find $\sup\{n/(n+1) : n \geq 1\}$.

Hint. Compare with 1. To prove a candidate supremum, show first that it is an upper bound and then that every smaller number fails to be one. ☐

Solution. The supremum is 1. ☐

Exercise 1.24. Find $\inf\{1/n : n \geq 1\}$.

Hint. Use Archimedean smallness. Use the Archimedean property to choose an integer large enough that its reciprocal lies below the prescribed tolerance. ☐

Solution. The infimum is 0. ☐

Exercise 1.25. Solve $|2x - 1| \leq 3$.

Hint. Use a double inequality. Rewrite the absolute-value inequality as a pair of ordinary inequalities before solving for the variable. ☐

Solution. The solution is $[-1, 2]$. ☐

Exercise 1.26. Show $|x + y| \geq ||x| - |y||$.

Hint. Apply reverse triangle inequality. Apply the reverse triangle inequality to compare two absolute values without expanding cases. ☐

Solution. It is the reverse triangle inequality with x and $-y$. ☐

Exercise 1.27. Give a bounded rational set with no rational supremum.

Hint. Use $q^2 < 3$. For incompleteness, use a bounded rational set whose real least upper bound is irrational. ☐

Solution. $\{q \in \mathbb{Q} : q^2 < 3, q \geq 0\}$ works. ☐

Exercise 1.28. Show every finite nonempty real set has a maximum.

Hint. Induct on the cardinality. Induct on the number of elements: adjoining one new element reduces the maximum question to comparing two numbers. ☐

Solution. Adjoin one element at a time and take the larger of it and the previous maximum. ☐

Exercise 1.29. If $A \subset B$, compare suprema.

Hint. Every upper bound of B bounds A. If $A \subseteq B$, start from an arbitrary upper bound of B and ask what it says about A. ☐

Solution. When both exist, $\sup A \leq \sup B$. ☐

Exercise 1.30. If $|x - y| < \varepsilon$ for every $\varepsilon > 0$, show $x = y$.

Hint. Assume positive distance. If the distance were positive, choose ε strictly smaller than that distance and contradict the hypothesis. ☐

Solution. Take $\varepsilon = |x - y|/2$ to obtain a contradiction. ☐

1.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 1.31 (Supremum of reciprocals)**.** Find $\sup\{1 - 1/n : n \geq 1\}$.

Hint. Show 1 is an upper bound and approximate it from below using large n. ☐

Solution. The supremum is 1, because $1 - (1 - 1/n) = 1/n \to 0$. ☐

Exercise 1.32 (Infimum of a quadratic set)**.** Find $\inf\{x^2 + 1 : x \in \mathbb{R}\}$.

Hint. Use $x^2 \geq 0$ and check whether the bound is attained. ☐

Solution. The infimum is 1, attained at $x = 0$. ☐

Exercise 1.33 (Absolute-value interval)**.** Solve $|3x - 2| < 4$.

Hint. Rewrite as $-4 < 3x - 2 < 4$. ☐

Solution. $-2/3 < x < 2$. ☐

Exercise 1.34 (Archimedean integer)**.** Find a sufficient condition on n guaranteeing $1/n < 10^{-6}$.

Hint. Take reciprocals of the positive inequality. ☐

Solution. Any integer $n > 10^6$ works. ☐

Exercise 1.35 (Sharp bound)**.** Find the maximum and supremum of $[0, 1)$.

Hint. Distinguish attainment from least upper bound. ☐

Solution. The supremum is 1; there is no maximum. ☐

Proofs

Exercise 1.36 (Supremum monotonicity). If $A \subseteq B$ and both are nonempty and bounded above, prove $\sup A \le \sup B$.

Hint. Every upper bound of B is an upper bound of A. □

Solution. $\sup B$ bounds A; by leastness of $\sup A$, $\sup A \le \sup B$. □

Exercise 1.37 (Reverse triangle inequality). Prove $||x| - |y|| \le |x - y|$.

Hint. Apply the triangle inequality to $x = (x - y) + y$ and then swap x, y. □

Solution. $|x| - |y| \le |x - y|$ and $|y| - |x| \le |x - y|$; combine them. □

Exercise 1.38 (Rational density). Prove there is a rational number between $x < y$.

Hint. Choose n with $1/n < y - x$, then trap an integer between nx and ny. □

Solution. Choose $m \in \mathbb{Z}$ with $nx < m \le nx + 1$; the width condition gives $m < ny$, hence $x < m/n < y$. □

Exercise 1.39 (No positive infinitesimal). Prove that if $a \ge 0$ and $a < 1/n$ for every n, then $a = 0$.

Hint. If $a > 0$, use Archimedean smallness with $\varepsilon = a$. □

Solution. Choose n with $1/n < a$, contradicting the hypothesis. □

Counterexamples and hypothesis testing

Exercise 1.40 (Rational incompleteness). Give a bounded nonempty subset of $\mathbb{Q}$ with no supremum in $\mathbb{Q}$.

Hint. Use rational $q \ge 0$ with $q^2 < 2$. □

Solution. Its real supremum is $\sqrt{2} \notin \mathbb{Q}$, so no rational supremum exists. □

Exercise 1.41 (Supremum need not belong). Give a bounded real set whose supremum is not an element.

Hint. Use an open interval. □

Solution. $(0, 1)$ has supremum $1 \notin (0, 1)$. □

Exercise 1.42 (Bounded need not have maximum). Why does completeness not imply every bounded set has a maximum?

Hint. Completeness gives a supremum, not attainment. □

Solution. The set $(0, 1)$ is bounded and complete as a subset of $\mathbb{R}$ only in the ambient supremum sense, but its supremum 1 is not in the set. □

Applications and investigations

Exercise 1.43 (Error scale)**.** To guarantee a discretization scale below 0.002, how large should n be if the scale is $1/n$?

Hint. Solve $1/n < 0.002$. □

Solution. Any $n > 500$ works. □

Exercise 1.44 (Bracketing a root)**.** Explain how completeness supports bisection once nested closed intervals containing a root have lengths tending to zero.

Hint. Use the nested-interval theorem and the vanishing-length uniqueness argument. □

Solution. The intervals have a common point, and shrinking lengths make it unique; continuity identifies that point as the root. □

Challenge problems

Exercise 1.45 (Supremum of a sum set)**.** For nonempty bounded-above $A, B \subset \mathbb{R}$, prove $\sup(A + B) = \sup A + \sup B$.

Hint. First prove the sum of suprema is an upper bound; then approximate each supremum within $\varepsilon/2$. □

Solution. Upper boundedness is immediate. Choose $a > \sup A - \varepsilon/2$, $b > \sup B - \varepsilon/2$; then $a + b > \sup A + \sup B - \varepsilon$. □

Exercise 1.46 (Square-root construction)**.** Let $S = \{x \geq 0 : x^2 < 3\}$. Explain the contradiction if $s = \sup S$ satisfies $s^2 < 3$.

Hint. Choose a sufficiently small positive h so $(s + h)^2 < 3$. □

Solution. Since $3 - s^2 > 0$, choose $h > 0$ with $2sh + h^2 < 3 - s^2$. Then $s + h \in S$, contradicting that s is an upper bound. □

Chapter summary

The central definitions, estimates, and structural theorems of this chapter will be used repeatedly in the remaining analysis volume.

Chapter 2

Euclidean and Normed Spaces

Normed spaces add quantitative geometry to vector spaces. The norm axioms isolate exactly the estimates needed to speak about distance, convergence, boundedness, and continuity.

Learning goals

After this chapter, the reader should be able to:

- compute norms and distances in Euclidean and normed spaces and verify the norm axioms in concrete examples;
- compare two norms by proving explicit upper and lower inequalities between them;
- describe open balls in several norms and explain how their geometry differs;
- use the triangle and reverse-triangle inequalities to bound perturbations and distances;
- decide whether a proposed formula defines a norm and identify the failed axiom when it does not;
- translate coordinate estimates into norm estimates using finite-dimensional norm equivalence;

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{structural theorem} \longrightarrow \text{application.}}$$

2.1 Euclidean norm

The Euclidean norm is induced by the standard inner product and measures ordinary length.

2.2 Norm axioms

A norm is positive definite, absolutely homogeneous, and subadditive.

2.3 Standard finite-dimensional norms

The 1-, 2-, and infinity norms emphasize total magnitude, Euclidean length, and largest coordinate.

Example 2.1 (Comparing three standard norms)**.** For $x = (1, -2, 2)$,

$$\|x\|_1 = 5, \qquad \|x\|_2 = 3, \qquad \|x\|_\infty = 2.$$

The general inequalities

$$\|x\|_\infty \le \|x\|_2 \le \sqrt{3}\,\|x\|_\infty$$

become $2 \le 3 \le 2\sqrt{3}$, which checks the computation. The example also shows that equivalent norms need not assign the same numerical length; they agree instead on the qualitative notions of convergence and openness.

2.4 Balls and spheres

Norm balls define neighborhoods and therefore the topology of a normed space.

2.5 Operator norms

The induced norm of a linear map measures its maximal stretching on the unit sphere.

Example 2.2 (Computing an induced operator norm)**.** Let $T(x, y) = (2x, 0)$ on Euclidean $\mathbb{R}^2$. For a unit vector $v = (x, y)$,

$$\|Tv\|_2 = 2|x| \le 2.$$

Equality occurs at $v = e_1$, so

$$\|T\|_{\text{op}} = 2.$$

The upper estimate and an attaining unit vector are both needed to certify the exact operator norm.

2.6 Equivalent norms

Comparable norms define the same convergence and open sets.

Example 2.3 (Norm equivalence transfers convergence)**.** Suppose on a finite-dimensional space

$$c\|v\|_a \le \|v\|_b \le C\|v\|_a$$

for positive constants c, C. If $v_n \to v$ in $\|\cdot\|_a$, then

$$\|v_n - v\|_b \le C\|v_n - v\|_a \to 0.$$

The reverse inequality gives the converse. Thus equivalent norms have exactly the same convergent sequences, even though their balls may have very different shapes.

2.7 Convexity

Every norm ball is convex by the triangle inequality.

2.8 Finite versus infinite dimensions

All norms are equivalent in finite dimensions, but this can fail in infinite-dimensional spaces.

2.9 Core structural results

Theorem 2.4 (Cauchy–Schwarz)**.** *For* $x, y \in \mathbb{R}^n$, $|\langle x, y\rangle| \le \|x\|_2\|y\|_2$.

Proof. Apply nonnegativity to $\|x - ty\|_2^2$ and require its quadratic discriminant to be nonpositive. ☐

Theorem 2.5 (Euclidean triangle inequality)**.** $\|x + y\|_2 \le \|x\|_2 + \|y\|_2$.

Proof. Expand the square and use Cauchy–Schwarz on the cross term. ☐

Theorem 2.6 (Norm comparison)**.** *On* $\mathbb{R}^n$, $\|x\|_\infty \le \|x\|_2 \le \sqrt{n}\|x\|_\infty$.

Proof. The largest coordinate square is bounded by the sum of squares, and each coordinate square is bounded by the largest one. ☐

2.10 Worked examples

Example 2.7 (Compare 1- and 2-norms)**.** Prove $\|x\|_2 \le \|x\|_1 \le \sqrt{n}\|x\|_2$. The first follows by expanding $(\sum |x_i|)^2$; the second is Cauchy–Schwarz against $(1, \dots, 1)$.

Example 2.8 (A seminorm, not a norm)**.** Is $p(x, y) = |x - y|$ a norm on $\mathbb{R}^2$? No: $p(1, 1) = 0$ although $(1, 1) \neq 0$.

Example 2.9 (Weighted Euclidean norm)**.** Show $\sqrt{4x^2 + y^2}$ is a norm. It is the Euclidean norm after the invertible linear change $(x, y) \mapsto (2x, y)$.

Example 2.10 (Distance to a coordinate axis)**.** Find the Euclidean distance from $(1, 2)$ to the x-axis. Minimizing $(1 - t)^2 + 4$ over $(t, 0)$ gives $t = 1$ and distance 2.

2.11 Solved dossiers

Each dossier is a substantial solved problem. Corpus mappings guide topic choice where explicit retained source blocks exist; otherwise the dossier is a fresh canonical completion.

Problem 2.11 (Compare 1- and 2-norms)**.** Prove $\|x\|_2 \le \|x\|_1 \le \sqrt{n}\|x\|_2$.

Solution. The first follows by expanding $(\sum |x_i|)^2$; the second is Cauchy–Schwarz against $(1, \dots, 1)$. ☐

Problem 2.12 (A seminorm, not a norm)**.** Is $p(x, y) = |x - y|$ a norm on $\mathbb{R}^2$?

Solution. No: $p(1, 1) = 0$ although $(1, 1) \neq 0$. ☐

Problem 2.13 (Weighted Euclidean norm)**.** Show $\sqrt{4x^2 + y^2}$ is a norm.

Solution. It is the Euclidean norm after the invertible linear change $(x, y) \mapsto (2x, y)$. ☐

Problem 2.14 (Distance to a coordinate axis)**.** Find the Euclidean distance from $(1, 2)$ to the x-axis.

Solution. Minimizing $(1 - t)^2 + 4$ over $(t, 0)$ gives $t = 1$ and distance 2. ☐

Problem 2.15 (Projection operator norm)**.** For $P(x, y) = (x, 0)$, find $\|P\|_2$.

Solution. $\|P(x, y)\| \le \|(x, y)\|$ and equality holds at $(1, 0)$, so $\|P\| = 1$. ☐

Problem 2.16 (Equivalent convergence)**.** Show $\|x_n - x\|_2 \to 0$ iff $\|x_n - x\|_\infty \to 0$.

Solution. Use $\|z\|_\infty \le \|z\|_2 \le \sqrt{n}\|z\|_\infty$. ☐

Problem 2.17 (Diameter of the unit sphere)**.** If $\|x\| = \|y\| = 1$, show $\|x - y\| \le 2$.

Solution. Use the triangle inequality on $x + (-y)$. ☐

Problem 2.18 (Reverse triangle inequality)**.** Prove $|\|x\| - \|y\|| \le \|x - y\|$.

Solution. Apply the triangle inequality to $x = (x - y) + y$ and then swap x, y. ☐

Problem 2.19 (Norm of a Euclidean functional)**.** For $f(x) = a \cdot x$, compute $\|f\|$.

Solution. Cauchy–Schwarz gives $\|f\| \le \|a\|_2$, with equality at $x = a/\|a\|_2$. ☐

Problem 2.20 (Convexity of closed balls)**.** Prove every closed norm ball is convex.

Solution. A convex combination of two radius-r displacement vectors has norm at most the same convex combination of their norms, hence at most r. ☐

Problem 2.21 (Infinite-dimensional warning)**.** Why is there no C with $\|f\|_\infty \le C\|f\|_1$ for every continuous f on $[0, 1]$?

Solution. Continuous spike functions can keep height 1 while their support and integral shrink to zero. ☐

Problem 2.22 (Equivalent-norm synthesis)**.** Why do two-sided norm bounds imply identical topology?

Solution. Each ball for one norm contains a suitably scaled ball for the other, in both directions. ☐

2.12 Exercises with complete solutions

The shorter exercise layer is kept separate from the dossiers.

Exercise 2.23. Compute all three standard norms of $(1, -1, 2)$.

Hint. Use definitions. Write out the norm axioms one at a time; homogeneity and the triangle inequality are usually the decisive tests. ☐

Solution. They are $4, \sqrt{6}, 2$. ☐

Exercise 2.24. Show $2|x| + 3|y|$ is a norm.

Hint. Check the axioms. Compute the distance from the norm definition before trying to visualize the geometry. ☐

Solution. Positive weighted sums of absolute values satisfy the norm axioms. ☐

Exercise 2.25. Is $\max(|x|, 2|y|)$ a norm?

Hint. View it as a weighted infinity norm. Describe the unit ball by solving the inequality $\|x\| < 1$ in coordinates. ☐

Solution. Yes. ☐

Exercise 2.26. Find the Euclidean distance from $(1, 2)$ to $(-2, 6)$.

Hint. Subtract first. Use the triangle inequality twice to obtain a reverse-triangle estimate for the difference of norms. □

Solution. The difference has length 5. □

Exercise 2.27. Prove $\|0\| = 0$.

Hint. Use homogeneity with scalar zero. In finite dimensions, compare each coordinate with the norm and then sum the coordinate bounds. □

Solution. $\|0x\| = 0\|x\| = 0$. □

Exercise 2.28. If $x_n \to x$, show $\|x_n\| \to \|x\|$.

Hint. Use reverse triangle inequality. To disprove a norm, test zero definiteness with a nonzero vector before attempting harder inequalities. □

Solution. The norm difference is bounded by $\|x_n - x\|$. □

Exercise 2.29. Find the operator norm of $T(x, y) = (3x, 0)$.

Hint. Bound and attain. Normalize a nonzero vector to reduce a general estimate to the unit sphere. □

Solution. The norm is 3. □

Exercise 2.30. Show the open unit ball is convex.

Hint. Use strict norm bounds. Track constants explicitly when proving norm equivalence; the direction of each inequality matters. □

Solution. A convex combination has norm below the corresponding convex combination of two numbers below 1. □

2.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 2.31 (Three norms)**.** Compute $\|(1, -2, 3)\|_1, \|\cdot\|_2, \|\cdot\|_\infty$.

Hint. Use the definitions sum, square-root of squares, and maximum coordinate magnitude. □

Solution. The values are $6, \sqrt{14}, 3$. □

Exercise 2.32 (Weighted norm)**.** Compute $\sqrt{4x^2 + y^2}$ at $(2, -3)$.

Hint. Substitute directly. □

Solution. The value is $\sqrt{16 + 9} = 5$. □

Exercise 2.33 (Ball geometry)**.** Describe the unit ball of $\|\cdot\|_\infty$ in $\mathbb{R}^2$.

Hint. Translate $\max(|x|, |y|) < 1$. □

Solution. It is the open square $(-1,1) \times (-1,1)$. □

Exercise 2.34 (Operator stretch)**.** For $T = \mathrm{diag}(3,1)$, compute the Euclidean operator norm.

Hint. Bound $\sqrt{9x^2 + y^2}$ on $x^2 + y^2 = 1$. □

Solution. The maximum is 3, attained at e_1. □

Exercise 2.35 (Distance)**.** Find the ℓ^1-distance between $(1,2)$ and $(-1,5)$.

Hint. Apply the norm to their difference. □

Solution. $|2| + |-3| = 5$. □

Proofs

Exercise 2.36 (Convexity of norm balls)**.** Prove every norm ball is convex.

Hint. For $0 \le t \le 1$, estimate the norm of $t(x-a) + (1-t)(y-a)$. □

Solution. The triangle inequality gives a value below $tr + (1-t)r = r$. □

Exercise 2.37 (Reverse norm inequality)**.** Prove $|\|x\| - \|y\|| \le \|x - y\|$.

Hint. Use $x = (x-y) + y$ and swap. □

Solution. $\|x\| - \|y\| \le \|x - y\|$ and the reversed inequality follow. □

Exercise 2.38 (Norm metric)**.** Prove $d(x,y) = \|x - y\|$ is a metric.

Hint. Match each metric axiom to a norm axiom. □

Solution. Nonnegativity/definiteness, symmetry via $\|-v\| = \|v\|$, and the triangle inequality follow directly. □

Exercise 2.39 (Equivalent norms give same open sets)**.** Prove comparable norms define the same topology.

Hint. Use the constants to put a ball for one norm inside a ball for the other. □

Solution. If $c\|v\|_a \le \|v\|_b \le C\|v\|_a$, then $B_a(x, r/C) \subset B_b(x,r)$ and conversely. □

Counterexamples and hypothesis testing

Exercise 2.40 (Seminorm)**.** Why is $p(x,y) = |x - y|$ not a norm?

Hint. Test positive definiteness. □

Solution. $p(1,1) = 0$ although $(1,1) \ne 0$. □

Exercise 2.41 (Wrong homogeneity)**.** Why is $p(x) = |x|^2$ not a norm on $\mathbb{R}$?

Hint. Test scaling by 2. □

Solution. $p(2x) = 4p(x)$, not $2p(x)$. □

Exercise 2.42 (Infinite-dimensional warning)**.** Explain why finite-dimensional norm equivalence cannot simply be assumed on $C[0,1]$.

Hint. Compare sup and integral-type norms on sharply peaked functions. □

Solution. One can build narrow peaks with fixed sup norm but arbitrarily small integral norm, preventing a uniform lower comparison constant. □

Applications and investigations

Exercise 2.43 (Perturbation bound)**.** If $\|x - y\| \leq 10^{-4}$, how much can the norm change?

Hint. Use the reverse triangle inequality. $\square$

Solution. At most 10^{-4}: $|\|x\| - \|y\|| \leq 10^{-4}$. $\square$

Exercise 2.44 (Condition estimate)**.** If $\|T\| = 7$ and $\|h\| \leq 0.01$, bound $\|Th\|$.

Hint. Use the operator norm definition. $\square$

Solution. $\|Th\| \leq 0.07$. $\square$

Challenge problems

Exercise 2.45 (One-dimensional norms)**.** Prove every norm on $\mathbb{R}$ has the form $c|x|$ with $c > 0$.

Hint. Use homogeneity and evaluate the norm of 1. $\square$

Solution. $\|x\| = |x|\|\operatorname{sgn}(x)\| = |x|\|1\|$; set $c = \|1\| > 0$. $\square$

Exercise 2.46 (Sharp norm comparison)**.** Show $\|x\|_1 \leq \sqrt{n}\|x\|_2$ and identify an equality case.

Hint. Apply Cauchy–Schwarz to $(|x_i|)$ and $(1, \dots, 1)$. $\square$

Solution. The inequality follows; equality occurs when all $|x_i|$ are equal, for example $x = (1, \dots, 1)$. $\square$

Chapter summary

The central definitions, estimates, and structural theorems of this chapter will be used repeatedly in the remaining analysis volume.

Chapter 3

Sequences and Cauchy Sequences

Sequences encode limiting behavior through quantifiers. Cauchy sequences compare late terms only with each other, and completeness is exactly the principle that such internal coherence produces a limit.

Learning goals

After this chapter, the reader should be able to:

- determine whether a sequence converges and prove the claimed limit with an explicit ε-N argument;
- test whether a sequence is Cauchy and use completeness to infer convergence when appropriate;
- construct subsequences with prescribed limiting behavior;
- apply monotone convergence and squeeze arguments to concrete sequences;
- distinguish convergence, boundedness, Cauchy behavior, and existence of convergent subsequences by counterexample;
- estimate convergence rates sufficiently sharply to choose an N for a prescribed tolerance;

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{structural theorem} \longrightarrow \text{application.}}$$

3.1 Convergence

A sequence converges when all sufficiently late terms lie in every prescribed neighborhood of a candidate limit.

Example 3.1 (An epsilon–N proof with a rate)**.** For $a_n = (3n+1)/(n+2)$, the candidate limit is 3. Indeed

$$|a_n - 3| = \frac{5}{n+2}.$$

Given $\varepsilon > 0$, it is enough to choose

$$N > \frac{5}{\varepsilon} - 2.$$

Then $n \geq N$ implies $|a_n - 3| < \varepsilon$. The explicit estimate does more than prove convergence: it tells how far into the sequence one must go for any requested tolerance.

3.2 Uniqueness

Metric separation makes limits unique.

3.3 Boundedness

Every convergent sequence is bounded, but boundedness alone does not force convergence.

3.4 Subsequences

Subsequences preserve order and are the basic tool for detecting hidden limiting behavior.

3.5 Cauchy sequences

A sequence is Cauchy when sufficiently late terms are mutually close.

Example 3.2 (Cauchy without knowing the limit first)**.** Let

$$s_n = \sum_{k=1}^{n} 2^{-k}.$$

For $m > n$,

$$|s_m - s_n| \leq \sum_{k=n+1}^{\infty} 2^{-k} = 2^{-n}.$$

Given $\varepsilon > 0$, choose N with $2^{-N} < \varepsilon$. Then $m, n \geq N$ makes the tail distance below ε. Thus (s_n) is Cauchy without first using its closed form. Completeness then guarantees a limit.

3.6 Completeness

A metric space is complete if every Cauchy sequence converges in the space.

3.7 Monotone real sequences

Bounded monotone real sequences converge by the supremum property.

3.8 Limsup and liminf

Tail suprema and infima capture asymptotic oscillation.

Example 3.3 (Limsup detects persistent oscillation)**.** For

$$a_n = (-1)^n + \frac{1}{n},$$

the even subsequence tends to 1 and the odd subsequence tends to -1. Hence

$$\limsup a_n = 1, \qquad \liminf a_n = -1.$$

Because these differ, the full sequence does not converge. Limsup and liminf therefore quantify the two persistent asymptotic edges of the oscillation.

3.9 Core structural results

Theorem 3.4 (Convergent implies Cauchy)**.** *Every convergent sequence in a metric space is Cauchy.*

Proof. Choose a tail within $\varepsilon/2$ of the limit and apply the triangle inequality. □

Theorem 3.5 (Monotone convergence)**.** *Every increasing real sequence bounded above converges to the supremum of its range.*

Proof. Supremum approximation supplies one late term within ε below the supremum; monotonicity traps every later term. □

Theorem 3.6 (Cauchy boundedness)**.** *Every Cauchy sequence is bounded.*

Proof. A tail lies in a unit ball about one tail term; combine this with the finite initial segment. □

3.10 Worked examples

Example 3.7 (Direct proof for $1/n$)**.** Prove $1/n \to 0$. Given $\varepsilon > 0$, choose $N > 1/\varepsilon$.

Example 3.8 (Uniqueness of limits)**.** Prove a metric sequence cannot converge to two distinct points. The triangle inequality would make the distance between the two proposed limits smaller than itself.

Example 3.9 (Convergent implies bounded)**.** Prove it without invoking a theorem name. Take a tail in a unit ball around the limit and enlarge the ball to contain the finite initial segment.

Example 3.10 (Subsequence inheritance)**.** Show every subsequence of a convergent sequence has the same limit. Subsequence indices eventually exceed any original convergence index.

3.11 Solved dossiers

Each dossier is a substantial solved problem. Corpus mappings guide topic choice where explicit retained source blocks exist; otherwise the dossier is a fresh canonical completion.

Problem 3.11 (Direct proof for $1/n$)**.** Prove $1/n \to 0$.

Solution. Given $\varepsilon > 0$, choose $N > 1/\varepsilon$. □

Problem 3.12 (Uniqueness of limits)**.** Prove a metric sequence cannot converge to two distinct points.

Solution. The triangle inequality would make the distance between the two proposed limits smaller than itself. □

Problem 3.13 (Convergent implies bounded)**.** Prove it without invoking a theorem name.

Solution. Take a tail in a unit ball around the limit and enlarge the ball to contain the finite initial segment. □

Problem 3.14 (Subsequence inheritance)**.** Show every subsequence of a convergent sequence has the same limit.

Solution. Subsequence indices eventually exceed any original convergence index. □

Problem 3.15 (A Cauchy sequence in $\mathbb{Q}$ without rational limit)**.** Use rational truncations of $\sqrt{2}$.

Solution. They converge in $\mathbb{R}$ and are therefore Cauchy, but a rational limit would equal $\sqrt{2}$ by uniqueness. □

Problem 3.16 (Geometric-series Cauchy criterion)**.** Show partial sums of $\sum r^k$ are Cauchy for $|r| < 1$.

Solution. Bound every tail by $|r|^{n+1}/(1 - |r|)$. □

Problem 3.17 (Recursive monotone convergence)**.** For $x_{n+1} = \sqrt{2 + x_n}$, $x_1 = 1$, prove $x_n \to 2$.

Solution. Inductively the sequence is increasing and bounded by 2; its limit satisfies $L^2 = L+2$ and positivity forces $L = 2$. □

Problem 3.18 (Bounded but divergent)**.** Analyze $(-1)^n$.

Solution. Even and odd subsequences converge to different values, so the full sequence diverges. □

Problem 3.19 (Limsup and liminf)**.** Find them for $(-1)^n + 1/n$.

Solution. Even and odd subsequences approach 1 and -1, giving limsup 1 and liminf -1. □

Problem 3.20 (Cauchy plus a convergent subsequence)**.** Prove the whole sequence converges to the subsequential limit.

Solution. Use one late subsequence term as a bridge between the Cauchy tail and the limit. □

Problem 3.21 (Completeness of reals via tail bounds)**.** Explain how a Cauchy real sequence can be trapped by tail infima and suprema.

Solution. Tail infima increase, tail suprema decrease, and their gap tends to zero; completeness gives a common limiting value. □

Problem 3.22 (Sequence synthesis)**.** Distinguish convergence from the Cauchy property conceptually.

Solution. Convergence references an external limit; Cauchy behavior is internal. Completeness connects the two. □

3.12 Exercises with complete solutions

The shorter exercise layer is kept separate from the dossiers.

Exercise 3.23. Prove $3/n \to 0$.

Hint. Choose $N > 3/\varepsilon$. For an ε-N proof, solve the desired inequality for n before choosing N. □

Solution. Then $n \geq N$ implies $3/n < \varepsilon$. □

Exercise 3.24. Does $(-1)^n/n$ converge?

Hint. Bound the absolute value. To show Cauchy behavior, estimate $|a_n - a_m|$ uniformly for all m, n beyond the same index. ☐

Solution. Yes, to 0. ☐

Exercise 3.25. Show every constant sequence is Cauchy.

Hint. Distances vanish. If monotonicity and boundedness are available, identify the bound and invoke monotone convergence rather than guessing from numerics. ☐

Solution. Every pairwise distance is zero. ☐

Exercise 3.26. If $x_n \to x$, prove $x_{n+1} \to x$.

Hint. It is a tail subsequence. Choose a subsequence by selecting indices where the sequence enters successively smaller neighborhoods of the proposed limit. ☐

Solution. The same epsilon estimate applies. ☐

Exercise 3.27. Find $\lim n/(n+1)$.

Hint. Rewrite the expression. Use the squeeze theorem only after producing upper and lower comparison sequences with the same limit. ☐

Solution. It is 1. ☐

Exercise 3.28. Give a bounded divergent sequence.

Hint. Use alternating signs. A convergent sequence is bounded; use this as a quick necessary-condition test for a proposed example. ☐

Solution. $(-1)^n$ works. ☐

Exercise 3.29. What happens to an increasing bounded real sequence?

Hint. Use monotone convergence. To separate convergence from Cauchy behavior, pay attention to whether the ambient space is complete. ☐

Solution. It converges to the supremum of its range. ☐

Exercise 3.30. Show a Cauchy tail is bounded.

Hint. Use epsilon 1. For an oscillating sequence, inspect even and odd subsequences before deciding whether a global limit exists. ☐

Solution. Eventually all terms lie in a unit ball around one fixed tail term. ☐

3.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 3.31 (Rational sequence limit)**.** Find the limit of $(2n-1)/(n+3)$.

Hint. Divide numerator and denominator by n. ☐

Solution. The limit is 2. ☐

Exercise 3.32 (Choose N)**.** For $a_n = 1/n$, find an N ensuring $|a_n| < 10^{-4}$ for all $n \geq N$.

Hint. Solve $1/n < 10^{-4}$. ☐

Solution. Any $N > 10^4$ works; for instance 10001. ☐

Exercise 3.33 (Subsequence limits)**.** Find subsequential limits of $(-1)^n$.

Hint. Separate even and odd indices. ☐

Solution. They are 1 and -1. ☐

Exercise 3.34 (Geometric Cauchy tail)**.** Bound $|s_m - s_n|$ for $s_n = \sum_{k=0}^{n} 3^{-k}$, $m > n$.

Hint. Use an infinite geometric tail. ☐

Solution. $|s_m - s_n| \leq \sum_{k=n+1}^{\infty} 3^{-k} = \frac{1}{2}\, 3^{-n}$. ☐

Exercise 3.35 (Limsup)**.** Compute limsup and liminf of $a_n = \sin(\pi n/2)$.

Hint. List the periodic values. ☐

Solution. The limsup is 1, liminf is -1. ☐

Proofs

Exercise 3.36 (Convergent implies bounded)**.** Prove every convergent sequence is bounded.

Hint. Put the tail in a unit ball around the limit and absorb finitely many initial terms. ☐

Solution. The tail has bounded distance from the limit; the finite initial segment has a finite maximum norm. ☐

Exercise 3.37 (Convergent implies Cauchy)**.** Prove convergence implies the Cauchy property.

Hint. Put two late terms within $\varepsilon/2$ of the limit. ☐

Solution. The triangle inequality gives $d(x_m, x_n) < \varepsilon$. ☐

Exercise 3.38 (Cauchy boundedness)**.** Prove every Cauchy sequence is bounded.

Hint. Use the Cauchy condition with tolerance 1 for the tail. ☐

Solution. The tail lies in a unit ball around one tail term; finitely many earlier terms add only a finite bound. ☐

Exercise 3.39 (Monotone convergence)**.** Prove an increasing bounded-above real sequence converges.

Hint. Let s be the supremum of its range and use supremum approximation. ☐

Solution. Choose N with $a_N > s - \varepsilon$; monotonicity traps $s - \varepsilon < a_n \leq s$ for $n \geq N$. ☐

Counterexamples and hypothesis testing

Exercise 3.40 (Bounded not convergent)**.** Give a bounded sequence that does not converge.

Hint. Use alternating signs. ☐

Solution. $(-1)^n$ is bounded but has two subsequential limits. ☐

Exercise 3.41 (Cauchy in incomplete space)**.** Give a Cauchy sequence in $\mathbb{Q}$ that does not converge in $\mathbb{Q}$.

Hint. Use rational decimal approximations to $\sqrt{2}$. ☐

Solution. Such approximations are Cauchy in the usual metric but their real limit $\sqrt{2}$ is not rational. ☐

Exercise 3.42 (Subsequence convergence insufficient)**.** Does one convergent subsequence force the full sequence to converge?

Hint. Use $(-1)^n$. ☐

Solution. No; the even subsequence converges to 1, but the full sequence does not converge. ☐

Applications and investigations

Exercise 3.43 (Stopping criterion)**.** If an iterative method has error bound $\|x_n - x_*\| \le 2^{-n}$, how large must n be for error below 10^{-6}?

Hint. Solve $2^{-n} < 10^{-6}$. ☐

Solution. It is enough to take $n \ge 20$, since $2^{-20} \approx 9.54 \times 10^{-7}$. ☐

Exercise 3.44 (Cauchy diagnostics)**.** Why is a computable bound on $\|x_m - x_n\|$ useful when the limit is unknown?

Hint. The Cauchy criterion uses only sequence terms. ☐

Solution. It can prove convergence in a complete space without knowing or being able to evaluate the limit in advance. ☐

Challenge problems

Exercise 3.45 (Cauchy subsequence criterion)**.** Prove a Cauchy sequence with a convergent subsequence converges to the same limit.

Hint. Bridge a late sequence term to a late subsequence term. ☐

Solution. For large n, choose a subsequence index $n_k \ge n$ with both $d(x_n, x_{n_k})$ and $d(x_{n_k}, x)$ below $\varepsilon/2$. ☐

Exercise 3.46 (Limsup convergence criterion)**.** For a bounded real sequence, prove convergence to L iff $\limsup a_n = \liminf a_n = L$.

Hint. Tail suprema and infima squeeze every late term. ☐

Solution. If both tail bounds converge to L, then eventually $L-\varepsilon < a_n < L+\varepsilon$. The converse is immediate from convergence. ☐

Chapter summary

The central definitions, estimates, and structural theorems of this chapter will be used repeatedly in the remaining analysis volume.

Chapter 4

Open and Closed Sets

Open and closed sets encode local and limiting structure without coordinates. Interiors, closures, boundaries, and limit points become the language in which continuity and topology are most naturally expressed.

Learning goals

After this chapter, the reader should be able to:

- decide whether a set is open, closed, both, or neither from the definitions;
- compute interiors, closures, boundaries, and limit-point sets for explicit subsets of metric spaces;
- prove that arbitrary unions of open sets and finite intersections of open sets are open;
- use complements to convert statements about open sets into statements about closed sets;
- construct neighborhood or sequence arguments showing that a point lies in a closure or boundary;
- produce counterexamples showing why infinite intersections of open sets or infinite unions of closed sets need not preserve type;

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{structural theorem} \longrightarrow \text{application.}}$$

4.1 Open sets

A set is open if every one of its points contains a ball lying entirely in the set.

4.2 Closed sets

A set is closed if its complement is open; in metric spaces this is equivalent to containing limits of convergent sequences from the set.

4.3 Interior

The interior is the largest open subset contained in a set.

Example 4.1 (Interior, closure, and boundary in one calculation)**.** For

$$A = [0, 1) \cup \{2\} \subset \mathbb{R},$$

the interior is $(0, 1)$: neither endpoint has a ball contained in A, and the isolated point 2 has none either. The closure is

$$\overline{A} = [0, 1] \cup \{2\},$$

because 1 is approached from the left and 2 is already included. Therefore

$$\partial A = \overline{A} \setminus A^\circ = \{0, 1, 2\}.$$

Checking each exceptional point against neighborhoods verifies the set identities.

4.4 Closure

The closure is the smallest closed set containing the set.

Example 4.2 (Closure by a sequence)**.** Let $A = (0, 1)$. To show $0 \in \overline{A}$, take

$$x_n = \frac{1}{n} \in A, \qquad x_n \to 0.$$

Likewise $1 - 1/n \to 1$, so both endpoints lie in the closure. No point outside $[0, 1]$ can lie in the closure because its positive distance from $[0, 1]$ produces a disjoint open ball. Hence $\overline{A} = [0, 1]$.

The sequence proves inclusion; the positive-distance ball proves exclusion.

4.5 Limit points

A point is a limit point if every punctured neighborhood meets the set.

4.6 Boundary

Boundary points have every neighborhood meeting both the set and its complement.

4.7 Set operations

Arbitrary unions and finite intersections preserve openness; dual statements hold for closed sets.

4.8 Relative topology and density

Relative openness is obtained by intersecting ambient open sets with a subspace, while density means the closure fills the ambient space.

Example 4.3 (Relative openness can differ from ambient openness)**.** Let $Y = [0, 2] \subset \mathbb{R}$ and $A = [0, 1) \subset Y$. The set A is not open in $\mathbb{R}$, but it is open in the subspace Y because

$$A = (-1, 1) \cap Y.$$

At the endpoint 0, a relative neighborhood is allowed to be cut off by the ambient subspace. Relative topology therefore records neighborhoods internal to the space under study, not to the larger containing space.

4.9 Core structural results

Theorem 4.4 (Open-set operations)**.** *Arbitrary unions and finite intersections of open sets are open.*

Proof. For unions inherit a ball from one containing member; for finite intersections take the minimum of finitely many available radii. □

Theorem 4.5 (Sequential closedness)**.** *A metric subset is closed iff it contains every limit of every convergent sequence drawn from it.*

Proof. Closedness prevents a convergent sequence from entering an open complement; conversely a point in the closure but outside the set yields points within $1/n$ converging to it. □

Theorem 4.6 (Closure by balls)**.** $x \in \overline{A}$ *iff every ball centered at* x *meets* A*.*

Proof. A ball missing A places x in the open complement of the closure; the converse is immediate from the same characterization. □

4.10 Worked examples

Example 4.7 (Interval anatomy)**.** Find the interior, closure, and boundary of $(0, 1]$. They are $(0, 1)$, $[0, 1]$, and $\{0, 1\}$.

Example 4.8 (Open balls are open)**.** Prove every open metric ball is open. For $x \in B(a, r)$ choose radius $r - d(a, x) > 0$ and use the triangle inequality.

Example 4.9 (Closed balls are closed)**.** Prove every closed metric ball is closed. If x_n lies in the ball and converges to x, continuity of $y \mapsto d(a, y)$ gives $d(a, x) \leq r$.

Example 4.10 (Sphere is closed)**.** Show $\{x : \|x\| = 1\}$ is closed in a normed space. The norm is continuous by the reverse triangle inequality, so the sphere is the inverse image of the closed singleton $\{1\}$.

4.11 Solved dossiers

Each dossier is a substantial solved problem. Corpus mappings guide topic choice where explicit retained source blocks exist; otherwise the dossier is a fresh canonical completion.

Problem 4.11 (Interval anatomy)**.** Find the interior, closure, and boundary of $(0, 1]$.

Solution. They are $(0, 1)$, $[0, 1]$, and $\{0, 1\}$. □

Problem 4.12 (Open balls are open)**.** Prove every open metric ball is open.

Solution. For $x \in B(a,r)$ choose radius $r - d(a,x) > 0$ and use the triangle inequality. □

Problem 4.13 (Closed balls are closed)**.** Prove every closed metric ball is closed.

Solution. If x_n lies in the ball and converges to x, continuity of $y \mapsto d(a,y)$ gives $d(a,x) \leq r$. □

Problem 4.14 (Sphere is closed)**.** Show $\{x : \|x\| = 1\}$ is closed in a normed space.

Solution. The norm is continuous by the reverse triangle inequality, so the sphere is the inverse image of the closed singleton $\{1\}$. □

Problem 4.15 (Boundary criterion)**.** Show $x \in \partial A$ iff every ball about x meets both A and its complement.

Solution. Membership in the closure gives intersection with A; failure to be interior gives intersection with the complement. □

Problem 4.16 (Closure of rationals)**.** Prove $\overline{\mathbb{Q}} = \mathbb{R}$.

Solution. Rational density says every real ball contains a rational point. □

Problem 4.17 (Interior of rationals)**.** Show $\mathbb{Q}^\circ = \varnothing$.

Solution. Every interval around a rational contains irrational points. □

Problem 4.18 (A clopen subspace piece)**.** In $Y = (-\infty, 0) \cup (0, \infty)$, show $(0, \infty)$ is clopen relative to Y.

Solution. It and its complement are intersections of Y with ambient open half-lines. □

Problem 4.19 (Closure of a finite union)**.** Prove $\overline{A \cup B} = \overline{A} \cup \overline{B}$.

Solution. The right side is closed and contains the union; monotonicity of closure gives the reverse inclusion. □

Problem 4.20 (Interior of an intersection)**.** Prove $(A \cap B)^\circ = A^\circ \cap B^\circ$.

Solution. The right side is open and contained in the intersection; every open subset of the intersection lies in both interiors. □

Problem 4.21 (Limit-point sequence)**.** If x is a limit point of A in a metric space, construct a sequence of points of $A \setminus \{x\}$ converging to x.

Solution. Choose $x_n \in A \cap (B(x, 1/n) \setminus \{x\})$; then $d(x_n, x) < 1/n$. □

Problem 4.22 (Topology synthesis)**.** Why can topology be specified by open sets without mentioning a metric?

Solution. Convergence, closedness, continuity, compactness, connectedness, interior, closure, and boundary can all be defined from the family of open sets. □

4.12 Exercises with complete solutions

The shorter exercise layer is kept separate from the dossiers.

Exercise 4.23. Is $(0, 1)$ open in $\mathbb{R}$?

Hint. Choose a point-dependent radius. Use the metric-ball definition directly: for each point of the set, find a radius whose ball stays inside. □

Solution. Yes. □

Exercise 4.24. Is $[0, 1]$ closed?

Hint. Inspect the complement. To prove closedness, work with the complement or show that every convergent sequence from the set has its limit in the set. □

Solution. Yes. □

Exercise 4.25. Find $\partial(0, 1)$.

Hint. Use closure minus interior. A point is in the closure exactly when every ball around it meets the set; test this neighborhood condition. □

Solution. It is $\{0, 1\}$. □

Exercise 4.26. Find $\overline{\mathbb{Z}}$ in $\mathbb{R}$.

Hint. Is $\mathbb{Z}$ closed? For a boundary point, show that every sufficiently small ball meets both the set and its complement. □

Solution. It equals $\mathbb{Z}$. □

Exercise 4.27. Show $\varnothing$ and X are clopen.

Hint. Use complements. Arbitrary unions preserve openness because one containing open set already supplies the needed neighborhood. □

Solution. Each is open and the complement of the other. □

Exercise 4.28. Are arbitrary intersections of open sets open?

Hint. Use shrinking intervals. Finite intersections preserve openness by taking the minimum of finitely many available radii. □

Solution. No; $\cap_n(-1/n, 1/n) = \{0\}$. □

Exercise 4.29. Find $[0, 1]^\circ$.

Hint. Endpoints fail the ball condition. For an infinite-intersection counterexample, let the open sets shrink toward a boundary point. □

Solution. It is $(0, 1)$. □

Exercise 4.30. Show $A^\circ \subset A \subset \overline{A}$.

Hint. Use the defining extremal properties. Compute interior, closure, and boundary separately; do not infer one from a sketch alone. □

Solution. Both inclusions are immediate from the definitions. □

4.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 4.31 (Interval anatomy)**.** Find interior, closure, and boundary of $(0, 2]$.

Hint. Test the two endpoints separately. ☐

Solution. The interior is $(0, 2)$, closure $[0, 2]$, boundary $\{0, 2\}$. ☐

Exercise 4.32 (Finite set)**.** Find the closure and interior of $\{1, 2, 3\} \subset \mathbb{R}$.

Hint. Finite sets are closed and contain no intervals. ☐

Solution. The closure is the set itself; the interior is empty. ☐

Exercise 4.33 (Rational closure)**.** Find the closure of $\mathbb{Q}$ in $\mathbb{R}$.

Hint. Use density of rationals. ☐

Solution. $\overline{\mathbb{Q}} = \mathbb{R}$. ☐

Exercise 4.34 (Boundary of a ball)**.** In $\mathbb{R}^n$, identify the boundary of $B(0, 1)$.

Hint. Points with norm below one are interior; points with norm above one are exterior. ☐

Solution. The boundary is the unit sphere $\{x : \|x\| = 1\}$. ☐

Exercise 4.35 (Relative topology)**.** In $Y = [0, 1]$, is $[0, 1/2)$ open relative to Y?

Hint. Write it as an intersection with an ambient open interval. ☐

Solution. Yes: $[0, 1/2) = (-1, 1/2) \cap Y$. ☐

Proofs

Exercise 4.36 (Union of open sets)**.** Prove arbitrary unions of open sets are open.

Hint. A point in the union lies in at least one member. ☐

Solution. Use an open ball contained in that member; it is then contained in the union. ☐

Exercise 4.37 (Finite intersection)**.** Prove a finite intersection of open sets is open.

Hint. Take the minimum of finitely many local radii. ☐

Solution. The minimum positive radius gives a ball lying in every member. ☐

Exercise 4.38 (Sequential closedness)**.** Prove a closed metric subset contains limits of its convergent sequences.

Hint. If the limit were outside, use an open complement neighborhood. ☐

Solution. Eventually the sequence would have to lie in the complement, contradicting that all terms lie in the set. ☐

Exercise 4.39 (Closure minimality)**.** Prove $\overline{A}$ is the smallest closed set containing A.

Hint. Show every closed superset contains all closure points. ☐

Solution. If closed $F \supset A$ omitted $x \in \overline{A}$, its open complement would give a neighborhood of x missing A, contradiction. ☐

Counterexamples and hypothesis testing

Exercise 4.40 (Infinite intersection)**.** Give open sets whose infinite intersection is not open.

Hint. Use shrinking intervals around 0. ☐

Solution. $\bigcap_n(-1/n, 1/n) = \{0\}$, not open in $\mathbb{R}$. ☐

Exercise 4.41 (Infinite union closed)**.** Give closed sets whose infinite union is not closed.

Hint. Use $[1/n, 1]$. ☐

Solution. $\bigcup_n[1/n, 1] = (0, 1]$, which is not closed. ☐

Exercise 4.42 (Neither open nor closed)**.** Give a subset of $\mathbb{R}$ that is neither open nor closed.

Hint. Use a half-open interval. ☐

Solution. $[0, 1)$ is neither. ☐

Applications and investigations

Exercise 4.43 (Feasible-set closure)**.** Why is a closed feasible set useful when taking limits of approximating solutions?

Hint. Use sequential closedness. ☐

Solution. If feasible approximants converge, their limit remains feasible. ☐

Exercise 4.44 (Dense sampling)**.** What does density of a sample set $D \subset X$ mean operationally?

Hint. Translate $\overline{D} = X$ into balls. ☐

Solution. Every neighborhood of every target point contains a sample point, so arbitrary local approximation is possible. ☐

Challenge problems

Exercise 4.45 (Boundary identity)**.** Prove $\partial A = \overline{A} \cap \overline{X \setminus A}$.

Hint. A boundary neighborhood must meet both A and its complement. ☐

Solution. The two closure conditions are exactly the two required neighborhood-intersection conditions. ☐

Exercise 4.46 (Closure of union)**.** Prove $\overline{A \cup B} = \overline{A} \cup \overline{B}$.

Hint. Use closedness/minimality or neighborhoods. ☐

Solution. The right side is closed and contains $A \cup B$, giving one inclusion; monotonicity of closure gives the reverse inclusions. ☐

Chapter summary

The central definitions, estimates, and structural theorems of this chapter will be used repeatedly in the remaining analysis volume.

Chapter 5

Metric Spaces and Continuity

Metric spaces isolate the role of distance in analysis. Continuity becomes preservation of local closeness, with equivalent epsilon-delta, sequential, and open-set formulations.

Learning goals

After this chapter, the reader should be able to:

- verify continuity at a point using ε-δ, sequential, and inverse-image criteria;
- prove continuity of compositions and algebraic combinations from structural properties rather than coordinate expansion alone;
- compute Lipschitz constants for explicit maps and use them to obtain continuity estimates;
- distinguish continuity from uniform continuity by constructing or analyzing counterexamples;
- identify which topological properties are preserved under continuous maps in the situations treated in the chapter;
- translate metric estimates into neighborhood statements and back again;

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{structural theorem} \longrightarrow \text{application.}}$$

5.1 Metrics

A metric is nonnegative, definite, symmetric, and satisfies the triangle inequality.

5.2 Metric examples

Norm metrics, discrete metrics, and bounded transforms of metrics show how one set can carry different geometries.

5.3 Metric convergence

A sequence converges exactly when its distance to the limit tends to zero.

5.4 Continuity at a point

Output tolerances are controlled by sufficiently small input tolerances.

Example 5.1 (Epsilon–delta continuity with an explicit local bound)**.** For $f(x) = x^2$, prove continuity at a. If $|x - a| < 1$, then

$$|x + a| \leq |x - a| + 2|a| < 1 + 2|a|.$$

Hence

$$|x^2 - a^2| = |x - a||x + a| < (1 + 2|a|)|x - a|.$$

Given $\varepsilon > 0$, choose

$$\delta = \min\left(1, \frac{\varepsilon}{1 + 2|a|}\right).$$

The preliminary bound on $|x + a|$ is what makes the epsilon choice valid.

5.5 Sequential continuity

In metric spaces continuity is equivalent to preserving sequential limits.

5.6 Open-set continuity

Continuity is equivalent to inverse images of open sets being open.

5.7 Uniform continuity

One input scale must work simultaneously at every point.

Example 5.2 (Continuity versus uniform continuity)**.** The function $f(x) = x^2$ is continuous on $\mathbb{R}$ but not uniformly continuous there. Take

$$x_n = n, \qquad y_n = n + \frac{1}{n}.$$

Then $|x_n - y_n| = 1/n \to 0$, while

$$|f(y_n) - f(x_n)| = 2 + \frac{1}{n^2} \to 2.$$

Thus arbitrarily close inputs can still have outputs separated by a fixed amount when the location drifts to infinity.

5.8 Lipschitz maps

A Lipschitz estimate gives quantitative control and automatically implies uniform continuity.

Example 5.3 (A Lipschitz estimate from the reverse triangle inequality)**.** Fix a nonempty set A in a metric space and define

$$d_A(x) = \inf_{a \in A} d(x, a).$$

For every $a \in A$,

$$d(x, a) \leq d(x, y) + d(y, a).$$

Taking infima gives $d_A(x) \leq d(x, y) + d_A(y)$. Swapping x, y yields

$$|d_A(x) - d_A(y)| \leq d(x, y).$$

Therefore distance to a set is 1-Lipschitz and hence uniformly continuous.

5.9 Core structural results

Theorem 5.4 (Sequential criterion). *A map between metric spaces is continuous iff it preserves limits of sequences.*

Proof. Continuity transfers epsilon estimates to a convergent sequence. Failure of continuity creates points within $1/n$ whose images stay a fixed distance away. ☐

Theorem 5.5 (Open-set criterion). *A map is continuous iff the inverse image of every open set is open.*

Proof. At a point in a preimage, continuity pulls back a small target ball; conversely apply openness to balls around the image point. ☐

Theorem 5.6 (Lipschitz implies uniform continuity). *If $d(f(x), f(y)) \leq Ld(x, y)$, then f is uniformly continuous.*

Proof. For $L > 0$ choose $\delta = \varepsilon/L$; for $L = 0$ the map is constant. ☐

5.10 Worked examples

Example 5.7 (Discrete metric topology). Show every subset is open in the discrete metric. For any point x, the ball $B(x, 1/2)$ is the singleton $\{x\}$.

Example 5.8 (Norm gives a metric). Prove $d(x, y) = \|x - y\|$ is a metric. Definiteness and symmetry come from norm axioms, and the metric triangle inequality is the norm triangle inequality.

Example 5.9 (Distance to a fixed point is Lipschitz). Show $x \mapsto d(x, a)$ is 1-Lipschitz. The triangle inequality gives $|d(x, a) - d(y, a)| \leq d(x, y)$.

Example 5.10 (A sequential discontinuity witness). Define $f(0) = 0$ and $f(x) = 1/x$ otherwise. Show discontinuity at zero. Take $x_n = 1/n \to 0$; then $f(x_n) = n$ does not converge to 0.

5.11 Solved dossiers

Each dossier is a substantial solved problem. Corpus mappings guide topic choice where explicit retained source blocks exist; otherwise the dossier is a fresh canonical completion.

Problem 5.11 (Discrete metric topology). Show every subset is open in the discrete metric.

Solution. For any point x, the ball $B(x, 1/2)$ is the singleton $\{x\}$. ☐

Problem 5.12 (Norm gives a metric). Prove $d(x, y) = \|x - y\|$ is a metric.

Solution. Definiteness and symmetry come from norm axioms, and the metric triangle inequality is the norm triangle inequality. ☐

Problem 5.13 (Distance to a fixed point is Lipschitz). Show $x \mapsto d(x, a)$ is 1-Lipschitz.

Solution. The triangle inequality gives $|d(x, a) - d(y, a)| \leq d(x, y)$. ☐

Problem 5.14 (A sequential discontinuity witness). Define $f(0) = 0$ and $f(x) = 1/x$ otherwise. Show discontinuity at zero.

Solution. Take $x_n = 1/n \to 0$; then $f(x_n) = n$ does not converge to 0. $\square$

Problem 5.15 (Square is not uniformly continuous on $\mathbb{R}$)**.** Produce nearby points with separated squares.

Solution. Take $x_n = n$, $y_n = n + 1/n$; input distance tends to zero while square difference tends to 2. $\square$

Problem 5.16 (Square on a bounded interval)**.** Show x^2 is uniformly continuous on $[-M, M]$.

Solution. $|x^2 - y^2| \leq 2M|x - y|$, so it is Lipschitz there. $\square$

Problem 5.17 (Composition)**.** Prove the composition of continuous maps is continuous.

Solution. Inverse images compose: $(g \circ f)^{-1}(U) = f^{-1}(g^{-1}(U))$. $\square$

Problem 5.18 (A bounded equivalent metric)**.** Show $\rho = d/(1 + d)$ has the same convergent sequences as d.

Solution. $\rho \leq d$, while for small ρ, $d = \rho/(1 - \rho) \leq 2\rho$. $\square$

Problem 5.19 (Closed-set criterion)**.** Show continuous maps pull closed sets back to closed sets.

Solution. Take complements and use the open-set criterion. $\square$

Problem 5.20 (Uniform continuity preserves Cauchy sequences)**.** Prove the claim.

Solution. A uniform delta converts a Cauchy input tail into an epsilon-close image tail. $\square$

Problem 5.21 (Lipschitz diameter control)**.** Show $\operatorname{diam} f(A) \leq L \operatorname{diam} A$.

Solution. Apply the Lipschitz inequality to every pair and take the supremum. $\square$

Problem 5.22 (Continuity synthesis)**.** Why do inverse images appear in the topological definition?

Solution. Inverse images preserve arbitrary unions and finite intersections exactly, matching the axioms for open sets. $\square$

5.12 Exercises with complete solutions

The shorter exercise layer is kept separate from the dossiers.

Exercise 5.23. Verify the triangle inequality for the discrete metric.

Hint. Consider whether endpoints agree. For ε-δ continuity, isolate $d(f(x), f(a))$ and bound it by a constant times $d(x, a)$ when possible. $\square$

Solution. If endpoints differ, at least one leg has length 1. $\square$

Exercise 5.24. Is $f(x) = 3x$ Lipschitz?

Hint. Compute differences. For sequential continuity, start with an arbitrary sequence $x_n \to a$ and push the convergence through the map. $\square$

Solution. Yes, with constant 3. $\square$

Exercise 5.25. Show a constant map is uniformly continuous.

Hint. Output distance is zero. To prove uniform continuity, the chosen δ must depend only on ε, not on the base point. ☐

Solution. Any positive delta works. ☐

Exercise 5.26. Are sums of continuous real functions continuous?

Hint. Use limits or epsilon bounds. A Lipschitz estimate $d(f(x), f(y)) \le Ld(x, y)$ immediately gives a uniform-continuity modulus. ☐

Solution. Yes. ☐

Exercise 5.27. Show $|x|$ is continuous.

Hint. Use reverse triangle inequality. For composition, feed the outer map's ε-requirement into the inner map's continuity estimate. ☐

Solution. It is 1-Lipschitz. ☐

Exercise 5.28. Characterize convergence in the discrete metric.

Hint. Use epsilon 1/2. To disprove uniform continuity, construct pairs whose inputs become arbitrarily close while their outputs stay separated. ☐

Solution. A convergent sequence is eventually constant. ☐

Exercise 5.29. Does uniform continuity imply continuity?

Hint. Fix a point. Use inverse images of open sets when the topology is easier to understand than direct metric estimates. ☐

Solution. Yes. ☐

Exercise 5.30. Give a continuous but not uniformly continuous map on $\mathbb{R}$.

Hint. Use quadratic growth. On compact domains, combine continuity with compactness when a global estimate is needed. ☐

Solution. x^2 is a standard example. ☐

5.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 5.31 (Delta for a linear map)**.** For $f(x) = 5x$, give δ in terms of ε.

Hint. Use $|f(x) - f(a)| = 5|x - a|$. ☐

Solution. Take $\delta = \varepsilon/5$. ☐

Exercise 5.32 (Lipschitz constant)**.** Find a Lipschitz constant for $f(x) = 3x - 7$ on $\mathbb{R}$.

Hint. Subtract function values. ☐

Solution. $|f(x) - f(y)| = 3|x - y|$, so $L = 3$. □

Exercise 5.33 (Discrete continuity)**.** Why is every map from a discrete metric space continuous?

Hint. Use the singleton ball $B(x, 1/2)$. □

Solution. Every point has an open singleton neighborhood, so every inverse image of an open set is open. □

Exercise 5.34 (Sequential test)**.** Show $f(x) = |x|$ is continuous using a metric estimate.

Hint. Use reverse triangle inequality. □

Solution. $||x| - |y|| \le |x - y|$, so f is 1-Lipschitz. □

Exercise 5.35 (Composition constant)**.** If f is L-Lipschitz and g is M-Lipschitz, what constant works for $g \circ f$?

Hint. Apply the two estimates in succession. □

Solution. ML works. □

Proofs

Exercise 5.36 (Composition continuity)**.** Prove a composition of continuous maps is continuous.

Hint. Use inverse images or nested epsilon choices. □

Solution. $(g \circ f)^{-1}(U) = f^{-1}(g^{-1}(U))$, and both inverse-image operations preserve openness. □

Exercise 5.37 (Sequential criterion forward)**.** Prove continuous maps preserve sequence limits.

Hint. Use the epsilon–delta condition after the sequence enters the delta-ball. □

Solution. For any epsilon choose delta from continuity; convergence provides N with inputs delta-close thereafter. □

Exercise 5.38 (Lipschitz uniform continuity)**.** Prove every Lipschitz map is uniformly continuous.

Hint. Choose one delta independent of the base point. □

Solution. For $L > 0$, take $\delta = \varepsilon/L$; constant maps are immediate. □

Exercise 5.39 (Uniformly continuous sends Cauchy to Cauchy)**.** Prove a uniformly continuous map sends Cauchy sequences to Cauchy sequences.

Hint. Use one delta for all late pairs. □

Solution. Choose delta from uniform continuity and then a Cauchy tail whose pairwise distances are below delta. □

Counterexamples and hypothesis testing

Exercise 5.40 (Continuous not uniform)**.** Show x^2 is not uniformly continuous on $\mathbb{R}$.

Hint. Use pairs $n, n + 1/n$. ☐

Solution. Input distances tend to zero while output differences tend to 2. ☐

Exercise 5.41 (Pointwise delta dependence)**.** Why does a proof with $\delta = \varepsilon/(1 + 2|a|)$ establish continuity but not uniform continuity?

Hint. Observe dependence on the base point a. ☐

Solution. Uniform continuity requires one delta working for all a; this choice varies with a. ☐

Exercise 5.42 (Continuous inverse not automatic)**.** Does a continuous bijection always have a continuous inverse?

Hint. Recall that extra hypotheses such as compact-to-Hausdorff are needed. ☐

Solution. No in general; continuity of the inverse is a separate topological property. ☐

Applications and investigations

Exercise 5.43 (Error propagation)**.** If f is 12-Lipschitz and the input error is at most 10^{-3}, bound output error.

Hint. Multiply by the Lipschitz constant. ☐

Solution. The output error is at most 0.012. ☐

Exercise 5.44 (Nearest-set distance)**.** Why is $d_A(x)$ stable under perturbing x?

Hint. Use its 1-Lipschitz property. ☐

Solution. Moving x by at most η changes its distance to A by at most η. ☐

Challenge problems

Exercise 5.45 (Uniform continuity on compact sets)**.** Sketch the contradiction proof that continuous $f : K \to Y$ on compact metric K is uniformly continuous.

Hint. Assume pairs x_n, y_n get arbitrarily close while images stay epsilon-separated; extract a convergent subsequence. ☐

Solution. Compactness gives $x_{n_k} \to x$; then $y_{n_k} \to x$ too. Continuity sends both image subsequences to $f(x)$, contradiction. ☐

Exercise 5.46 (Equivalent metric)**.** If $\rho(x, y) = d(x, y)/(1 + d(x, y))$, explain why d- and ρ-convergence are equivalent.

Hint. Study the scalar transform $t \mapsto t/(1 + t)$ near zero and invert it. ☐

Solution. $\rho \to 0$ iff $d = \rho/(1 - \rho) \to 0$; thus they have the same convergent sequences. ☐

Chapter summary

The central definitions, estimates, and structural theorems of this chapter will be used repeatedly in the remaining analysis volume.

Chapter 6

Compactness

Compactness is the finiteness principle of analysis. It converts infinite covers into finite subcovers, arbitrary sequences into convergent subsequences, and local continuity into global boundedness and uniform control.

Learning goals

After this chapter, the reader should be able to:

- prove compactness using finite-subcover, sequential, or Heine–Borel criteria in the setting where each applies;
- extract convergent subsequences from bounded sequences in Euclidean space;
- use compactness to prove attainment of maxima and minima of continuous functions;
- show that continuous images of compact sets are compact and use this to obtain boundedness or closedness conclusions;
- construct open covers or sequences that demonstrate failure of compactness;
- distinguish compactness from closedness and boundedness outside finite-dimensional Euclidean settings;

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{structural theorem} \longrightarrow \text{application.}}$$

6.1 Open-cover compactness

Every open cover of a compact space has a finite subcover.

6.2 Sequential compactness

In metric spaces compactness is equivalent to every sequence having a convergent subsequence with limit in the space.

Example 6.1 (Sequential compactness extracts useful structure)**.** Consider any sequence (x_n) in the closed unit disk of $\mathbb{R}^2$. The disk is bounded, so Bolzano–Weierstrass supplies a convergent subsequence $x_{n_k} \to x$ in $\mathbb{R}^2$. Because the disk is closed, x remains in the disk. This verifies sequential compactness.

Both hypotheses in Heine–Borel appear explicitly: boundedness extracts a subsequence, while closedness keeps its limit inside the set.

6.3 Heine–Borel

In Euclidean space compact sets are exactly the closed and bounded sets.

6.4 Bolzano–Weierstrass

Every bounded sequence in Euclidean space has a convergent subsequence.

6.5 Continuous images

Continuous images of compact spaces are compact.

6.6 Extreme values

Continuous real-valued functions on compact sets attain maxima and minima.

Example 6.2 (Extreme values turn a supremum into an attained value)**.** Let $f(x) = x(1-x)$ on $[0,1]$. Compactness and continuity guarantee that a maximum exists before we compute it. Completing the square,

$$f(x) = \frac{1}{4} - \left(x - \frac{1}{2}\right)^2 \leq \frac{1}{4}.$$

Equality occurs at $x = 1/2$, so the maximum is $1/4$.

The theorem supplies attainment; the algebra identifies the attaining point.

6.7 Uniform continuity

Continuous maps on compact metric domains are uniformly continuous.

6.8 Completeness and total boundedness

A metric space is compact exactly when it is complete and totally bounded.

Example 6.3 (Total boundedness is stronger than boundedness)**.** In an infinite-dimensional normed space, the closed unit ball is bounded but need not be totally bounded. If it contained an infinite sequence of unit vectors separated by a fixed positive distance, then no finite family of sufficiently small balls could cover it.

This is why compactness is characterized by completeness plus total boundedness, not merely completeness plus ordinary boundedness.

6.9 Core structural results

Theorem 6.4 (Continuous image theorem)**.** *If K is compact and f is continuous, then $f(K)$ is compact.*

Proof. Pull an open cover of the image back to K, extract a finite subcover, and push the corresponding finite family forward. □

Theorem 6.5 (Extreme-value theorem)**.** *A continuous real function on nonempty compact K attains its maximum and minimum.*

Proof. Its image is compact in $\mathbb{R}$, hence closed and bounded; its supremum and infimum belong to the image. □

Theorem 6.6 (Heine–Cantor)**.** *A continuous map on a compact metric domain is uniformly continuous.*

Proof. Otherwise choose pairs arbitrarily close with images separated by a fixed epsilon; a convergent subsequence of one side forces both sides to the same limit, contradicting continuity. □

6.10 Worked examples

Example 6.7 (Open interval is not compact)**.** Use an open cover to show $(0, 1)$ is not compact. The sets $(1/n, 1)$ for $n \geq 2$ cover $(0, 1)$ but no finite subfamily covers points near zero.

Example 6.8 (Sequential noncompactness)**.** Show $(0, 1]$ is not compact using a sequence. The sequence $1/n$ has no subsequence converging to a point of $(0, 1]$.

Example 6.9 (Compact sets are closed)**.** Prove this in metric spaces. For $x \notin K$, the continuous distance to x attains a positive minimum on compact K, leaving a ball around x disjoint from K.

Example 6.10 (Compact sets are bounded)**.** Prove it directly from an open cover. The increasing balls $B(x_0, n)$ cover the space; finitely many reduce to one largest ball.

6.11 Solved dossiers

Each dossier is a substantial solved problem. Corpus mappings guide topic choice where explicit retained source blocks exist; otherwise the dossier is a fresh canonical completion.

Problem 6.11 (Open interval is not compact)**.** Use an open cover to show $(0, 1)$ is not compact.

Solution. The sets $(1/n, 1)$ for $n \geq 2$ cover $(0, 1)$ but no finite subfamily covers points near zero. □

Problem 6.12 (Sequential noncompactness)**.** Show $(0, 1]$ is not compact using a sequence.

Solution. The sequence $1/n$ has no subsequence converging to a point of $(0, 1]$. □

Problem 6.13 (Compact sets are closed)**.** Prove this in metric spaces.

Solution. For $x \notin K$, the continuous distance to x attains a positive minimum on compact K, leaving a ball around x disjoint from K. □

Problem 6.14 (Compact sets are bounded)**.** Prove it directly from an open cover.

Solution. The increasing balls $B(x_0, n)$ cover the space; finitely many reduce to one largest ball. □

Problem 6.15 (Continuous real functions are bounded)**.** Prove boundedness on compact domains.

Solution. The compact image in $\mathbb{R}$ is bounded. □

Problem 6.16 (Positive distance between disjoint compact sets)**.** Prove disjoint compact K, L have positive distance.

Solution. The continuous function $d(x, y)$ on compact $K \times L$ attains a minimum; zero would force a common point. □

Problem 6.17 (Finite intersection property)**.** Prove compact spaces have it for closed sets.

Solution. If the total closed intersection were empty, their complements would give an open cover with a finite subcover, contradicting the finite intersection hypothesis. □

Problem 6.18 (Nested compact sets)**.** Show decreasing nonempty compact sets have nonempty intersection.

Solution. They satisfy the finite intersection property inside the first compact set. □

Problem 6.19 (Heine–Cantor contradiction sequence)**.** Describe the standard sequence proof.

Solution. Failure gives $d(x_n, y_n) < 1/n$ but image distance at least ε_0; compactness and continuity contradict this. □

Problem 6.20 (Bolzano–Weierstrass by nested boxes)**.** Sketch the construction.

Solution. Repeatedly choose a smaller closed box containing infinitely many sequence terms and diameters tending to zero; choose one term from each stage. □

Problem 6.21 (Closed subset of compact is compact)**.** Prove it by covers.

Solution. Add the open complement of the closed subset to any cover, extract a finite cover of the ambient compact set, then discard the complement. □

Problem 6.22 (Compactness synthesis)**.** Why is compactness a finite-extraction principle?

Solution. Its cover definition extracts finitely many neighborhoods, and its sequential form extracts one convergent subsequence from infinitely many terms. □

6.12 Exercises with complete solutions

The shorter exercise layer is kept separate from the dossiers.

Exercise 6.23. Is $\{1/n : n \geq 1\} \cup \{0\}$ compact?

Hint. Check closed and bounded. In $\mathbb{R}^n$, check closedness and boundedness first; Heine–Borel then converts them to compactness. □

Solution. Yes. □

Exercise 6.24. Is $\mathbb{Z}$ compact in $\mathbb{R}$?

Hint. Check boundedness. For sequential compactness, begin with an arbitrary sequence and construct a convergent subsequence. □

Solution. No. ☐

Exercise 6.25. Show every finite metric space is compact.

Hint. Choose one cover member per point. To show a continuous function attains a maximum, apply compactness to its image or to a maximizing sequence. ☐

Solution. Finitely many choices suffice. ☐

Exercise 6.26. Does compactness imply completeness in metric spaces?

Hint. Use a convergent subsequence of a Cauchy sequence. A continuous image of a compact set is compact; use this before trying to prove boundedness directly. ☐

Solution. Yes. ☐

Exercise 6.27. Is $(0, 1)$ compact?

Hint. Use Heine–Borel. To disprove compactness, give either an open cover with no finite subcover or a sequence with no convergent subsequence. ☐

Solution. No. ☐

Exercise 6.28. Is a continuous image of compact closed in a metric codomain?

Hint. Compact subsets of metric spaces are closed. Closed subsets of compact spaces are compact; isolate the ambient compact set if one is available. ☐

Solution. Yes. ☐

Exercise 6.29. Why is $[a, b]$ compact?

Hint. Use Heine–Borel. Compact subsets of metric spaces are closed; separate a point outside the set using a positive distance argument. ☐

Solution. It is closed and bounded. ☐

Exercise 6.30. Must every sequence in a compact set converge?

Hint. Only a subsequence is guaranteed. Do not use Heine–Borel outside finite-dimensional Euclidean space without checking that the theorem still applies. ☐

Solution. No; $(-1)^n$ in $\{-1, 1\}$ is a counterexample. ☐

6.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 6.31 (Heine–Borel test)**.** Is $[0,1] \times [-2,3] \subset \mathbb{R}^2$ compact?

Hint. Check closedness and boundedness. □

Solution. Yes, by Heine–Borel. □

Exercise 6.32 (Noncompact interval)**.** Is $(0,1]$ compact in $\mathbb{R}$?

Hint. Check closedness or use the sequence $1/n$. □

Solution. No; $1/n$ converges to $0 \notin (0,1]$. □

Exercise 6.33 (Continuous image)**.** What can be said about $f(K)$ if K is compact and f continuous?

Hint. Apply the continuous-image theorem. □

Solution. $f(K)$ is compact. □

Exercise 6.34 (Extreme value)**.** Find the maximum of x^2 on $[-2,1]$.

Hint. Compactness guarantees attainment; compare endpoint values. □

Solution. The maximum is 4 at $x = -2$. □

Exercise 6.35 (Finite epsilon-net)**.** Give a $1/2$-net for $[0,1]$.

Hint. Choose finitely many evenly spaced centers. □

Solution. For example $\{0, 1/2, 1\}$ covers $[0,1]$ by radius-$1/2$ balls. □

Proofs

Exercise 6.36 (Continuous image compact)**.** Prove continuous images of compact spaces are compact.

Hint. Pull an open cover back to the domain. □

Solution. The pullback covers K; choose a finite subcover and push the corresponding finite family forward. □

Exercise 6.37 (Compact metric sets bounded)**.** Prove a compact metric space is bounded.

Hint. Cover it by balls $B(x_0, n)$, $n \in \mathbb{N}$. □

Solution. A finite subcover is contained in the largest selected ball. □

Exercise 6.38 (Compact metric sets closed)**.** Prove a compact subset of a metric space is closed.

Hint. For $x \notin K$, use the positive minimum of $d(x,\cdot)$ on K or a finite-ball argument. □

Solution. Distance from x to compact K attains a positive minimum; a smaller ball around x misses K. □

Exercise 6.39 (Heine–Cantor)**.** Prove a continuous map on a compact metric domain is uniformly continuous.

Hint. Argue by sequences of increasingly close pairs and compact subsequences. □

Solution. A compact subsequence converges to one point on both sides; continuity contradicts any fixed output separation. □

Counterexamples and hypothesis testing

Exercise 6.40 (Closed bounded outside Euclidean finite dimension)**.** Why can closed and bounded fail to imply compactness in infinite-dimensional normed spaces?

Hint. Look for an infinite uniformly separated sequence in the unit ball. ☐

Solution. Such a sequence has no Cauchy, hence no convergent, subsequence; the closed unit ball is then not compact. ☐

Exercise 6.41 (Complete not compact)**.** Give a complete metric space that is not compact.

Hint. Use $\mathbb{R}$. ☐

Solution. $\mathbb{R}$ is complete but unbounded, hence not compact. ☐

Exercise 6.42 (Bounded not compact)**.** Give a bounded noncompact subset of $\mathbb{R}$.

Hint. Use an open interval. ☐

Solution. $(0, 1)$ is bounded but not closed and not compact. ☐

Applications and investigations

Exercise 6.43 (Optimization guarantee)**.** Why is compactness valuable before numerically maximizing a continuous objective?

Hint. Use the extreme-value theorem. ☐

Solution. It guarantees a true maximizer exists in the feasible set rather than only a supremum approached by escaping iterates. ☐

Exercise 6.44 (Subsequence extraction)**.** Why does compactness help prove existence from a sequence of approximations?

Hint. Extract a convergent subsequence and pass properties to the limit. ☐

Solution. It turns bounded/controlled approximations into a candidate limit inside the space. ☐

Challenge problems

Exercise 6.45 (Compact iff complete and totally bounded)**.** Explain the difficult direction: complete and totally bounded implies compact in a metric space.

Hint. Use successively finer finite covers to construct a Cauchy subsequence from any sequence. ☐

Solution. Nested selections give a subsequence Cauchy at scales 2^{-k}; completeness gives a limit, hence sequential compactness and compactness. ☐

Exercise 6.46 (Nested compact sets)**.** Prove a decreasing sequence of nonempty compact sets K_n has nonempty intersection if all lie in one compact space.

Hint. Assume empty intersection and use complements as an open cover. ☐

Solution. The complements cover K_1; a finite subcover implies some K_N is empty by nestedness, contradiction. ☐

Chapter summary

The central definitions, estimates, and structural theorems of this chapter will be used repeatedly in the remaining analysis volume.

Chapter 7

Connectedness and Path Connectedness

Connectedness excludes decompositions into separated open pieces, while path connectedness asks for explicit continuous curves between points. These notions capture global cohesion that compactness and completeness do not.

Learning goals

After this chapter, the reader should be able to:

- decide whether an explicitly described set is connected or path connected;
- construct explicit paths joining two points when path connectedness holds;
- use separation arguments or continuous images to prove connectedness;
- prove that intervals in $\mathbb{R}$ are connected and apply the result to intermediate-value arguments;
- construct examples showing that connectedness and path connectedness are not equivalent in complete generality;
- identify connected components or path components in elementary examples;

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{structural theorem} \longrightarrow \text{application.}}$$

7.1 Connected spaces

A space is connected if it admits no separation into disjoint nonempty open pieces.

7.2 Clopen criterion

Connected spaces have no nontrivial subset that is both open and closed.

7.3 Connected subsets of the line

A subset of $\mathbb{R}$ is connected exactly when it is an interval in the order-convex sense.

Example 7.1 (Connected subsets of the line cannot have gaps)**.** Let $A \subset \mathbb{R}$ be connected and suppose $x < y$ lie in A. Take z with $x < z < y$. If $z \notin A$, then

$$A \cap (-\infty, z), \qquad A \cap (z, \infty)$$

are disjoint, nonempty, relatively open subsets whose union is A. That is a separation, contradicting connectedness. Therefore $z \in A$.

This proves the order-convexity mechanism behind the theorem that connected subsets of $\mathbb{R}$ are intervals.

7.4 Continuous images

Continuous maps preserve connectedness.

7.5 Intermediate values

The intermediate-value theorem follows because continuous images of intervals are intervals.

7.6 Path connectedness

A path is a continuous map from $[0, 1]$ joining two prescribed endpoints.

Example 7.2 (Constructing a path, not merely asserting one)**.** The annulus

$$A = \{x \in \mathbb{R}^2 : 1 < \|x\| < 2\}$$

is path connected. Given $x, y \in A$, first move radially from x to a point on the circle of radius $3/2$, travel along that circle to the radial direction of y, then move radially to y. Each piece stays in the annulus, and the pieces can be reparameterized and concatenated continuously.

The explicit construction verifies the path never crosses either forbidden boundary circle.

7.7 Path connected implies connected

A path cannot cross a separation without separating the interval parameter.

7.8 Components

Connected and path components are maximal subsets with the corresponding cohesion property.

Example 7.3 (Components partition the space)**.** For

$$X = \mathbb{R} \setminus \{0, 1\},$$

the connected components are

$$(-\infty, 0), \qquad (0, 1), \qquad (1, \infty).$$

Each interval is connected, and no connected set can cross either missing point because that would create a gap in a connected subset of the line. Thus the components are maximal connected pieces and form a partition of X.

7.9 Core structural results

Theorem 7.4 (Continuous image theorem)**.** *The continuous image of a connected space is connected.*

Proof. A separation of the image pulls back to a separation of the domain. □

Theorem 7.5 (Intervals are connected)**.** *Every real interval is connected.*

Proof. A hypothetical separation and a supremum argument at the interface produce a point that cannot consistently belong to either relatively open side. □

Theorem 7.6 (Path connected implies connected)**.** *Every path-connected space is connected.*

Proof. A separation of the space would pull back along a path joining opposite sides to a separation of $[0, 1]$. □

7.10 Worked examples

Example 7.7 (Clopen criterion)**.** Prove connectedness is equivalent to having only trivial clopen subsets. A nontrivial clopen set and its complement form a separation, and every separation makes each side clopen.

Example 7.8 (Punctured line)**.** Show $\mathbb{R} \setminus \{0\}$ is disconnected. The negative and positive half-lines are disjoint nonempty relatively open sets covering it.

Example 7.9 (Connected subsets of the line)**.** If $A \subset \mathbb{R}$ is connected and $x < z < y$ with $x, y \in A$, show $z \in A$. If $z \notin A$, split A at z into points below and above, creating a separation.

Example 7.10 (Intermediate-value theorem)**.** Derive it from connectedness. The continuous image of $[a, b]$ is connected in $\mathbb{R}$, hence an interval containing all values between endpoint values.

7.11 Solved dossiers

Each dossier is a substantial solved problem. Corpus mappings guide topic choice where explicit retained source blocks exist; otherwise the dossier is a fresh canonical completion.

Problem 7.11 (Clopen criterion)**.** Prove connectedness is equivalent to having only trivial clopen subsets.

Solution. A nontrivial clopen set and its complement form a separation, and every separation makes each side clopen. □

Problem 7.12 (Punctured line)**.** Show $\mathbb{R} \setminus \{0\}$ is disconnected.

Solution. The negative and positive half-lines are disjoint nonempty relatively open sets covering it. □

Problem 7.13 (Connected subsets of the line)**.** If $A \subset \mathbb{R}$ is connected and $x < z < y$ with $x, y \in A$, show $z \in A$.

Solution. If $z \notin A$, split A at z into points below and above, creating a separation. □

Problem 7.14 (Intermediate-value theorem)**.** Derive it from connectedness.

Solution. The continuous image of $[a, b]$ is connected in $\mathbb{R}$, hence an interval containing all values between endpoint values. $\square$

Problem 7.15 (Convex implies path connected)**.** Prove every convex subset of a normed vector space is path connected.

Solution. The straight-line path $(1 - t)x + ty$ remains in the convex set. $\square$

Problem 7.16 (Path components are disjoint)**.** Show two path components are equal or disjoint.

Solution. If they meet, concatenate paths through a common point to show their union is path connected, contradicting maximality unless they coincide. $\square$

Problem 7.17 (Union through a common point)**.** Show a union of connected sets with a common point is connected.

Solution. Any separation would place the common point on one side; each connected member meeting that side must lie entirely there. $\square$

Problem 7.18 (Closure preserves connectedness)**.** Prove the closure of a connected set is connected.

Solution. A separation of the closure would meet the original dense connected set on both sides and induce a separation of it. $\square$

Problem 7.19 (Continuous image of a path)**.** Show continuous maps preserve path connectedness.

Solution. Compose the original path with the continuous map. $\square$

Problem 7.20 (Components are closed)**.** Prove every connected component is closed.

Solution. The closure of a connected set is connected and contains the component; maximality forces equality. $\square$

Problem 7.21 (Products of path-connected spaces)**.** Show $X \times Y$ is path connected when both factors are.

Solution. Pair paths coordinatewise: $t \mapsto (\alpha(t), \beta(t))$. $\square$

Problem 7.22 (Connectedness synthesis)**.** Contrast connectedness with compactness.

Solution. Connectedness excludes a global split into separated regions; compactness excludes infinite escape by finite extraction. Neither implies the other in general. $\square$

7.12 Exercises with complete solutions

The shorter exercise layer is kept separate from the dossiers.

Exercise 7.23. Is $[0, 1] \cup [2, 3]$ connected?

Hint. Find a separation. To prove path connectedness, write an explicit continuous path $\gamma : [0, 1] \to X$ joining arbitrary endpoints. $\square$

Solution. No. ☐

Exercise 7.24. Is $\mathbb{R}$ path connected?

Hint. Use straight lines. Path connectedness implies connectedness, so use a path construction if it is easier than a separation argument. ☐

Solution. Yes. ☐

Exercise 7.25. Are continuous images of connected sets connected?

Hint. Use the theorem. To prove disconnectedness, exhibit two disjoint nonempty relatively open sets whose union is the whole set. ☐

Solution. Yes. ☐

Exercise 7.26. Are connected subsets of $\mathbb{R}$ intervals?

Hint. Use an intermediate-point argument. Continuous images of connected sets are connected; map an interval onto the set when possible. ☐

Solution. Yes. ☐

Exercise 7.27. Show a singleton is connected.

Hint. There is no nontrivial split. An interval cannot be separated because an assumed gap would contradict the least-upper-bound property. ☐

Solution. Its only clopen subsets are trivial. ☐

Exercise 7.28. Is $\mathbb{Q}$ connected?

Hint. Cut at an irrational number. For intermediate-value arguments, apply connectedness to the continuous image and use that connected subsets of $\mathbb{R}$ are intervals. ☐

Solution. No. ☐

Exercise 7.29. Does path connected imply connected?

Hint. Use interval connectedness. To identify components, find maximal connected subsets rather than merely visually separate clusters. ☐

Solution. Yes. ☐

Exercise 7.30. Does connected imply path connected?

Hint. Recall a classical plane counterexample. For a non-path-connected example, test whether every attempted path would force passage through a missing limit point. ☐

Solution. No; the topologist's sine curve is a standard counterexample. ☐

7.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 7.31 (Punctured line)**.** Find the connected components of $\mathbb{R} \setminus \{0\}$.

Hint. Split at the missing point. □

Solution. They are $(-\infty, 0)$ and $(0, \infty)$. □

Exercise 7.32 (Interval path)**.** Give a path in $[a, b]$ from x to y.

Hint. Use linear interpolation. □

Solution. $\gamma(t) = (1-t)x + ty$. □

Exercise 7.33 (Circle path)**.** Give a path on the unit circle from angle α to β.

Hint. Interpolate the angle. □

Solution. $\gamma(t) = (\cos((1-t)\alpha + t\beta), \sin((1-t)\alpha + t\beta))$. □

Exercise 7.34 (Disconnected finite set)**.** Is $\{0, 1\} \subset \mathbb{R}$ connected?

Hint. Use relative singleton neighborhoods. □

Solution. No; the two singletons form a separation. □

Exercise 7.35 (Continuous image)**.** If X is connected and $f : X \to \mathbb{R}$ continuous, what shape is $f(X)$?

Hint. Connected subsets of $\mathbb{R}$ are intervals. □

Solution. $f(X)$ is an interval (possibly a point). □

Proofs

Exercise 7.36 (Continuous image connected)**.** Prove continuous images preserve connectedness.

Hint. Pull back a hypothetical separation. □

Solution. The inverse images would be disjoint nonempty relatively open sets covering the connected domain. □

Exercise 7.37 (Path connected implies connected)**.** Prove path connectedness implies connectedness.

Hint. If a separation existed, choose one point in each side and pull it back along a joining path. □

Solution. The interval $[0, 1]$ would be separated by the inverse images, impossible. □

Exercise 7.38 (Union lemma)**.** Prove a union of connected subsets all sharing one common point is connected.

Hint. A separation of the union would have to split one connected member containing the common point. □

Solution. Every member must lie in the same side as the common point, leaving the other side empty. □

Exercise 7.39 (Intermediate value theorem)**.** Derive the IVT from connectedness.

Hint. Use continuity of $f : [a, b] \to \mathbb{R}$. ☐

Solution. The image is connected, hence an interval containing all values between $f(a)$ and $f(b)$. ☐

Counterexamples and hypothesis testing

Exercise 7.40 (Connected not path connected)**.** Name a standard type of example showing connectedness need not imply path connectedness.

Hint. Think of a topologist's sine curve. ☐

Solution. The closure of the graph $y = \sin(1/x)$ for $x > 0$, with its vertical limit segment, is connected but not path connected. ☐

Exercise 7.41 (Compact not connected)**.** Give a compact disconnected subset of $\mathbb{R}$.

Hint. Use two separated closed intervals. ☐

Solution. $[0, 1] \cup [2, 3]$ is compact and disconnected. ☐

Exercise 7.42 (Complete not connected)**.** Give a complete disconnected metric space.

Hint. Use a finite discrete space. ☐

Solution. $\{0, 1\}$ with the usual or discrete metric is complete but disconnected. ☐

Applications and investigations

Exercise 7.43 (Root existence)**.** Why does connectedness support root-finding when a continuous function changes sign?

Hint. Use IVT on the interval between the sign-change points. ☐

Solution. Zero lies between the two function values, so it belongs to the connected image. ☐

Exercise 7.44 (Configuration spaces)**.** What does path connectedness mean for a configuration space in motion planning?

Hint. Interpret a path as a continuous feasible motion. ☐

Solution. Any two configurations in one path component can be joined by a collision-free continuous trajectory. ☐

Challenge problems

Exercise 7.45 (Product paths)**.** Prove the product of two path-connected spaces is path connected.

Hint. Join coordinates independently. ☐

Solution. If γ_X, γ_Y join the respective coordinates, $t \mapsto (\gamma_X(t), \gamma_Y(t))$ joins the product points. ☐

Exercise 7.46 (Continuous image of component)**.** If C is a connected component of X and $f : X \to Y$ continuous, what can be said about $f(C)$?

Hint. Use continuous-image connectedness, but be careful about maximality. □

Solution. $f(C)$ is connected and therefore lies inside a connected component of Y, though it need not equal the whole component. □

Chapter summary

The central definitions, estimates, and structural theorems of this chapter will be used repeatedly in the remaining analysis volume.

Part II

Calculus

Chapter 8

Differentiability in Several Variables

Multivariable differentiability replaces a one-dimensional slope by the best linear approximation. The derivative is therefore an operator, not merely a list of partial derivatives, and the remainder estimate is the central invariant statement.

Learning goals

After this chapter, the reader should be able to:

- compute Fréchet derivatives and Jacobian matrices of maps between finite-dimensional normed spaces;
- verify differentiability from a linear approximation estimate rather than from partial derivatives alone;
- apply the multivariable chain rule to compositions and coordinate changes;
- compute directional derivatives and distinguish their existence from full differentiability;
- use derivative bounds to obtain local error estimates;
- construct examples where partial or directional derivatives exist but differentiability fails;

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

8.1 Directional and partial derivatives

Partial derivatives probe coordinate directions, while directional derivatives probe arbitrary directions; neither collection alone automatically guarantees differentiability.

8.2 Fréchet differentiability

A map $f : \mathbb{R}^n \to \mathbb{R}^m$ is differentiable at a when $f(a+h) = f(a) + Df(a)h + o(\|h\|)$.

Example 8.1 (Verifying differentiability from the remainder)**.** Let $f(x, y) = x^2 + y^2$. At $a = (a_1, a_2)$,

$$f(a+h) - f(a) - 2a \cdot h = \|h\|_2^2.$$

Therefore

$$\frac{|f(a+h)-f(a)-2a\cdot h|}{\|h\|_2}=\|h\|_2\to 0.$$

Thus

$$Df(a)h=2a\cdot h.$$

The exact remainder $\|h\|^2$ verifies Fréchet differentiability directly, rather than merely listing partial derivatives.

8.3 Jacobian matrices

After choosing standard bases, the derivative is represented by the Jacobian matrix.

8.4 Chain rule

Derivatives compose as linear maps: $D(g\circ f)(a)=Dg(f(a))Df(a)$.

Example 8.2 (Chain rule as matrix composition)**.** Let

$$f(u,v)=(u+v,u-v),\qquad g(x,y)=x^2+y^2.$$

Then

$$Df=\begin{pmatrix}1&1\\1&-1\end{pmatrix},\qquad Dg(x,y)=\begin{pmatrix}2x&2y\end{pmatrix}.$$

At (u,v), the chain rule gives

$$D(g\circ f)=Dg(f(u,v))Df=\begin{pmatrix}4u&4v\end{pmatrix}.$$

Directly $g\circ f=2u^2+2v^2$, whose derivative is the same. The independent calculation checks the order of matrix multiplication.

8.5 Gradient and level sets

For scalar-valued functions, $Df(a)h=\nabla f(a)\cdot h$; gradients are normal to regular level sets.

8.6 Higher derivatives

Second derivatives assemble into the Hessian, a bilinear form governing second-order behavior.

8.7 Taylor approximation

Taylor's formula organizes linear and quadratic approximations plus a controlled remainder.

Example 8.3 (Second-order Taylor approximation with a remainder scale)**.** For $f(x,y)=e^{x+y}$ near $(0,0)$,

$$f(h,k)=1+(h+k)+\frac{1}{2}(h+k)^2+R_3(h,k),$$

where $R_3=O((|h|+|k|)^3)$. The gradient at the origin is $(1,1)$ and the Hessian is

$$\begin{pmatrix}1&1\\1&1\end{pmatrix}.$$

The quadratic term predicts curvature only in the $h+k$ direction, matching the fact that the function depends on $x+y$ alone.

8.8 Critical points

At interior extrema differentiable scalar functions have zero gradient; Hessian definiteness gives second-order classification.

8.9 Core structural results

Theorem 8.4 (Differentiability implies continuity)**.** *If f is differentiable at a, then f is continuous at a.*

Proof. Write $f(a+h) - f(a) = Df(a)h + r(h)$ with $\|r(h)\|/\|h\| \to 0$. The bounded linear term and the remainder both tend to zero. ☐

Theorem 8.5 (Chain rule)**.** *If f is differentiable at a and g at $f(a)$, then $g \circ f$ is differentiable at a with derivative $Dg(f(a))Df(a)$.*

Proof. Insert the linear-plus-remainder expansions for f and g; boundedness of Dg and the smallness of both remainders produce an $o(\|h\|)$ total error. ☐

Theorem 8.6 (Symmetry of mixed partials)**.** *If second partial derivatives are continuous near a point, then mixed partials agree there.*

Proof. Apply the one-variable mean-value theorem twice to a rectangular difference quotient and pass to the limit using continuity. ☐

8.10 Worked examples

Example 8.7 (A Jacobian by linearization)**.** For $f(x, y) = (x^2y, e^{x+y})$, compute $Df(1, 0)$. The Jacobian is $\begin{pmatrix} 2xy & x^2 \\ e^{x+y} & e^{x+y} \end{pmatrix}$, hence at $(1, 0)$ it is $\begin{pmatrix} 0 & 1 \\ e & e \end{pmatrix}$.

Example 8.8 (Directional derivatives without differentiability)**.** Show $f(x, y) = |xy|^{1/2}$ has directional derivatives at the origin but is not differentiable there. Along (tu, tv), $f = |t|\sqrt{|uv|}$, so one-sided behavior depends linearly only in magnitude and fails a genuine linear approximation; along $y = x$, $f(t, t) = |t|$, which is not $o(\|(t, t)\|)$.

Example 8.9 (Gradient normal to a level set)**.** For $F(x, y) = x^2 + 2y^2$, find the normal direction to $F = 3$ at $(1, 1)$. $\nabla F = (2x, 4y)$, so at $(1, 1)$ a normal vector is $(2, 4)$.

Example 8.10 (Chain rule in coordinates)**.** Let $f(t) = (t, t^2)$ and $g(x, y) = xe^y$. Compute $(g \circ f)'(1)$ using the chain rule. $Df(1) = (1, 2)^T$ and $Dg(1, 1) = (e, e)$, so the derivative is $e + 2e = 3e$.

8.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 8.11 (A Jacobian by linearization)**.** For $f(x, y) = (x^2y, e^{x+y})$, compute $Df(1, 0)$.

Solution. The Jacobian is $\begin{pmatrix} 2xy & x^2 \\ e^{x+y} & e^{x+y} \end{pmatrix}$, hence at $(1, 0)$ it is $\begin{pmatrix} 0 & 1 \\ e & e \end{pmatrix}$. ☐

Problem 8.12 (Directional derivatives without differentiability)**.** Show $f(x,y) = |xy|^{1/2}$ has directional derivatives at the origin but is not differentiable there.

Solution. Along (tu, tv), $f = |t|\sqrt{|uv|}$, so one-sided behavior depends linearly only in magnitude and fails a genuine linear approximation; along $y = x$, $f(t,t) = |t|$, which is not $o(\|(t,t)\|)$. □

Problem 8.13 (Gradient normal to a level set)**.** For $F(x,y) = x^2 + 2y^2$, find the normal direction to $F = 3$ at $(1,1)$.

Solution. $\nabla F = (2x, 4y)$, so at $(1,1)$ a normal vector is $(2,4)$. □

Problem 8.14 (Chain rule in coordinates)**.** Let $f(t) = (t, t^2)$ and $g(x,y) = xe^y$. Compute $(g \circ f)'(1)$ using the chain rule.

Solution. $Df(1) = (1,2)^T$ and $Dg(1,1) = (e,e)$, so the derivative is $e + 2e = 3e$. □

Problem 8.15 (Quadratic approximation)**.** Find the Hessian of $f(x,y) = x^2 + xy + 3y^2$.

Solution. $H = \begin{pmatrix} 2 & 1 \\ 1 & 6 \end{pmatrix}$, constant everywhere. □

Problem 8.16 (Critical-point classification)**.** Classify the origin for $f(x,y) = x^2 + 4y^2$.

Solution. The gradient vanishes and the Hessian is positive definite, so the origin is a strict local minimum. □

Problem 8.17 (A saddle)**.** Classify the origin for $f(x,y) = x^2 - y^2$.

Solution. The Hessian has eigenvalues 2 and -2, hence is indefinite; the origin is a saddle. □

Problem 8.18 (Differentiability from continuous partials)**.** Explain why continuous first partial derivatives near a point imply differentiability there.

Solution. Move from the point to a nearby point along coordinate segments and apply the one-variable mean-value theorem; continuity of the partials controls the remainder uniformly by $o(\|h\|)$. □

Problem 8.19 (Derivative of a norm square)**.** For $f(x) = \|x\|_2^2$ on $\mathbb{R}^n$, compute $Df(x)$.

Solution. $Df(x)h = 2x \cdot h$, so the gradient is $2x$. □

Problem 8.20 (Derivative of a matrix map)**.** For $F(A) = A^2$ on square matrices, compute $DF(A)H$.

Solution. Expand $(A+H)^2 - A^2 = AH + HA + H^2$; the linear part is $AH + HA$ and $\|H^2\| = o(\|H\|)$. □

Problem 8.21 (Second-order Taylor formula)**.** Write the quadratic Taylor approximation of e^{x+y} at $(0,0)$.

Solution. Since $e^{x+y} = 1 + (x+y) + \frac{1}{2}(x+y)^2 + o(\|(x,y)\|^2)$, the quadratic approximation is $1 + x + y + \frac{1}{2}(x^2 + 2xy + y^2)$. □

Problem 8.22 (Differentiability synthesis)**.** Why are partial derivatives not the definition of differentiability?

Solution. They test only coordinate directions. Differentiability demands one linear map approximate all small directions simultaneously with an error negligible relative to distance. □

8.12 Exercises with complete solutions

Exercise 8.23. Compute the gradient of $x^2y + y^3$.

Hint. Differentiate coordinatewise. Propose the derivative linear map first, then divide the remainder by $\|h\|$ and show the quotient tends to zero. ☐

Solution. It is $(2xy, x^2 + 3y^2)$. ☐

Exercise 8.24. Find the Jacobian of $(x + y, xy)$.

Hint. Differentiate each component. The Jacobian matrix represents the Fréchet derivative in standard coordinates; compute partial derivatives only after fixing the point. ☐

Solution. $\begin{pmatrix} 1 & 1 \\ y & x \end{pmatrix}$. ☐

Exercise 8.25. Show linear maps are differentiable.

Hint. Try the map itself as derivative. For the chain rule, compose the derivative maps in the same order as the functions. ☐

Solution. For L, $L(a + h) - L(a) - Lh = 0$. ☐

Exercise 8.26. Find the Hessian of $x^2 + xy + y^2$.

Hint. Differentiate twice. A directional derivative examines one line only; full differentiability requires one linear map to control all directions simultaneously. ☐

Solution. $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. ☐

Exercise 8.27. What is $D(f + g)$?

Hint. Use linearity of the approximation. Use a norm estimate on the remainder term rather than checking coordinate limits separately. ☐

Solution. $Df + Dg$. ☐

Exercise 8.28. State the critical-point necessary condition.

Hint. Assume an interior extremum. To disprove differentiability, compare the function along two paths or show the candidate linear approximation leaves a first-order remainder. ☐

Solution. The gradient must vanish. ☐

Exercise 8.29. If $Df(a) = 0$, is a necessarily an extremum?

Hint. Think of a cubic. For a scalar-valued function, identify the derivative with the gradient only after choosing the Euclidean inner product. ☐

Solution. No; $f(x) = x^3$ at 0 is a counterexample. ☐

Exercise 8.30. Compute $D(x \mapsto Ax)$.

Hint. It is linear already. If partial derivatives are continuous near the point, invoke the standard sufficient differentiability theorem instead of rebuilding the proof. ☐

Solution. The derivative is the constant map A. ☐

8.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 8.31 (Jacobian)**.** Compute the Jacobian of $f(x, y) = (xy, x^2 - y)$.

Hint. Differentiate each output component with respect to each input. □

Solution. $Df = \begin{pmatrix} y & x \\ 2x & -1 \end{pmatrix}$. □

Exercise 8.32 (Directional derivative)**.** For $f(x, y) = x^2 + y^2$, compute the directional derivative at $(1, 2)$ in direction $v = (3, 4)$.

Hint. Use $\nabla f \cdot v$. □

Solution. $(2, 4) \cdot (3, 4) = 22$. □

Exercise 8.33 (Gradient)**.** Find the gradient of $f(x, y, z) = xe^{yz}$.

Hint. Differentiate componentwise. □

Solution. $\nabla f = (e^{yz}, xze^{yz}, xye^{yz})$. □

Exercise 8.34 (Chain rule)**.** If $f(t) = (t, t^2)$ and $g(x, y) = x + y^2$, compute $(g \circ f)'(1)$.

Hint. Use $Dg(f(1))Df(1)$. □

Solution. $Dg(1, 1) = (1, 2)$, $Df(1) = (1, 2)^T$, so the derivative is 5. □

Exercise 8.35 (Hessian)**.** Find the Hessian of $f(x, y) = x^2 + 3xy + 2y^2$.

Hint. Differentiate the gradient. □

Solution. $\begin{pmatrix} 2 & 3 \\ 3 & 4 \end{pmatrix}$. □

Proofs

Exercise 8.36 (Differentiability implies continuity)**.** Prove Fréchet differentiability at a implies continuity there.

Hint. Write the linear term plus $o(\|h\|)$. □

Solution. $\|Df(a)h\| \leq \|Df(a)\|\|h\| \to 0$, and the remainder also tends to zero. □

Exercise 8.37 (Derivative uniqueness)**.** Prove the Fréchet derivative at a point is unique.

Hint. Subtract two candidate linear approximations and evaluate along $h = tv$. □

Solution. If L, M both work, $\|(L - M)v\| = \|(L - M)(tv)\|/|t| \to 0$, hence $(L - M)v = 0$ for all v. □

Exercise 8.38 (Gradient normality)**.** Prove the gradient is orthogonal to tangent directions of a regular level set.

Hint. Differentiate $F(\gamma(t)) = \text{constant}$. □

Solution. $\nabla F(\gamma(0)) \cdot \gamma'(0) = 0$. □

Exercise 8.39 (Mixed partial symmetry hypothesis)**.** Why does continuity of second partials matter in the standard mixed-partial theorem?

Hint. The rectangle-difference proof passes nearby mixed derivatives to the limit. □

Solution. Continuity ensures the intermediate mixed derivatives selected by mean-value arguments converge to the value at the point. □

Counterexamples and hypothesis testing

Exercise 8.40 (Partial derivatives insufficient)**.** Why do existing partial derivatives at a point not by themselves imply differentiability?

Hint. Look for a function with different behavior along nonlinear paths. □

Solution. Classical examples have all coordinate partials at the origin but a remainder not $o(\|(x, y)\|)$; directional/coordinate information alone misses joint behavior. □

Exercise 8.41 (Directional derivatives insufficient)**.** Explain why existence of all directional derivatives need not imply Fréchet differentiability.

Hint. Directional derivatives inspect fixed lines only. □

Solution. A function can behave regularly along every line but badly along curved approaches, so no uniform linear remainder estimate exists. □

Exercise 8.42 (Critical point not extremum)**.** Give a differentiable function with zero gradient at a point that is not a local extremum.

Hint. Use a saddle. □

Solution. $f(x, y) = x^2 - y^2$ has $\nabla f(0, 0) = 0$ but changes sign near the origin. □

Applications and investigations

Exercise 8.43 (Linearized error)**.** If $\|Df(a)\| \leq 10$, what first-order output scale corresponds to input perturbation $\|h\| \leq 10^{-3}$?

Hint. Use $\|Df(a)h\| \leq \|Df(a)\|\|h\|$. □

Solution. The linearized output is at most 10^{-2}, plus the smaller-order remainder. □

Exercise 8.44 (Level-set normal)**.** Find a normal to $x^2 + y^2 + z^2 = 9$ at $(1, 2, 2)$.

Hint. Use the gradient. □

Solution. $(2, 4, 4)$, or any nonzero scalar multiple. □

Challenge problems

Exercise 8.45 (Quadratic remainder criterion)**.** Show a map with remainder bound $\|r(h)\| \le C\|h\|^2$ is differentiable with the stated linear part.

Hint. Divide by $\|h\|$. $\square$

Solution. $\|r(h)\|/\|h\| \le C\|h\| \to 0$, which is exactly the Fréchet condition. $\square$

Exercise 8.46 (Derivative of norm squared)**.** In an inner-product space, derive $D(\|x\|^2)_a(h) = 2\langle a, h\rangle$ in the real case.

Hint. Expand $\|a + h\|^2$. $\square$

Solution. $\|a + h\|^2 - \|a\|^2 = 2\langle a, h\rangle + \|h\|^2$, and the final term is $o(\|h\|)$. $\square$

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.

Chapter 9

Inverse and Implicit Function Principles

The inverse and implicit function principles are local nonlinear analogues of solving an invertible linear system. This chapter is a fresh canonical construction because the source atlas contains no mapped legacy chapter for II/09.

Learning goals

After this chapter, the reader should be able to:

- verify the hypotheses of the inverse function theorem by computing the derivative and checking its invertibility at a specified point;
- distinguish local invertibility from global injectivity and construct examples where the former holds without the latter;
- compute the derivative of a local inverse from $D(f^{-1})(f(a)) = Df(a)^{-1}$;
- identify which variables can be solved for in $F(x, y) = 0$ by checking the appropriate derivative block;
- derive and evaluate the implicit-derivative formula $Dg = -D_yF^{-1}D_xF$ in concrete systems;
- analyze failure cases where a singular derivative or rank loss prevents the theorem from applying, without concluding more than the hypotheses justify;

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

9.1 Local invertibility

A nonlinear map with an invertible derivative should behave like its linearization on a sufficiently small neighborhood.

9.2 Inverse function theorem

Invertibility of $Df(a)$ gives a local differentiable inverse near a.

9.3 Derivative of the inverse

The inverse derivative is the inverse linear map: $D(f^{-1})(f(a)) = Df(a)^{-1}$.

Example 9.1 (A genuinely nonlinear local inverse with a computable inverse derivative)**.** Define

$$f(x,y) = (x + y^2,\ y + x^2).$$

At the origin,

$$Df(0,0) = I,$$

so the inverse function theorem gives a local C^1 inverse near 0. Moreover

$$D(f^{-1})(0) = Df(0,0)^{-1} = I.$$

To see the nonlinear content, note that solving $f(x,y) = (u,v)$ is not a linear system: $x = u - y^2$, $y = v - x^2$. The theorem guarantees that these coupled equations have one nearby solution for every nearby (u,v), and that the solution depends smoothly on the data.

9.4 Implicit equations

A system $F(x,y) = 0$ can be solved locally for y as a function of x when the y-Jacobian is invertible.

Example 9.2 (Solving a coupled implicit system for two variables)**.** Let

$$F(t,x,y) = \begin{pmatrix} x + y - t \\ x - y^2 \end{pmatrix}.$$

At $(t,x,y) = (0,0,0)$,

$$D_{(x,y)}F = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \det = -1 \neq 0.$$

Hence there are unique smooth functions $x(t), y(t)$ near 0 solving $F = 0$. Differentiating,

$$D_{(x,y)}F \begin{pmatrix} x' \\ y' \end{pmatrix} + D_tF = 0, \qquad D_tF = \begin{pmatrix} -1 \\ 0 \end{pmatrix}.$$

Solving at 0 gives $x'(0) = 0$, $y'(0) = 1$. This is the matrix-valued implicit derivative formula in a concrete system.

9.5 Implicit differentiation

The derivative of the implicit function is obtained by solving the differentiated linear system.

9.6 Regular level sets

Full-rank derivatives make level sets locally look like Euclidean subspaces.

9.7 Contraction proof strategy

One proof route reformulates the nonlinear equation as a contraction on a small ball.

9.8 Failure modes

Singular derivative, noninjectivity, or rank loss can destroy local solvability or uniqueness.

Example 9.3 (A singular derivative means theorem failure, not automatic impossibility)**.** Consider

$$F(x, y) = y^3 - x.$$

At the origin $F_y(0, 0) = 0$, so the implicit function theorem cannot solve smoothly for y as a function of x there. Nevertheless the equation has the unique continuous solution

$$y = x^{1/3}.$$

The solution is not differentiable at the origin. Thus a singular derivative means the theorem's smooth-local-solvability conclusion is unavailable; it does not by itself mean that no solution or no uniqueness exists.

9.9 Core structural results

Theorem 9.4 (Inverse function theorem)**.** *If $f : \mathbb{R}^n \to \mathbb{R}^n$ is C^1 near a and $Df(a)$ is invertible, then f is a C^1 diffeomorphism between neighborhoods of a and $f(a)$.*

Proof. After composing with $Df(a)^{-1}$, reduce to derivative near the identity. On a sufficiently small ball the nonlinear error has Lipschitz constant below one, so solving $f(x) = y$ becomes a contraction problem. Differentiability of the inverse follows from the linearized equation. □

Theorem 9.5 (Derivative of inverse)**.** *Under the hypotheses of the inverse theorem, $D(f^{-1})(f(a)) = Df(a)^{-1}$.*

Proof. Differentiate $f^{-1} \circ f = \mathrm{id}$ and use the chain rule. □

Theorem 9.6 (Implicit function theorem)**.** *If $F : \mathbb{R}^{n+m} \to \mathbb{R}^m$ is C^1, $F(a, b) = 0$, and $D_yF(a, b)$ is invertible, then near (a, b) the equation $F(x, y) = 0$ has a unique C^1 solution $y = g(x)$.*

Proof. Apply the inverse function theorem to $\Phi(x, y) = (x, F(x, y))$, whose derivative is block triangular with invertible diagonal blocks. □

9.10 Worked examples

Example 9.7 (One-dimensional inverse theorem)**.** Explain why $f'(a) \neq 0$ yields local invertibility for a C^1 real function. Continuity keeps f' of one sign near a, so f is strictly monotone there. The inverse exists locally and differentiating $f(f^{-1}(y)) = y$ gives the inverse derivative.

Example 9.8 (A locally invertible map)**.** For $f(x, y) = (e^x \cos y, e^x \sin y)$, determine where the derivative is invertible. The Jacobian determinant is e^{2x}, always positive, so the derivative is invertible everywhere; global injectivity fails because of 2π-periodicity in y.

Example 9.9 (Local versus global inverse)**.** Why does the inverse function theorem not imply a global inverse for the previous map? The theorem is local. Distinct points differing by $(0, 2\pi k)$ have the same image despite nonsingular derivative everywhere.

Example 9.10 (Implicit circle graph)**.** Near $(0, 1)$ solve $x^2 + y^2 = 1$ for y and compute $y'(0)$. Take $F = x^2 + y^2 - 1$; $F_y = 2y = 2$ at $(0, 1)$, so $y = \sqrt{1 - x^2}$ locally and $y' = -x/y$, hence $y'(0) = 0$.

9.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 9.11 (One-dimensional inverse theorem)**.** Explain why $f'(a) \neq 0$ yields local invertibility for a C^1 real function.

Solution. Continuity keeps f' of one sign near a, so f is strictly monotone there. The inverse exists locally and differentiating $f(f^{-1}(y)) = y$ gives the inverse derivative. ☐

Problem 9.12 (A locally invertible map)**.** For $f(x, y) = (e^x \cos y, e^x \sin y)$, determine where the derivative is invertible.

Solution. The Jacobian determinant is e^{2x}, always positive, so the derivative is invertible everywhere; global injectivity fails because of 2π-periodicity in y. ☐

Problem 9.13 (Local versus global inverse)**.** Why does the inverse function theorem not imply a global inverse for the previous map?

Solution. The theorem is local. Distinct points differing by $(0, 2\pi k)$ have the same image despite nonsingular derivative everywhere. ☐

Problem 9.14 (Implicit circle graph)**.** Near $(0, 1)$ solve $x^2 + y^2 = 1$ for y and compute $y'(0)$.

Solution. Take $F = x^2 + y^2 - 1$; $F_y = 2y = 2$ at $(0, 1)$, so $y = \sqrt{1 - x^2}$ locally and $y' = -x/y$, hence $y'(0) = 0$. ☐

Problem 9.15 (Implicit derivative formula)**.** Derive $Dg(a) = -D_yF(a, b)^{-1}D_xF(a, b)$.

Solution. Differentiate $F(x, g(x)) = 0$: $D_xF + D_yF\, Dg = 0$, then solve for Dg. ☐

Problem 9.16 (Regular level curve)**.** Show $x^2 + y^2 = 1$ is locally a smooth curve at every point.

Solution. The gradient $(2x, 2y)$ never vanishes on the unit circle, so one partial derivative is nonzero and the implicit theorem applies. ☐

Problem 9.17 (A singular failure)**.** What happens to $F(x, y) = y^2 - x^3 = 0$ at the origin?

Solution. $D_yF(0, 0) = 0$, so the theorem does not apply. The cusp cannot be represented as a single smooth function $y = g(x)$ through the origin. ☐

Problem 9.18 (Coordinate straightening)**.** Explain geometrically what the implicit theorem says about a regular level set.

Solution. After a local C^1 change of coordinates, the equation becomes the coordinate condition $y = 0$, so the level set is locally flat. ☐

Problem 9.19 (Newton linearization link)**.** Relate inverse-theorem solvability to Newton's method.

Solution. Both use invertibility of the derivative to solve a linearized correction equation. The theorem supplies local uniqueness and stability; Newton iterates attempt to realize that correction computationally. ☐

Problem 9.20 (Parameter dependence)**.** If $F(x, y) = y^3 + y - x$, show y is a smooth local function of x everywhere on the solution set.

Solution. $F_y = 3y^2 + 1 > 0$ everywhere, so the implicit theorem applies at every solution point. ☐

Problem 9.21 (Rank perspective)**.** Why is the determinant condition coordinate-dependent notation for an invariant statement?

Solution. Nonzero determinant means the derivative linear map is an isomorphism. Under basis changes the determinant is multiplied by nonzero factors, so invertibility itself is invariant. ☐

Problem 9.22 (IFT synthesis)**.** Summarize inverse and implicit theorems as nonlinear linear algebra.

Solution. An invertible derivative lets one solve all variables locally; an invertible derivative block lets one solve selected variables locally. Both are perturbative versions of solving nonsingular linear systems. ☐

9.12 Exercises with complete solutions

Exercise 9.23. Can $x \mapsto x^3$ use the inverse theorem at 0?

Hint. Check the derivative. Compute $Df(a)$ first and check whether it is an isomorphism; in coordinates this is the nonzero-Jacobian-determinant test. ☐

Solution. No; $f'(0) = 0$, although a continuous inverse exists globally. ☐

Exercise 9.24. Find the inverse derivative formula in one dimension.

Hint. Differentiate an identity. Differentiate $f^{-1} \circ f = \mathrm{id}$ at the base point and solve the resulting linear identity for $D(f^{-1})$. ☐

Solution. $(f^{-1})'(f(a)) = 1/f'(a)$. ☐

Exercise 9.25. For $F(x, y) = x + y + xy$, compute $F_y(0, 0)$.

Hint. Differentiate in y. For $F(x, y) = 0$, compute the derivative only with respect to the variables you want to solve for and check that block for invertibility. ☐

Solution. It is 1, so y is locally a function of x. ☐

Exercise 9.26. Solve the previous equation explicitly.

Hint. Factor in y. After verifying D_yF is invertible, differentiate $F(x, g(x)) = 0$ and solve the linear equation for Dg. ☐

Solution. $y = -x/(1 + x)$ near 0. ☐

Exercise 9.27. Why does nonsingular Jacobian imply local openness?

Hint. Use local diffeomorphism. A nonsingular derivative gives only local invertibility; test global injectivity separately, for example by looking for periodicity or symmetry. ☐

Solution. A local diffeomorphism maps neighborhoods to neighborhoods. ☐

Exercise 9.28. State a failure of global injectivity with nonsingular derivative.

Hint. Use periodicity. If the derivative is singular, conclude only that the theorem does not apply; then inspect the equation directly before claiming nonexistence or nonuniqueness. ☐

Solution. The exponential-polar map is a standard example. ☐

Exercise 9.29. What derivative block matters for solving $F(x, y) = 0$ for y?

Hint. Differentiate with respect to the solved variables. For a regular level set, check surjectivity of the full derivative and identify which coordinate block can be used to graph the set locally. ☐

Solution. $D_y F$. ☐

Exercise 9.30. What is a regular value condition?

Hint. Think surjectivity. To connect the theorem with contraction mapping, normalize the derivative to the identity and estimate the nonlinear remainder on a sufficiently small ball. ☐

Solution. At every point of the level set, the derivative is surjective. ☐

9.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 9.31 (Jacobian invertibility)**.** For $f(x, y) = (x + y, x - y)$, compute Df and decide whether the inverse theorem applies.

Hint. Compute the determinant. ☐

Solution. $Df = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ with determinant -2, so it applies everywhere. ☐

Exercise 9.32 (Inverse derivative)**.** If $Df(a) = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$, compute $D(f^{-1})(f(a))$.

Hint. Invert the triangular matrix. ☐

Solution. $\begin{pmatrix} 1/2 & -1/6 \\ 0 & 1/3 \end{pmatrix}$. ☐

Exercise 9.33 (Implicit slope)**.** For $F(x, y) = x^2 + xy + y^2 - 3$, compute dy/dx where $F_y \neq 0$.

Hint. Use $y' = -F_x/F_y$. ☐

Solution. $y' = -(2x + y)/(x + 2y)$. ☐

Exercise 9.34 (Solve-for block)**.** For $F : \mathbb{R}^3 \to \mathbb{R}$, which derivative must be nonzero to solve $F(x, y, z) = 0$ locally for z?

Hint. Differentiate with respect to the variable being solved for. ☐

Solution. $F_z \neq 0$. ☐

Exercise 9.35 (Regular level set)**.** Is 0 a regular value of $F(x, y) = x^2 + y^2 - 1$?

Hint. Check the gradient on the zero set. ☐

Solution. Yes; $\nabla F = (2x, 2y)$ never vanishes on the unit circle. ☐

Proofs

Exercise 9.36 (Inverse derivative formula)**.** Derive $D(f^{-1})(f(a)) = Df(a)^{-1}$.

Hint. Differentiate $f^{-1} \circ f = \mathrm{id}$. □

Solution. The chain rule gives $D(f^{-1})(f(a))Df(a) = I$, so multiply by the inverse. □

Exercise 9.37 (Implicit derivative formula)**.** Derive $Dg = -D_yF^{-1}D_xF$ from $F(x, g(x)) = 0$.

Hint. Differentiate the identity and solve the resulting linear equation. □

Solution. $D_xF + D_yF\,Dg = 0$, hence the formula. □

Exercise 9.38 (Local openness)**.** Why does an invertible derivative imply f maps a sufficiently small neighborhood of a onto a neighborhood of $f(a)$?

Hint. Use the local diffeomorphism conclusion of the inverse theorem. □

Solution. The local inverse is defined on an open neighborhood of $f(a)$, so that neighborhood lies in the image. □

Exercise 9.39 (Regular graph dimension)**.** If $F : \mathbb{R}^{n+m} \to \mathbb{R}^m$ has full y-rank, why is the solution set locally n-dimensional?

Hint. The implicit theorem graphs $y = g(x)$ over n free x-variables. □

Solution. A local parametrization $x \mapsto (x, g(x))$ uses exactly n parameters. □

Counterexamples and hypothesis testing

Exercise 9.40 (Local not global)**.** Give a smooth map with invertible derivative everywhere that is not globally injective.

Hint. Use exponential polar coordinates. □

Solution. $(x, y) \mapsto (e^x \cos y, e^x \sin y)$ has determinant $e^{2x} > 0$ but is 2π-periodic in y. □

Exercise 9.41 (Inverse exists but theorem fails)**.** Show $f(x) = x^3$ has a global inverse although the inverse theorem fails at 0.

Hint. Compute $f'(0)$ and the inverse. □

Solution. $f'(0) = 0$, so the theorem does not apply; nevertheless $f^{-1}(y) = y^{1/3}$ exists, though its derivative blows up at 0. □

Exercise 9.42 (Singular implicit block)**.** For $F(x, y) = y^2 - x^2$, explain what fails at the origin when solving for y.

Hint. Compute $F_y(0, 0)$ and inspect branches. □

Solution. $F_y = 2y = 0$; the zero set has two branches $y = \pm x$, so local uniqueness as one graph fails. □

Applications and investigations

Exercise 9.43 (Sensitivity of inverse)**.** What does a large norm of $Df(a)^{-1}$ suggest about local inversion?

Hint. The inverse derivative controls first-order amplification. ☐

Solution. Small output perturbations can require large input changes, indicating local ill-conditioning. ☐

Exercise 9.44 (Newton correction)**.** How is Newton's step related to the inverse theorem's linearization?

Hint. Solve the linearized equation $Df(x_k)\Delta = -f(x_k)$. ☐

Solution. Newton uses the derivative inverse to correct the current residual, mirroring the theorem's local linear solvability mechanism. ☐

Challenge problems

Exercise 9.45 (Polar map local charts)**.** For $f(x,y) = (e^x \cos y, e^x \sin y)$, describe a neighborhood on which f is injective.

Hint. Restrict y to an interval shorter than 2π and stay away from its endpoints. ☐

Solution. Any sufficiently small rectangle around a point, with y-width below 2π, gives a local chart; the inverse theorem guarantees such a neighborhood. ☐

Exercise 9.46 (Implicit second derivative)**.** For $x^2 + y^2 = 1$ near $(0,1)$, compute $y''(0)$.

Hint. From $y' = -x/y$, differentiate once more or use $y = \sqrt{1 - x^2}$. ☐

Solution. $y''(0) = -1$. ☐

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.

Chapter 10

Riemann Integration

Riemann integration quantifies accumulation by controlled finite sums. The theory rests on upper and lower sums, oscillation, and the principle that sufficiently fine partitions suppress local variation.

Learning goals

After this chapter, the reader should be able to:

- compute upper and lower sums for explicit partitions and bounded functions;
- prove Riemann integrability by controlling $U(f, P) - L(f, P)$;
- evaluate elementary Riemann integrals from partitions, antiderivatives, or comparison arguments;
- show that monotone or continuous functions on compact intervals are Riemann integrable;
- construct bounded functions that fail to be Riemann integrable and identify the obstruction;
- use additivity and comparison properties of the integral to derive estimates;

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

10.1 Partitions and sums

Tagged partitions produce Riemann sums; upper and lower sums eliminate dependence on tags.

10.2 Riemann integrability

A bounded function is integrable when upper and lower integrals coincide.

10.3 Darboux criterion

Integrability is equivalent to making the gap between upper and lower sums arbitrarily small.

Example 10.1 (A Darboux-gap proof for a Lipschitz function)**.** Suppose f is L-Lipschitz on $[a, b]$. On a partition interval of length Δx_i, its oscillation is at most $L\Delta x_i$. Therefore

$$U(f, P) - L(f, P) \leq L \sum_i (\Delta x_i)^2 \leq L \|P\| \sum_i \Delta x_i = L(b - a)\|P\|.$$

Choosing mesh

$$\|P\| < \frac{\varepsilon}{L(b - a)}$$

makes the Darboux gap below ε, proving integrability with an explicit quantitative criterion.

10.4 Continuous functions

Uniform continuity on a compact interval makes continuous functions Riemann integrable.

Example 10.2 (Uniform continuity drives continuous integrability)**.** Let f be continuous on $[a, b]$. Given $\varepsilon > 0$, uniform continuity supplies $\delta > 0$ such that intervals shorter than δ have oscillation below $\varepsilon/(b - a)$. Choose a partition with mesh below δ. Then

$$U - L \leq \frac{\varepsilon}{b - a} \sum_i \Delta x_i = \varepsilon.$$

The proof reveals exactly where compactness enters: it upgrades pointwise continuity to one uniform spatial scale.

10.5 Monotone functions

Monotonicity controls total oscillation and also guarantees integrability.

10.6 Algebra of integrable functions

Sums, scalar multiples, products, and absolute values preserve integrability under standard hypotheses.

10.7 Fundamental theorem

Integration and differentiation are inverse operations for continuous integrands.

Example 10.3 (Fundamental theorem checked by a difference quotient)**.** Let

$$F(x) = \int_0^x t^2 \, dt.$$

Then

$$\frac{F(x + h) - F(x)}{h} = \frac{1}{h} \int_x^{x+h} t^2 \, dt.$$

For $h > 0$, the average lies between the minimum and maximum of t^2 on the shrinking interval; both tend to x^2 as $h \to 0$. The same holds for $h < 0$. Therefore $F'(x) = x^2$, illustrating the local-average mechanism of the fundamental theorem.

10.8 Improper limits preview

Classical Riemann integration on compact intervals is the base case for later extension to unbounded domains or singular integrands.

10.9 Core structural results

Theorem 10.4 (Darboux criterion)**.** *A bounded function on $[a, b]$ is Riemann integrable iff for every $\varepsilon > 0$ there is a partition P with $U(f, P) - L(f, P) < \varepsilon$.*

Proof. One direction is immediate from equality of upper and lower integrals. Conversely a partition with small gap traps all finer upper and lower sums within epsilon, forcing the two integrals to coincide. □

Theorem 10.5 (Continuous integrability)**.** *Every continuous function on a compact interval is Riemann integrable.*

Proof. Uniform continuity makes oscillation on sufficiently short subintervals uniformly small; summing oscillation times interval length controls the Darboux gap. □

Theorem 10.6 (Fundamental theorem, Part I)**.** *If f is continuous on $[a, b]$ and $F(x) = \int_a^x f(t)dt$, then $F'(x) = f(x)$.*

Proof. The difference quotient is an average value of f over a shrinking interval; continuity makes that average converge to $f(x)$. □

10.10 Worked examples

Example 10.7 (Integral of a step function)**.** Compute the integral of $f = 2$ on $[0, 1)$ and $f = 5$ on $[1, 3]$. The integral is area sum $2 \cdot 1 + 5 \cdot 2 = 12$.

Example 10.8 (Darboux gap estimate)**.** If oscillation of f on every partition interval is at most η, bound $U - L$. The gap is at most $\eta \sum \Delta x_i = \eta(b - a)$.

Example 10.9 (Continuous integrability mechanism)**.** Explain where compactness enters the proof. Continuity on $[a, b]$ becomes uniform continuity, so one partition scale works simultaneously across the whole interval.

Example 10.10 (Monotone integrability)**.** Show an increasing bounded function on $[a, b]$ is integrable. For an equal partition into n pieces, the upper-lower gap telescopes to $(b - a)(f(b) - f(a))/n$.

10.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 10.11 (Integral of a step function)**.** Compute the integral of $f = 2$ on $[0, 1)$ and $f = 5$ on $[1, 3]$.

Solution. The integral is area sum $2 \cdot 1 + 5 \cdot 2 = 12$. □

Problem 10.12 (Darboux gap estimate)**.** If oscillation of f on every partition interval is at most η, bound $U - L$.

Solution. The gap is at most $\eta \sum \Delta x_i = \eta(b - a)$. □

Problem 10.13 (Continuous integrability mechanism)**.** Explain where compactness enters the proof.

Solution. Continuity on $[a, b]$ becomes uniform continuity, so one partition scale works simultaneously across the whole interval. $\square$

Problem 10.14 (Monotone integrability)**.** Show an increasing bounded function on $[a, b]$ is integrable.

Solution. For an equal partition into n pieces, the upper-lower gap telescopes to $(b-a)(f(b)-f(a))/n$. $\square$

Problem 10.15 (Integral of x)**.** Compute $\int_0^1 x\,dx$ from equal Riemann sums.

Solution. Right-endpoint sums are $n^{-2}\sum_{k=1}^{n} k = (n+1)/(2n) \to 1/2$. $\square$

Problem 10.16 (A single discontinuity)**.** Why is a bounded function with one jump discontinuity still Riemann integrable?

Solution. Isolate the jump in an interval of arbitrarily small length; on the remaining compact pieces continuity or small oscillation controls the Darboux gap. $\square$

Problem 10.17 (Dirichlet function)**.** Why is $1_{\mathbb{Q}}$ not Riemann integrable on any nontrivial interval?

Solution. Every subinterval contains rationals and irrationals, so every upper sum is the interval length and every lower sum is zero. $\square$

Problem 10.18 (Absolute-value inequality)**.** Prove $|\int f| \le \int |f|$ for Riemann-integrable f.

Solution. Pointwise $-|f| \le f \le |f|$; monotonicity of the integral gives the estimate. $\square$

Problem 10.19 (Mean value for integrals)**.** If f is continuous on $[a, b]$, show $\int_a^b f = f(c)(b-a)$ for some c.

Solution. The integral average lies between the minimum and maximum of f; the intermediate-value theorem gives a point attaining that average. $\square$

Problem 10.20 (FTC computation)**.** Evaluate $\int_0^1 3x^2\,dx$.

Solution. An antiderivative is x^3, so the integral is 1. $\square$

Problem 10.21 (Additivity)**.** Explain why $\int_a^c f + \int_c^b f = \int_a^b f$.

Solution. Combine partitions on the two subintervals or compare Riemann sums; the common endpoint has no effect on the sum. $\square$

Problem 10.22 (Integration synthesis)**.** Why is Riemann integration a limit of finite-dimensional data?

Solution. Each sum uses finitely many sample values and interval lengths; integrability asserts that all sufficiently fine such finite approximations converge to one common number. $\square$

10.12 Exercises with complete solutions

Exercise 10.23. Compute $\int_0^2 1\,dx$.

Hint. Area of a rectangle. Write the upper and lower sums from the suprema and infima on each subinterval; keep the partition lengths explicit. □

Solution. It is 2. □

Exercise 10.24. Compute $\int_0^1 x^2\,dx$.

Hint. Use FTC. To prove integrability, aim directly at $U(f, P) - L(f, P) < \varepsilon$ rather than at the value of the integral. □

Solution. It is 1/3. □

Exercise 10.25. Is every continuous function on $[a, b]$ integrable?

Hint. Use compactness. For a monotone function, bound the oscillation on each subinterval by endpoint differences and sum the resulting telescoping estimate. □

Solution. Yes. □

Exercise 10.26. Is every bounded function integrable?

Hint. Recall Dirichlet's example. For a continuous function on a compact interval, use uniform continuity to make every subinterval oscillation small. □

Solution. No. □

Exercise 10.27. State additivity over adjacent intervals.

Hint. Split at c. Use additivity over adjacent intervals to reduce a complicated integral to simpler pieces. □

Solution. $\int_a^b = \int_a^c + \int_c^b$. □

Exercise 10.28. If $f \geq 0$, what can be said about the integral?

Hint. Use monotonicity. Comparison follows from comparing upper or lower sums term by term. □

Solution. The integral is nonnegative. □

Exercise 10.29. Does changing one function value change the Riemann integral?

Hint. Think of partition sums. To show nonintegrability, prove that every partition leaves a fixed positive gap between upper and lower sums. □

Solution. No, for an integrable function a finite modification does not change the integral. □

Exercise 10.30. What is $\int_a^a f$?

Hint. Zero interval length. For a tagged Riemann-sum computation, separate the mesh-size estimate from the sample-point choice. □

Solution. It is 0. □

10.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 10.31 (Step integral)**.** Integrate the function equal to 2 on $[0, 1)$ and 5 on $[1, 2]$.

Hint. Sum rectangle areas. ☐

Solution. The integral is $2 \cdot 1 + 5 \cdot 1 = 7$. ☐

Exercise 10.32 (Upper lower sums)**.** For $f(x) = x$ on $[0, 1]$ with partition $\{0, 1/2, 1\}$, compute Darboux upper and lower sums.

Hint. Use monotonicity on each subinterval. ☐

Solution. $L = 0(1/2) + (1/2)(1/2) = 1/4$; $U = (1/2)(1/2) + 1(1/2) = 3/4$. ☐

Exercise 10.33 (Integral bound)**.** If $|f| \le 3$ on $[1, 5]$, bound $|\int_1^5 f|$.

Hint. Use interval length times sup bound. ☐

Solution. At most $4 \cdot 3 = 12$. ☐

Exercise 10.34 (Monotone gap)**.** For increasing f and equal n-partition of $[a, b]$, what Darboux-gap bound arises?

Hint. The differences telescope. ☐

Solution. $(b - a)(f(b) - f(a))/n$. ☐

Exercise 10.35 (FTC)**.** If $F(x) = \int_0^x \cos t \, dt$, compute $F'(x)$.

Hint. Use the fundamental theorem. ☐

Solution. $F'(x) = \cos x$. ☐

Proofs

Exercise 10.36 (Darboux refinement)**.** Explain why refining a partition cannot increase the lower sum or decrease the upper sum.

Hint. On smaller intervals, infima can only rise and suprema can only fall. ☐

Solution. Summing the refined contributions gives $L(P) \le L(P') \le U(P') \le U(P)$. ☐

Exercise 10.37 (Continuous integrability)**.** Prove continuous functions on compact intervals are Riemann integrable.

Hint. Use uniform continuity to control oscillations on a fine partition. ☐

Solution. Choose mesh small enough that each oscillation is below $\varepsilon/(b-a)$, making $U - L < \varepsilon$. ☐

Exercise 10.38 (Monotone integrability)**.** Prove a monotone bounded function on $[a, b]$ is Riemann integrable.

Hint. Use equal partitions and telescope the oscillation sum. ☐

Solution. The Darboux gap is bounded by $(b-a)(f(b)-f(a))/n \to 0$. ☐

Exercise 10.39 (Integral comparison)**.** If $f \le g$ and both are integrable, prove $\int f \le \int g$.

Hint. Compare lower/upper sums or integrate $g - f \ge 0$. ☐

Solution. Nonnegativity of $g - f$ gives $\int(g - f) \ge 0$, hence the result. ☐

Counterexamples and hypothesis testing

Exercise 10.40 (Dirichlet function)**.** Why is the indicator of $\mathbb{Q} \cap [0, 1]$ not Riemann integrable?

Hint. Every interval contains rational and irrational points. ☐

Solution. Every lower sum is 0 and every upper sum is 1, so the Darboux gap never shrinks. ☐

Exercise 10.41 (Bounded insufficient)**.** Does boundedness alone imply Riemann integrability?

Hint. Use the rational indicator. ☐

Solution. No; the Dirichlet function is bounded but nowhere locally low-oscillation. ☐

Exercise 10.42 (Unbounded classical case)**.** Why is $1/\sqrt{x}$ on $(0, 1]$ not covered by the classical bounded Riemann theory on $[0, 1]$?

Hint. Check boundedness near zero. ☐

Solution. It is unbounded at 0; its integral is treated as an improper limit, not a classical bounded Riemann integral on the closed interval. ☐

Applications and investigations

Exercise 10.43 (Numerical quadrature error intuition)**.** Why does small oscillation on partition cells support accurate Riemann-sum quadrature?

Hint. Upper and lower sums squeeze all tagged sums. ☐

Solution. If $U - L$ is small, every tagged sum lies in a narrow interval around the integral. ☐

Exercise 10.44 (Accumulated error)**.** If a uniform integrand approximation has error at most η on $[a, b]$, bound integral error.

Hint. Integrate the absolute difference. ☐

Solution. The error is at most $(b - a)\eta$. ☐

Challenge problems

Exercise 10.45 (Lipschitz integrability rate)**.** Derive a mesh bound guaranteeing Darboux gap below ε for an L-Lipschitz function.

Hint. Oscillation on a cell is at most $L\Delta x_i$. ☐

Solution. $U - L \le L(b - a)\|P\|$, so take $\|P\| < \varepsilon/[L(b - a)]$. ☐

Exercise 10.46 (Zero integral nonnegative function)**.** If continuous $f \ge 0$ on $[a, b]$ and $\int f = 0$, prove $f \equiv 0$.

Hint. If $f(x_0) > 0$, continuity gives a neighborhood with a positive lower bound. ☐

Solution. That neighborhood would contribute positive area, contradicting zero integral. ☐

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.

Part III

Sequences of Functions

Chapter 11

Pointwise and Uniform Convergence

Pointwise convergence controls each input separately; uniform convergence controls all inputs at one common scale. This distinction determines which global properties survive passage to limits.

Learning goals

After this chapter, the reader should be able to:

- distinguish pointwise from uniform convergence using definitions and explicit examples;
- prove uniform convergence by estimating $\sup_x |f_n(x) - f(x)|$;
- use uniform convergence to pass continuity from f_n to the limit;
- construct examples where pointwise convergence does not preserve continuity or bounded error control;
- apply the uniform Cauchy criterion to sequences of functions;
- determine whether convergence is uniform on a whole domain or only on restricted subsets;

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

11.1 Pointwise convergence

For each fixed x, the numerical sequence $f_n(x)$ converges.

11.2 Uniform convergence

One index N works for every x simultaneously.

Example 11.1 (Supremum error gives a uniform convergence rate)**.** On $[0, 1]$, let

$$f_n(x) = \frac{x}{n + x}.$$

Then

$$0 \leq f_n(x) \leq \frac{1}{n + 1},$$

and the maximum occurs at $x = 1$. Therefore

$$\|f_n\|_\infty = \frac{1}{n+1} \to 0.$$

Given $\varepsilon > 0$, one index $N > 1/\varepsilon$ works for every x. This is the quantitative signature of uniform convergence: a global error bound independent of the evaluation point.

11.3 Supremum norm

Uniform convergence is exactly convergence in the sup norm when that norm is finite.

11.4 Continuity under uniform limits

Uniform limits of continuous functions are continuous.

Example 11.2 (The three-term continuity estimate)**.** Suppose $f_n \to f$ uniformly and every f_n is continuous at a. Given $\varepsilon > 0$, choose N so

$$\|f - f_N\|_\infty < \varepsilon/3.$$

Then choose δ from continuity of f_N at a. For $d(x, a) < \delta$,

$$|f(x) - f(a)| \leq |f - f_N|(x) + |f_N(x) - f_N(a)| + |f_N - f|(a) < \varepsilon.$$

Uniformity is what controls the first and third terms with the same N.

11.5 Cauchy criterion

Uniform convergence is equivalent to a uniform Cauchy condition in complete codomains.

11.6 Examples of failure

Pointwise limits can lose continuity or exchange of limits.

11.7 Uniform approximation

Uniform error bounds quantify global approximation quality.

11.8 Compact-domain tests

Monotonicity and compactness can sometimes upgrade pointwise convergence to uniform convergence.

Example 11.3 (Dini's theorem converts monotone pointwise convergence into uniform convergence)**.** Let $f_n(x) = x^n$ on $[0, a]$ with $0 < a < 1$. The functions decrease pointwise to 0, and both the f_n and the limit are continuous on the compact interval. Dini's theorem therefore gives uniform convergence. Directly,

$$\|f_n\|_\infty = a^n \to 0,$$

which checks the theorem in this case. The compact-domain hypothesis is what prevents slow convergence from escaping to a moving point.

11.9 Core structural results

Theorem 11.4 (Uniform limit theorem)**.** *A uniform limit of continuous functions is continuous.*

Proof. Choose one approximant uniformly close to the limit and use its continuity to bridge values of the limit at nearby points. □

Theorem 11.5 (Uniform Cauchy criterion)**.** *For complete codomain Y, a sequence $f_n : X \to Y$ converges uniformly iff for every ε the tail is uniformly epsilon-Cauchy.*

Proof. Uniform convergence implies uniform Cauchy by the triangle inequality. Conversely each pointwise tail is Cauchy and converges; the same uniform tail estimate survives in the limit. □

Theorem 11.6 (Dini-type principle)**.** *If continuous f_n on compact K decrease pointwise to continuous f, then convergence is uniform.*

Proof. The open sets where $f_n - f < \varepsilon$ increase and cover compact K; a finite subcover reduces to one sufficiently large index. □

11.10 Worked examples

Example 11.7 (Powers on $[0, 1]$)**.** Analyze $f_n(x) = x^n$ pointwise and uniformly. Pointwise the limit is 0 on $[0, 1)$ and 1 at 1, discontinuous. Uniform convergence fails because $\sup |f_n - f| = 1$.

Example 11.8 (Powers on $[0, a]$)**.** For $0 < a < 1$, show $x^n \to 0$ uniformly. $\sup_{[0,a]} x^n = a^n \to 0$.

Example 11.9 (Uniform implies pointwise)**.** Prove the implication directly. A uniform epsilon index works in particular at any fixed point.

Example 11.10 (Continuity of the uniform limit)**.** Give the three-term epsilon estimate. $|f(x) - f(a)| \le |f - f_N|(x) + |f_N(x) - f_N(a)| + |f_N - f|(a)$, and choose the three terms below thirds of epsilon.

11.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 11.11 (Powers on $[0, 1]$)**.** Analyze $f_n(x) = x^n$ pointwise and uniformly.

Solution. Pointwise the limit is 0 on $[0, 1)$ and 1 at 1, discontinuous. Uniform convergence fails because $\sup |f_n - f| = 1$. □

Problem 11.12 (Powers on $[0, a]$)**.** For $0 < a < 1$, show $x^n \to 0$ uniformly.

Solution. $\sup_{[0,a]} x^n = a^n \to 0$. □

Problem 11.13 (Uniform implies pointwise)**.** Prove the implication directly.

Solution. A uniform epsilon index works in particular at any fixed point. □

Problem 11.14 (Continuity of the uniform limit)**.** Give the three-term epsilon estimate.

Solution. $|f(x) - f(a)| \le |f - f_N|(x) + |f_N(x) - f_N(a)| + |f_N - f|(a)$, and choose the three terms below thirds of epsilon. ☐

Problem 11.15 (Sup-norm reformulation)**.** Show $f_n \to f$ uniformly iff $\|f_n - f\|_\infty \to 0$.

Solution. This is just the definition of supremum norm and uniform epsilon control. ☐

Problem 11.16 (Nonuniform pointwise convergence)**.** Analyze $f_n(x) = x/(1 + nx)$ on $[0, 1]$.

Solution. For each fixed x, the limit is 0. The maximum occurs near the endpoint and is $1/(n + 1)$, so in fact convergence is uniform; this example illustrates the need to compute the supremum rather than guess. ☐

Problem 11.17 (A genuine nonuniform sequence)**.** Analyze $f_n(x) = nx(1 - x^2)^n$ on $[0, 1]$.

Solution. For fixed $x > 0$ the exponential factor wins and the limit is zero, also at 0. But maxima remain of order $\sqrt{n}$, so convergence is not uniform. ☐

Problem 11.18 (Uniform Cauchy estimate)**.** If $\|f_{n+1} - f_n\|_\infty \le 2^{-n}$, prove uniform convergence in $C(K)$.

Solution. The tail is bounded by a geometric series, hence uniformly Cauchy; completeness of the sup-norm space gives uniform convergence. ☐

Problem 11.19 (Uniform approximation preserves boundedness)**.** If f_n are uniformly close to bounded f_N, show the limit is bounded.

Solution. For one large N, $|f| \le |f - f_N| + |f_N|$, and the uniform error is globally bounded. ☐

Problem 11.20 (Interchanging an outer limit)**.** If $x_m \to x$ and $f_n \to f$ uniformly with continuous f_n, explain why $f(x_m) \to f(x)$.

Solution. Uniform convergence gives continuity of f, then ordinary continuity handles the sequence x_m. ☐

Problem 11.21 (Dini mechanism)**.** Why is compactness decisive in Dini's theorem?

Solution. Pointwise convergence gives one local index per point; compactness extracts finitely many local neighborhoods and hence one maximum index working globally. ☐

Problem 11.22 (Convergence synthesis)**.** Why is uniform convergence the natural topology for preserving continuity?

Solution. Continuity is a local statement needing one approximation to work at two nearby points simultaneously. Uniform error control supplies that common approximant everywhere. ☐

11.12 Exercises with complete solutions

Exercise 11.23. Does uniform convergence imply pointwise?

Hint. Use definitions. Pointwise convergence fixes x first; uniform convergence requires one index that works simultaneously for every x. ☐

Solution. Yes. ☐

Exercise 11.24. Does pointwise imply uniform?

Hint. Use x^n on $[0,1]$. Compute or estimate $\sup_x |f_n(x)-f(x)|$; convergence of this supremum to zero is the direct uniform test. □

Solution. No. □

Exercise 11.25. If $\|f_n - f\|_\infty \leq 1/n$, what follows?

Hint. Take the bound to zero. To disprove uniform convergence, choose points x_n depending on n where the error stays bounded away from zero. □

Solution. Uniform convergence. □

Exercise 11.26. Can a uniform limit of continuous functions be discontinuous?

Hint. Use the uniform limit theorem. Uniform limits of continuous functions are continuous; if the proposed limit is discontinuous, uniform convergence is impossible. □

Solution. No. □

Exercise 11.27. Find the pointwise limit of x/n.

Hint. Fix x. Use the uniform Cauchy criterion when the candidate limit is difficult to identify explicitly. □

Solution. It is 0. □

Exercise 11.28. Is $x/n \to 0$ uniform on $[0,1]$?

Hint. Take the supremum. Restricting the domain can turn nonuniform convergence into uniform convergence; inspect where the bad points accumulate. □

Solution. Yes. □

Exercise 11.29. Is $x/n \to 0$ uniform on $\mathbb{R}$?

Hint. Supremum is infinite. For monotone sequences of functions, do not infer uniform convergence without an additional theorem such as Dini under its hypotheses. □

Solution. No. □

Exercise 11.30. State the uniform Cauchy condition.

Hint. One index for all points and tail pairs. Keep endpoint behavior separate from interior behavior when the convergence deteriorates near a boundary. □

Solution. For every epsilon, $d(f_m(x), f_n(x)) < \varepsilon$ for all x and all large m, n. □

11.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 11.31 (Sup error)**.** Compute $\|x^n\|_\infty$ on $[0, a]$, $0 < a < 1$.

Hint. Use monotonicity in x. ☐

Solution. The sup norm is a^n. ☐

Exercise 11.32 (Uniform rational sequence)**.** For $f_n(x) = x/(n + x)$ on $[0, 1]$, bound the sup norm.

Hint. Show the function increases in x. ☐

Solution. $\|f_n\|_\infty = 1/(n + 1)$. ☐

Exercise 11.33 (Pointwise limit)**.** Find the pointwise limit of x^n on $[0, 1]$.

Hint. Separate $x < 1$ from $x = 1$. ☐

Solution. The limit is 0 on $[0, 1)$ and 1 at 1. ☐

Exercise 11.34 (Uniform Cauchy)**.** If $\|f_n - f_m\|_\infty \le 2^{-n}$ for $m \ge n$, show the sequence is uniformly Cauchy.

Hint. Choose N with $2^{-N} < \varepsilon$. ☐

Solution. For $m, n \ge N$, reorder if needed and apply the bound. ☐

Exercise 11.35 (Restricted uniformity)**.** Is $x^n \to 0$ uniformly on $[0, 0.9]$?

Hint. Compute the supremum. ☐

Solution. Yes; the sup error is $0.9^n \to 0$. ☐

Proofs

Exercise 11.36 (Uniform implies pointwise)**.** Prove uniform convergence implies pointwise convergence.

Hint. A uniform N works at each fixed point. ☐

Solution. For any fixed x, the global inequality $|f_n(x) - f(x)| < \varepsilon$ holds after the uniform index. ☐

Exercise 11.37 (Uniform limit continuity)**.** Prove a uniform limit of continuous functions is continuous.

Hint. Use the three-term epsilon estimate with one approximant. ☐

Solution. Choose N for two uniform error terms and delta for the middle continuous term. ☐

Exercise 11.38 (Uniform Cauchy forward)**.** Prove a uniformly convergent sequence is uniformly Cauchy.

Hint. Compare each tail function with the common limit. ☐

Solution. $\|f_n - f_m\|_\infty \le \|f_n - f\|_\infty + \|f_m - f\|_\infty$. ☐

Exercise 11.39 (Sup-norm equivalence)**.** Show $f_n \to f$ uniformly iff $\|f_n - f\|_\infty \to 0$ when the sup norm is finite.

Hint. Translate the quantifiers. ☐

Solution. Both statements say that after one index, the error is below epsilon for every point. ☐

Counterexamples and hypothesis testing

Exercise 11.40 (Pointwise not uniform)**.** Show x^n on $[0, 1]$ does not converge uniformly to its pointwise limit.

Hint. Use discontinuity of the limit or compute the sup error. ☐

Solution. The limit is discontinuous while all x^n are continuous; equivalently the sup error remains 1. ☐

Exercise 11.41 (Moving spike)**.** Explain how continuous spikes can converge pointwise to zero without uniform convergence.

Hint. Let the support move/shrink while the height stays 1. ☐

Solution. Each fixed point eventually misses the spike, but the supremum of each function remains 1. ☐

Exercise 11.42 (Noncompact Dini failure)**.** Why is compactness essential in Dini-type results?

Hint. Construct monotone convergence whose slow region escapes to infinity. ☐

Solution. On a noncompact domain, the index needed for a given epsilon can drift with location, preventing one global N. ☐

Applications and investigations

Exercise 11.43 (Approximation certificate)**.** What does $\|f_n - f\|_\infty \le 10^{-3}$ certify?

Hint. Interpret the sup norm. ☐

Solution. Every pointwise approximation error is at most 10^{-3} simultaneously. ☐

Exercise 11.44 (Property preservation)**.** Why is uniform convergence a natural standard for approximating continuous functions?

Hint. Use the uniform-limit theorem. ☐

Solution. It preserves continuity and supplies a domain-wide error certificate. ☐

Challenge problems

Exercise 11.45 (Dini proof)**.** Prove the decreasing Dini theorem using compactness.

Hint. For fixed epsilon, sets $U_n = \{x : f_n(x) - f(x) < \varepsilon\}$ are increasing open sets covering K. ☐

Solution. Compactness gives a finite subcover; because the sets increase, one largest index already covers all of K. ☐

Exercise 11.46 (Uniform convergence of products)**.** If $f_n \to f$ and $g_n \to g$ uniformly and both sequences are uniformly bounded, prove $f_n g_n \to fg$ uniformly.

Hint. Add and subtract fg_n. □

Solution. $|f_n g_n - fg| \leq |f_n - f||g_n| + |f||g_n - g|$; uniform bounds make both sup terms tend to zero. □

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.

Chapter 12

Interchanging Limits, Derivatives and Integrals

Passing a limit through a derivative or integral requires more than pointwise convergence. The correct hypotheses combine uniform control with convergence of derivative data or domination of integral errors.

Learning goals

After this chapter, the reader should be able to:

- verify hypotheses that justify interchanging a limit with an integral or derivative;
- prove convergence of derivatives by combining uniform convergence of derivatives with convergence at one base point;
- construct examples showing that pointwise convergence alone does not justify differentiating or integrating termwise;
- estimate remainders uniformly enough to pass a limit through an operation;
- distinguish sufficient hypotheses for integral interchange from those needed for derivative interchange;
- use a theorem only after checking its domain, regularity, and convergence assumptions explicitly;

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

12.1 Limits and integrals

Uniform convergence allows Riemann integration to commute with limits on compact intervals.

Example 12.1 (Uniform convergence justifies integral interchange quantitatively)**.** Suppose $f_n \to f$ uniformly on $[a, b]$. Then

$$\left| \int_a^b f_n - \int_a^b f \right| \le \int_a^b |f_n - f| \le (b-a)\|f_n - f\|_\infty.$$

If the sup error is at most 10^{-4} on an interval of length 7, then the integral error is at most 7×10^{-4}. Thus the interchange theorem comes with a direct stability estimate, not merely a qualitative limit.

12.2 Integral error estimates

The difference of integrals is controlled by interval length times the sup-norm error.

12.3 Limits and derivatives

Uniform convergence of derivatives plus convergence at one base point yields a differentiable limit.

Example 12.2 (Derivative convergence from one anchor value)**.** Let

$$f_n(x) = c_n + \int_0^x g_n(t)\,dt$$

on $[0, 1]$, where $c_n \to c$ and $g_n \to g$ uniformly. Then

$$|f_n(x) - f_m(x)| \leq |c_n - c_m| + \|g_n - g_m\|_\infty$$

uniformly in x. Hence f_n converges uniformly to

$$f(x) = c + \int_0^x g(t)\,dt,$$

and $f' = g$. The single anchor value fixes the additive constant that derivative information alone cannot determine.

12.4 Why derivative exchange is harder

Differentiation amplifies small-scale oscillation and therefore needs stronger hypotheses.

12.5 Series versions

The same principles apply to term-by-term integration and differentiation of function series.

12.6 Uniform convergence on compacta

Local uniform convergence is often the right compromise on noncompact domains.

12.7 Counterexamples

Pointwise convergence alone can fail spectacularly for derivatives and integrals.

Example 12.3 (A narrow spike defeats pointwise integral interchange)**.** Define a triangular continuous spike f_n on $[0, 1]$ supported in $[0, 2/n]$, with height n at $1/n$. Its area is

$$\frac{1}{2} \cdot \frac{2}{n} \cdot n = 1.$$

For every fixed $x > 0$, eventually x lies outside the support, while $f_n(0) = 0$. Thus $f_n(x) \to 0$ pointwise everywhere, but

$$\int_0^1 f_n = 1$$

for all n. Pointwise convergence alone gives no uniform control over the moving concentration.

12.8 Stability viewpoint

Interchange theorems are continuity statements for integral or derivative operators in suitable function-space topologies.

12.9 Core structural results

Theorem 12.4 (Uniform limit under the integral)**.** *If $f_n \to f$ uniformly on $[a, b]$ and each f_n is Riemann integrable, then f is integrable and $\int f_n \to \int f$.*

Proof. The integral difference is at most $(b-a)\|f_n - f\|_\infty$, which tends to zero; integrability follows by approximating f uniformly with an integrable f_n. ☐

Theorem 12.5 (Derivative limit theorem)**.** *If $f_n \in C^1[a, b]$, $f_n(x_0)$ converges at one point, and f_n' converges uniformly to g, then f_n converges uniformly to a differentiable f with $f' = g$.*

Proof. Write $f_n(x) = f_n(x_0) + \int_{x_0}^x f_n'$. Uniform convergence of derivatives gives uniform Cauchy control of the integrals, and the fundamental theorem identifies the derivative of the limit. ☐

Theorem 12.6 (Termwise integration)**.** *If $\sum f_n$ converges uniformly and each f_n is integrable, then the sum is integrable and $\int \sum f_n = \sum \int f_n$.*

Proof. Apply the uniform-limit integral theorem to the partial sums. ☐

12.10 Worked examples

Example 12.7 (Integral interchange estimate)**.** Prove $|\int_a^b f_n - \int_a^b f| \le (b-a)\|f_n - f\|_\infty$. Apply $|\int g| \le \int |g|$ and bound the integrand by its supremum.

Example 12.8 (A pointwise failure for integrals)**.** Construct continuous spikes f_n on $[0, 1]$ converging pointwise to 0 but with integrals not tending to 0. Use triangular spikes of height n and width $2/n$ centered near zero so area stays 1 while each fixed positive point eventually lies outside the support.

Example 12.9 (Termwise integration of a geometric series)**.** For $|x| \le r < 1$, integrate $\sum_{n \ge 0} x^n$ termwise. The series is uniformly convergent by the M-test, so $\int_0^r \sum x^n dx = \sum r^{n+1}/(n+1) = -\log(1-r)$.

Example 12.10 (Derivative exchange mechanism)**.** Why is convergence at one base point needed? Derivatives determine functions only up to additive constants. Convergence at one point fixes those constants.

12.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 12.11 (Integral interchange estimate)**.** Prove $|\int_a^b f_n - \int_a^b f| \le (b-a)\|f_n - f\|_\infty$.

Solution. Apply $|\int g| \le \int |g|$ and bound the integrand by its supremum. ☐

Problem 12.12 (A pointwise failure for integrals)**.** Construct continuous spikes f_n on $[0,1]$ converging pointwise to 0 but with integrals not tending to 0.

Solution. Use triangular spikes of height n and width $2/n$ centered near zero so area stays 1 while each fixed positive point eventually lies outside the support. ☐

Problem 12.13 (Termwise integration of a geometric series)**.** For $|x| \le r < 1$, integrate $\sum_{n\ge 0} x^n$ termwise.

Solution. The series is uniformly convergent by the M-test, so $\int_0^r \sum x^n dx = \sum r^{n+1}/(n+1) = -\log(1-r)$. ☐

Problem 12.14 (Derivative exchange mechanism)**.** Why is convergence at one base point needed?

Solution. Derivatives determine functions only up to additive constants. Convergence at one point fixes those constants. ☐

Problem 12.15 (Derivative counterexample)**.** Let $f_n(x) = \sin(nx)/n$. What happens to f_n and f_n'?

Solution. $f_n \to 0$ uniformly, but $f_n' = \cos(nx)$ does not converge pointwise at most points, showing uniform convergence of functions alone does not control derivatives. ☐

Problem 12.16 (Uniform derivative convergence)**.** If $\|f_n' - g\|_\infty \to 0$, estimate the derivative-integral error.

Solution. For any x, $|\int_{x_0}^x (f_n' - g)| \le |b-a|\|f_n' - g\|_\infty$. ☐

Problem 12.17 (Series of derivatives)**.** Suppose $\sum f_n'$ converges uniformly and $\sum f_n(x_0)$ converges. What follows?

Solution. The series $\sum f_n$ converges uniformly to a differentiable function whose derivative is $\sum f_n'$. ☐

Problem 12.18 (Local uniform convergence)**.** Why is uniform convergence on every compact interval enough for local integral interchange?

Solution. Any fixed finite integration interval is compact, so the uniform theorem applies there. ☐

Problem 12.19 (Continuity of integration operator)**.** View $I(f) = \int_a^b f$. What is its operator norm on $C[a,b]$ with sup norm?

Solution. $|I(f)| \le (b-a)\|f\|_\infty$, with equality for the constant function 1, so the norm is $b-a$. ☐

Problem 12.20 (Differentiation is not bounded in sup norm)**.** Use $f_n(x) = \sin(nx)/n$ to explain.

Solution. $\|f_n\|_\infty = 1/n \to 0$ while $\|f_n'\|_\infty = 1$, so no fixed bound $\|f'\|_\infty \le C\|f\|_\infty$ can hold. ☐

Problem 12.21 (M-test bridge)**.** Why does the Weierstrass M-test often precede termwise integration?

Solution. It supplies uniform convergence of the function series, exactly the hypothesis needed to exchange sum and integral. ☐

Problem 12.22 (Interchange synthesis)**.** What unifies limit/integral and limit/derivative theorems?

Solution. Both ask whether an operator is continuous under a chosen function-space topology. Integration is continuous in sup norm; differentiation requires a stronger topology controlling derivatives. ☐

12.12 Exercises with complete solutions

Exercise 12.23. If $f_n \to f$ uniformly, do integrals converge?

Hint. On a compact interval, yes. For interchanging limit and integral, identify a theorem whose hypotheses give uniform or dominated control before passing the limit. ☐

Solution. Yes, for Riemann-integrable terms. ☐

Exercise 12.24. Bound the integral error by sup error.

Hint. Use interval length. For derivative interchange, check uniform convergence of the derivative sequence and convergence at one base point. ☐

Solution. At most $(b-a)\|f_n - f\|_\infty$. ☐

Exercise 12.25. Does uniform convergence imply derivative convergence?

Hint. Use $\sin(nx)/n$. Integrate the derivative convergence from a fixed base point to reconstruct convergence of the functions. ☐

Solution. No. ☐

Exercise 12.26. Why one base point in derivative theorem?

Hint. Constants. To build a counterexample, look for pointwise convergence with spikes or rapidly changing slopes that defeat uniform control. ☐

Solution. Derivative data omit additive constants. ☐

Exercise 12.27. Can a uniformly convergent series be integrated termwise?

Hint. Under integrability. Estimate the remainder uniformly in the variable before moving a limit through an operation. ☐

Solution. Yes. ☐

Exercise 12.28. What topology makes integration continuous?

Hint. Sup norm on a compact interval. Termwise differentiation is stronger than termwise integration; check the derivative hypotheses separately. ☐

Solution. The sup norm. ☐

Exercise 12.29. Is differentiation bounded in the sup norm?

Hint. Use high-frequency functions. Write the theorem's hypotheses next to the sequence at hand and verify each one explicitly. ☐

Solution. No. ☐

Exercise 12.30. If derivatives converge uniformly to g, what candidate derivative has the limit?

Hint. Use the theorem. If only pointwise convergence is known, avoid exchanging operations until an additional bound or compactness argument supplies control. ☐

Solution. g. ☐

12.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 12.31 (Integral error)**.** If $\|f_n - f\|_\infty \le 1/n$ on $[0, 3]$, bound the integral error.

Hint. Multiply the sup error by interval length. ☐

Solution. At most $3/n$. ☐

Exercise 12.32 (Base-point constants)**.** If $f_n' = 0$ for all n but $f_n(0) = n$, do the functions converge?

Hint. Derivatives determine functions only up to constants. ☐

Solution. No; $f_n \equiv n$ diverges despite identical derivatives. ☐

Exercise 12.33 (Termwise integral)**.** For $|x| \le r < 1$, justify integrating $\sum x^n$ termwise.

Hint. Use the M-test with r^n. ☐

Solution. The series is uniformly convergent, so $\int_0^r \sum x^n dx = \sum r^{n+1}/(n+1)$. ☐

Exercise 12.34 (Uniform derivative limit)**.** If $f_n'(x) = x/n$ on $[0, 1]$, what is the uniform limit of derivatives?

Hint. Take the sup norm. ☐

Solution. $\|f_n'\|_\infty = 1/n \to 0$, so the limit is 0. ☐

Exercise 12.35 (Local uniformity)**.** Does $1/(1 + x/n) \to 1$ uniformly on every compact interval?

Hint. Bound x on a fixed compact set. ☐

Solution. Yes; on $[-R, R]$ for large n, the deviation is bounded by a constant times R/n. ☐

Proofs

Exercise 12.36 (Integral interchange)**.** Prove uniform convergence permits passing a limit through the Riemann integral.

Hint. Use the sup-norm integral estimate. ☐

Solution. The integral differences are bounded by $(b-a)\|f_n - f\|_\infty \to 0$. ☐

Exercise 12.37 (Derivative theorem anchor)**.** Explain why convergence of $f_n(x_0)$ at one point is needed in the derivative-limit theorem.

Hint. All derivatives are unchanged by additive constants. ☐

Solution. Without anchoring constants, functions can drift vertically and fail to converge even when derivatives converge perfectly. ☐

Exercise 12.38 (Termwise integration of series)**.** Prove a uniformly convergent series of integrable functions can be integrated termwise.

Hint. Apply the uniform integral theorem to partial sums. ☐

Solution. If $S_N \to S$ uniformly, then $\int S_N \to \int S$, while $\int S_N = \sum_{n\le N} \int f_n$. ☐

Exercise 12.39 (Uniform derivative Cauchy estimate)**.** If $f_n(x_0)$ is Cauchy and f_n' is uniformly Cauchy on $[a, b]$, prove f_n is uniformly Cauchy.

Hint. Write differences using the fundamental theorem from x_0. ☐

Solution. $|f_n(x) - f_m(x)| \le |f_n(x_0) - f_m(x_0)| + (b-a)\|f_n' - f_m'\|_\infty$. ☐

Counterexamples and hypothesis testing

Exercise 12.40 (Pointwise integral failure)**.** Give a pointwise-zero sequence of continuous functions whose integrals stay 1.

Hint. Use narrowing triangular spikes with increasing height. ☐

Solution. Height n, base $2/n$ triangular spikes have area 1 and converge pointwise to zero if placed at the endpoint appropriately. ☐

Exercise 12.41 (Derivative interchange failure)**.** Why can $f_n(x) = \sin(nx)/n$ converge uniformly while derivatives fail to converge?

Hint. Compute f_n'. ☐

Solution. $\|f_n\|_\infty \le 1/n \to 0$, but $f_n'(x) = \cos(nx)$ has no pointwise limit at many points and no uniform limit. ☐

Exercise 12.42 (Pointwise series manipulation)**.** Why is pointwise convergence of a function series insufficient for termwise integration in general?

Hint. Partial sums may concentrate error on moving sets. ☐

Solution. Without uniform or another dominating control, integrals of the tails need not tend to zero even if tails vanish pointwise. ☐

Applications and investigations

Exercise 12.43 (Certified integral approximation)**.** If a numerical surrogate $\tilde{f}$ has uniform error η on an interval of length L, what integral certificate follows?

Hint. Integrate the absolute error. ☐

Solution. The integral error is at most $L\eta$. ☐

Exercise 12.44 (Parameter differentiation)**.** What kind of control should you seek before differentiating a limit of approximating functions?

Hint. Pointwise convergence is too weak; control derivatives uniformly and anchor function values. ☐

Solution. A standard sufficient pattern is uniform convergence of derivatives plus convergence at one base point. ☐

Challenge problems

Exercise 12.45 (Series derivative theorem)**.** State a practical sufficient condition for differentiating $\sum f_n$ termwise on $[a, b]$.

Hint. Control the derivative series uniformly and make the original series converge at one point. ☐

Solution. If $\sum f_n'(x)$ converges uniformly and $\sum f_n(x_0)$ converges at one point, then the original series converges uniformly to a differentiable sum with derivative $\sum f_n'$. ☐

Exercise 12.46 (Interchange as operator continuity)**.** Explain why integration is continuous in the sup norm but differentiation is not.

Hint. Compare the bounds for the two operators. ☐

Solution. Integration satisfies $|\int g| \le (b-a)\|g\|_\infty$. No bound of $\|f'\|_\infty$ by a constant times $\|f\|_\infty$ holds on C^1; small-amplitude high-frequency functions show differentiation can amplify oscillation. ☐

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.

Chapter 13

Infinite Series of Functions

Infinite series of functions are controlled most effectively by uniform comparison tests. Absolute and uniform convergence turn termwise algebra, integration, and sometimes differentiation into rigorous operations.

Learning goals

After this chapter, the reader should be able to:

- test convergence of numerical and function series using comparison, ratio, root, or uniform criteria;
- apply the Weierstrass M-test to prove uniform and absolute convergence of a function series;
- justify termwise integration or differentiation of a series when the required uniform hypotheses hold;
- determine the interval or region of convergence of a power series;
- estimate tails of convergent series sufficiently to meet a prescribed error tolerance;
- construct examples separating pointwise, uniform, absolute, and conditional convergence;

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

13.1 Partial sums

A function series is studied through its partial-sum sequence.

13.2 Uniform convergence

Uniform convergence of partial sums is the key global notion.

13.3 Weierstrass M-test

Uniform domination by a summable numerical series gives uniform absolute convergence.

Example 13.1 (M-test with a computable truncation guarantee)**.** On $[-1, 1]$, consider

$$\sum_{n=1}^{\infty} \frac{x^n}{3^n}.$$

Since $|x^n/3^n| \leq 3^{-n}$, the M-test gives uniform absolute convergence. Moreover the tail after N satisfies

$$\sup_{|x|\leq 1} \left| \sum_{n>N} \frac{x^n}{3^n} \right| \leq \sum_{n>N} 3^{-n} = \frac{1}{2 \cdot 3^N}.$$

Thus N can be chosen directly from a requested uniform tolerance. The majorant proves convergence and supplies an error certificate at the same time.

13.4 Power series

Inside the radius of convergence, power series converge uniformly on compact subintervals.

Example 13.2 (Power-series convergence is locally uniform inside the radius)**.** For the geometric power series $\sum_{n\geq 0} x^n$, fix $0 < r < 1$. On $[-r, r]$,

$$|x^n| \leq r^n,$$

and $\sum r^n$ converges. Hence the series converges uniformly there and

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}.$$

The same argument works on every compact subinterval strictly inside a general power series' radius of convergence: the compact set stays a positive distance from the boundary where convergence may deteriorate.

13.5 Algebra of series

Uniformly convergent series can be added and scaled termwise.

13.6 Integration

Uniform convergence justifies termwise integration.

13.7 Differentiation

Uniform convergence of derivative series plus one-point convergence justifies termwise differentiation.

13.8 Remainder bounds

Majorant tails provide explicit uniform truncation error estimates.

Example 13.3 (Turning a majorant tail into a stopping rule)**.** Suppose $|f_n(x)| \leq 2^{-n}$ for every $x \in X$. Then the remainder after the N-th term obeys

$$\|R_N\|_\infty \leq \sum_{n=N+1}^{\infty} 2^{-n} = 2^{-N}.$$

To guarantee uniform error below 10^{-6}, it is enough to take $N = 20$. This is stronger than knowing only that the series converges: it converts the proof into a practical truncation rule.

13.9 Core structural results

Theorem 13.4 (Weierstrass M-test). *If $|f_n(x)| \le M_n$ for all x and $\sum M_n < \infty$, then $\sum f_n$ converges uniformly and absolutely.*

Proof. The numerical tail bound $\sum_{n>N} M_n$ controls every function tail uniformly, so the partial sums are uniformly Cauchy. □

Theorem 13.5 (Uniform termwise integration). *A uniformly convergent series of integrable functions may be integrated termwise.*

Proof. Apply the uniform-limit integral theorem to partial sums. □

Theorem 13.6 (Power-series local uniformity). *A power series converges uniformly on every closed subinterval strictly inside its radius of convergence.*

Proof. Choose a larger interior radius and dominate coefficients times r^n by a convergent geometric comparison from the root or ratio test. □

13.10 Worked examples

Example 13.7 (M-test with geometric majorant). Show $\sum x^n/n^2$ converges uniformly on $[-1, 1]$. $|x^n/n^2| \le 1/n^2$ and $\sum 1/n^2$ converges.

Example 13.8 (Uniform tail bound). If $|f_n| \le 2^{-n}$, bound the tail after N. The sup tail is at most $\sum_{n>N} 2^{-n} = 2^{-N}$.

Example 13.9 (Harmonic obstruction). Why does the M-test fail for x^n/n on $[0, 1]$? The natural majorant $1/n$ diverges; indeed at $x = 1$ the series is harmonic.

Example 13.10 (Power-series compact convergence). Why does $\sum x^n$ converge uniformly on $[-r, r]$ for $r < 1$? $|x^n| \le r^n$ and the geometric series converges.

13.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 13.11 (M-test with geometric majorant). Show $\sum x^n/n^2$ converges uniformly on $[-1, 1]$.

Solution. $|x^n/n^2| \le 1/n^2$ and $\sum 1/n^2$ converges. □

Problem 13.12 (Uniform tail bound). If $|f_n| \le 2^{-n}$, bound the tail after N.

Solution. The sup tail is at most $\sum_{n>N} 2^{-n} = 2^{-N}$. □

Problem 13.13 (Harmonic obstruction). Why does the M-test fail for x^n/n on $[0, 1]$?

Solution. The natural majorant $1/n$ diverges; indeed at $x = 1$ the series is harmonic. □

Problem 13.14 (Power-series compact convergence)**.** Why does $\sum x^n$ converge uniformly on $[-r, r]$ for $r < 1$?

Solution. $|x^n| \le r^n$ and the geometric series converges. ☐

Problem 13.15 (Termwise integral)**.** Integrate $\sum_{n\ge 0} x^n$ on $[0, r]$, $r < 1$.

Solution. Uniform convergence gives $\sum r^{n+1}/(n+1) = -\log(1-r)$. ☐

Problem 13.16 (A derivative series)**.** For $\sum x^n/n!$, justify termwise differentiation on bounded intervals.

Solution. Derivative terms are $x^{n-1}/(n-1)!$, uniformly dominated on $[-R, R]$ by $R^{n-1}/(n-1)!$. ☐

Problem 13.17 (Absolute versus uniform)**.** Does pointwise absolute convergence imply uniform convergence?

Solution. No. Uniformity requires tail control independent of x; absolute convergence can deteriorate as x varies. ☐

Problem 13.18 (Uniform Cauchy criterion)**.** State it for a series.

Solution. For every epsilon there is N such that every finite tail $\sum_{n=p}^{q} f_n(x)$ has magnitude below epsilon for all x whenever $q \ge p \ge N$. ☐

Problem 13.19 (Remainder estimate)**.** If $|f_n| \le M_n$, how can one certify truncation accuracy?

Solution. Bound the sup error by the numerical tail $\sum_{n>N} M_n$. ☐

Problem 13.20 (Uniform convergence preserves continuity)**.** Why is a uniformly convergent series of continuous functions continuous?

Solution. Its partial sums are continuous and converge uniformly, so the uniform-limit theorem applies. ☐

Problem 13.21 (Local uniformity on noncompact domains)**.** What does it mean for a series to converge locally uniformly?

Solution. It converges uniformly on every compact subset of the domain. ☐

Problem 13.22 (Series synthesis)**.** Why are majorants central?

Solution. They replace a difficult family of function tails by one numerical tail independent of the point, converting pointwise estimates into uniform convergence. ☐

13.12 Exercises with complete solutions

Exercise 13.23. Apply M-test to $\sum x^n/2^n$ on $[-1, 1]$.

Hint. Use $(1/2)^n$. For the M-test, dominate the absolute value of each term by a numerical sequence whose sum is known to converge. ☐

Solution. It converges uniformly. ☐

Exercise 13.24. Does $\sum x^n$ converge uniformly on $[0, 1]$?

Hint. Check $x = 1$. To find a power-series radius, apply the ratio or root test to the coefficient sequence and then inspect endpoints separately. □

Solution. No; it does not even converge there. □

Exercise 13.25. Uniform convergence of $\sum f_n$ means what for partial sums?

Hint. Use the definition. For termwise integration, uniform convergence on the interval is the key safe route in this chapter. □

Solution. Partial sums converge uniformly. □

Exercise 13.26. If $\sum M_n$ converges, what about its tails?

Hint. Cauchy criterion. For termwise differentiation, control the derivative series uniformly and anchor the original series at one point. □

Solution. They tend to zero. □

Exercise 13.27. Can a uniformly convergent series of continuous functions be discontinuous?

Hint. Estimate a series tail by comparison with a geometric or integral bound rather than summing many terms explicitly. □

Solution. Its sum is continuous. □

Exercise 13.28. What is a power-series radius?

Hint. Boundary between convergence and divergence. Absolute convergence implies ordinary convergence, but the converse can fail; test signs and absolute values separately. □

Solution. A number R such that convergence holds for $|x - a| < R$ and divergence for $|x - a| > R$. □

Exercise 13.29. Can the M-test prove absolute convergence?

Hint. To disprove uniform convergence, locate points where the tail does not become uniformly small. □

Solution. It gives uniform absolute convergence. □

Exercise 13.30. How do you estimate a remainder under M-test?

Hint. Use majorant tail. Use the Cauchy criterion for series when no closed-form sum is available. □

Solution. By $\sum_{n>N} M_n$. □

13.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 13.31 (Geometric function series)**.** Find the uniform sum of $\sum_{n\geq 0}(x/2)^n$ on $[-1,1]$.

Hint. Use the geometric-series formula; $|x/2| \leq 1/2$. ☐

Solution. The sum is $1/(1-x/2)$, and convergence is uniform by the M-test. ☐

Exercise 13.32 (Tail size)**.** If $|f_n| \leq 4^{-n}$, bound the tail after N.

Hint. Sum the geometric majorant from $N+1$ onward. ☐

Solution. The tail is at most $\frac{4^{-N}}{3}$. ☐

Exercise 13.33 (Power-series radius)**.** Find the radius of convergence of $\sum nx^n$.

Hint. Use the ratio test on $n|x|^n$. ☐

Solution. The ratio tends to $|x|$, so the radius is 1. ☐

Exercise 13.34 (Termwise integral)**.** Integrate $\sum_{n\geq 0} x^n$ termwise from 0 to $r<1$.

Hint. Use uniform convergence on $[0,r]$. ☐

Solution. $\sum_{n\geq 0} r^{n+1}/(n+1) = -\log(1-r)$. ☐

Exercise 13.35 (Uniform absolute convergence)**.** Does $\sum x^n/n^2$ converge uniformly on $[-1,1]$?

Hint. Compare with $\sum 1/n^2$. ☐

Solution. Yes, absolutely and uniformly by the M-test. ☐

Proofs

Exercise 13.36 (M-test proof)**.** Prove the Weierstrass M-test from the uniform Cauchy criterion.

Hint. Bound every function tail by the corresponding numerical tail. ☐

Solution. For $m>n$, $\sup_x |\sum_{k=n+1}^m f_k(x)| \leq \sum_{k=n+1}^m M_k$, which becomes arbitrarily small. ☐

Exercise 13.37 (Uniform termwise integration)**.** Prove a uniformly convergent series of Riemann-integrable functions can be integrated termwise.

Hint. Apply the uniform integral theorem to partial sums. ☐

Solution. The partial sums converge uniformly to the sum, and integration is continuous in sup norm. ☐

Exercise 13.38 (Power-series compact uniformity)**.** Prove $\sum a_n x^n$ converges uniformly on every $[-r,r]$ with $r<R$, where R is the radius.

Hint. Choose ρ with $r<\rho<R$ and dominate $|a_n|r^n$ by a convergent comparison based on ρ. ☐

Solution. Convergence at ρ bounds $|a_n|\rho^n$; multiplying by $(r/\rho)^n$ gives a geometric majorant. ☐

Exercise 13.39 (Tail criterion)**.** Show a uniformly absolutely convergent series has uniform tails tending to zero.

Hint. Use the numerical majorant tail. □

Solution. The sup norm of each tail is bounded by a scalar tail that tends to zero. □

Counterexamples and hypothesis testing

Exercise 13.40 (M-test not necessary)**.** Give a uniformly convergent series for which the obvious absolute M-test is unavailable.

Hint. Think of cancellation or conditional convergence with a common factor. □

Solution. For example a uniformly convergent scalar conditional series multiplied by a fixed bounded function can converge uniformly without an absolutely summable coefficient majorant. □

Exercise 13.41 (Boundary failure)**.** Why does $\sum x^n$ fail to converge uniformly on $[0, 1)$?

Hint. The point producing the largest tail can move toward 1. □

Solution. The tail supremum over $[0, 1)$ remains 1, so uniform convergence fails despite pointwise convergence. □

Exercise 13.42 (Termwise differentiation warning)**.** Why may a uniformly convergent series not be differentiated termwise?

Hint. Uniform convergence of the original series does not control derivative amplitudes. □

Solution. Derivative series can diverge or fail uniform convergence; extra derivative hypotheses are required. □

Applications and investigations

Exercise 13.43 (Certified truncation)**.** If a function series has majorant tail 2^{-N}, how should N be chosen for 10^{-8} accuracy?

Hint. Solve $2^{-N} < 10^{-8}$. □

Solution. Any $N \geq 27$ works. □

Exercise 13.44 (Spectral expansion viewpoint)**.** Why are uniform tail bounds useful in numerical expansions?

Hint. They control worst-case error over the whole domain. □

Solution. They turn a formal infinite representation into a finite approximation with a rigorous global error guarantee. □

Challenge problems

Exercise 13.45 (Cauchy product)**.** Under absolute uniform convergence of two function series, explain why their Cauchy product is well behaved.

Hint. Use absolute numerical majorants and the scalar Cauchy-product theorem. □

Solution. The product coefficients are dominated by the convolution of two summable majorants, so the product series converges uniformly and equals the pointwise product. □

Exercise 13.46 (Differentiate a power series)**.** Why may a power series be differentiated termwise strictly inside its radius of convergence?

Hint. On a smaller compact interval, dominate the derivative series using a slightly larger interior radius. □

Solution. The derivative series converges uniformly on compacta inside the radius; the derivative-limit theorem then justifies termwise differentiation. □

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.

Chapter 14

Trigonometric Series

Trigonometric series represent periodic functions in orthogonal sine and cosine modes. The first layer of Fourier analysis concerns coefficients, orthogonality, partial sums, and the distinction between formal expansions and genuine convergence.

Learning goals

After this chapter, the reader should be able to:

- compute Fourier coefficients of elementary periodic functions;
- use symmetry to predict vanishing sine or cosine coefficients before integration;
- determine pointwise convergence values at continuity points and jump discontinuities under the stated hypotheses;
- apply Parseval-type identities in concrete coefficient computations;
- estimate partial sums or tails from coefficient information;
- distinguish convergence of a Fourier series from convergence of an arbitrary trigonometric series;

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

14.1 Periodic functions

Periodicity identifies functions on the line with functions on a circle.

14.2 Trigonometric system

Sines and cosines form an orthogonal family on a full period.

14.3 Fourier coefficients

Coefficients are obtained by inner products with the trigonometric modes.

Example 14.1 (Fourier coefficients of a simple odd function)**.** Let $f(x) = x$ on $[-\pi, \pi]$, extended periodically. Odd symmetry kills the constant and cosine coefficients. For $n \geq 1$,

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} x\sin(nx)\,dx = \frac{2}{\pi}\int_{0}^{\pi} x\sin(nx)\,dx = \frac{2(-1)^{n+1}}{n}.$$

Thus the Fourier series is

$$2\sum_{n\geq 1}\frac{(-1)^{n+1}}{n}\sin(nx).$$

The parity prediction provides a check before any integration is done.

14.4 Partial sums

Finite Fourier sums are trigonometric polynomials.

14.5 Bessel inequality

Orthogonality bounds the square-sum of coefficients by the L^2 energy.

14.6 Dirichlet kernel

Partial sums can be written as convolution with the Dirichlet kernel.

Example 14.2 (Dirichlet kernel as the mechanism of a partial sum)**.** The N-th Fourier partial sum can be written

$$S_N f(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x-t)D_N(t)\,dt,$$

where

$$D_N(t) = 1 + 2\sum_{n=1}^{N}\cos(nt) = \frac{\sin((N+\frac{1}{2})t)}{\sin(t/2)}.$$

This formula shows why convergence is not merely coefficient bookkeeping: D_N concentrates near $t = 0$ but also oscillates strongly, so local regularity and cancellation determine the limiting value.

14.7 Pointwise convergence ideas

Smoothness and one-sided limits influence pointwise convergence.

14.8 Parseval preview

When the system is complete, coefficient energy equals function energy.

Example 14.3 (Parseval turns coefficient decay into an energy identity)**.** For $f(x) = \cos(3x)$ on $[-\pi, \pi]$, all Fourier coefficients vanish except the $n = 3$ cosine coefficient, which equals 1 in the standard normalization. Parseval therefore gives

$$\frac{1}{\pi}\int_{-\pi}^{\pi} |f(x)|^2\,dx = 1.$$

Directly, the integral of $\cos^2(3x)$ over a full period is π, so the same normalized energy is 1. The identity checks the coefficient normalization and exhibits orthogonal energy decomposition.

14.9 Core structural results

Theorem 14.4 (Orthogonality relations). *Distinct sine/cosine modes are orthogonal on* $[-\pi, \pi]$.

Proof. Use product-to-sum identities; integrals of nonconstant integer-frequency sines and cosines over a full period vanish. □

Theorem 14.5 (Bessel inequality). *For an orthonormal family, the sum of squared Fourier coefficients is bounded by the squared norm of the function.*

Proof. Subtract the finite orthogonal projection and expand its nonnegative squared norm. □

Theorem 14.6 (Dirichlet midpoint principle). *Under standard piecewise smooth hypotheses, Fourier sums converge at a jump to the midpoint of one-sided limits.*

Proof. Rewrite the partial sum using the Dirichlet kernel and use oscillatory cancellation away from the origin plus local one-sided regularity. □

14.10 Worked examples

Example 14.7 (Coefficient of a constant). Find the Fourier coefficients of $f(x) = 1$. Only the constant coefficient is nonzero; all sine and positive cosine modes integrate to zero.

Example 14.8 (Odd symmetry). What coefficients vanish for an odd function? The constant and cosine coefficients vanish; only sine coefficients may remain.

Example 14.9 (Even symmetry). What coefficients vanish for an even function? All sine coefficients vanish.

Example 14.10 (Orthogonality computation). Compute $\int_{-\pi}^{\pi} \cos(nx)\cos(mx)dx$ for $n \neq m$. Product-to-sum reduces it to integrals of nonconstant cosine modes, hence zero.

14.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 14.11 (Coefficient of a constant). Find the Fourier coefficients of $f(x) = 1$.

Solution. Only the constant coefficient is nonzero; all sine and positive cosine modes integrate to zero. □

Problem 14.12 (Odd symmetry). What coefficients vanish for an odd function?

Solution. The constant and cosine coefficients vanish; only sine coefficients may remain. □

Problem 14.13 (Even symmetry). What coefficients vanish for an even function?

Solution. All sine coefficients vanish. □

Problem 14.14 (Orthogonality computation). Compute $\int_{-\pi}^{\pi} \cos(nx)\cos(mx)dx$ for $n \neq m$.

Solution. Product-to-sum reduces it to integrals of nonconstant cosine modes, hence zero. □

Problem 14.15 (Sawtooth structure). Why does an odd sawtooth have a sine-only series?

Solution. Odd symmetry kills the even cosine modes and constant term. □

Problem 14.16 (Finite projection)**.** Why is the Nth Fourier partial sum the best L^2 approximation among trigonometric polynomials of degree at most N?

Solution. It is the orthogonal projection onto that finite-dimensional subspace. □

Problem 14.17 (Bessel estimate)**.** If $\|f\|_2 = 1$, what can be said about Fourier coefficients?

Solution. Their squared magnitudes have total finite partial sums bounded by 1. □

Problem 14.18 (Dirichlet kernel integral)**.** Why must the Dirichlet kernel have total mass 2π under the standard normalization?

Solution. The partial-sum operator reproduces constants, so integrating the kernel against the constant function must return that constant. □

Problem 14.19 (Jump value)**.** What Fourier value is expected at a jump from L_- to L_+?

Solution. The midpoint $(L_- + L_+)/2$ under Dirichlet hypotheses. □

Problem 14.20 (Gibbs warning)**.** What does Gibbs phenomenon say qualitatively?

Solution. Near jumps, partial sums exhibit persistent overshoot whose width shrinks but relative amplitude does not vanish. □

Problem 14.21 (Parseval meaning)**.** Interpret Parseval's identity.

Solution. It says the orthogonal mode coefficients preserve total quadratic energy. □

Problem 14.22 (Trigonometric synthesis)**.** Why are Fourier coefficients canonical?

Solution. Orthogonality isolates each mode by an inner product, making coefficients invariantly determined by projection rather than by solving a coupled system. □

14.12 Exercises with complete solutions

Exercise 14.23. What modes survive for an even function?

Hint. Parity. Exploit parity first: even functions kill sine coefficients and odd functions kill cosine coefficients. □

Solution. Constant and cosines. □

Exercise 14.24. What modes survive for an odd function?

Hint. Parity. Compute Fourier coefficients from their defining integrals over one period and simplify symmetry before integrating. □

Solution. Sines. □

Exercise 14.25. Are $\sin nx$ and $\cos mx$ orthogonal?

Hint. Parity/product identities. At a jump, the standard convergence theorem gives the midpoint of the one-sided limits rather than either side value. □

Solution. Yes on a full symmetric period. □

Exercise 14.26. What is a trigonometric polynomial?

Hint. Finite linear combination. Parseval converts an L^2 norm into a sum of squared coefficients; verify normalization constants before applying it. ☐

Solution. A finite combination of constant, sine, and cosine modes. ☐

Exercise 14.27. What is Bessel inequality about?

Hint. Coefficient energy. Integrate by parts to obtain coefficient decay when the function has additional smoothness. ☐

Solution. It bounds coefficient square-sums by function energy. ☐

Exercise 14.28. At a smooth point, what value is expected from Fourier convergence?

Hint. The function value. Compare a trigonometric polynomial with a partial Fourier sum by matching coefficients. ☐

Solution. The ordinary value. ☐

Exercise 14.29. At a jump, what value is expected?

Hint. Average of one-sided limits. Check the period and normalization convention before copying any coefficient formula. ☐

Solution. Their midpoint. ☐

Exercise 14.30. What operator is the Fourier partial sum?

Hint. Orthogonal projection in L^2. For tail estimates, use coefficient decay or an ℓ^2 identity rather than pointwise term estimates alone. ☐

Solution. Projection onto the first modes. ☐

14.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 14.31 (Odd symmetry)**.** For an odd 2π-periodic integrable function, which Fourier coefficients vanish?

Hint. Check parity of the integrands. ☐

Solution. The constant and all cosine coefficients vanish. ☐

Exercise 14.32 (Even symmetry)**.** For an even function, which Fourier coefficients vanish?

Hint. Sine is odd. ☐

Solution. All sine coefficients vanish. ☐

Exercise 14.33 (Orthogonality)**.** Compute $\int_{-\pi}^{\pi} \sin(nx)\sin(mx)\,dx$ for $n \neq m$.

Hint. Use product-to-sum. ☐

Solution. The integral is 0. □

Exercise 14.34 (Low Chebyshev-like mode)**.** Find the Fourier series of $f(x) = \cos(2x)$.

Hint. It is already one trigonometric basis mode. □

Solution. Only the $n = 2$ cosine coefficient is nonzero and equals 1. □

Exercise 14.35 (Jump value)**.** Under Dirichlet hypotheses, to what value does the Fourier series converge at a jump with one-sided limits 2 and 6?

Hint. Use the midpoint principle. □

Solution. It converges to 4. □

Proofs

Exercise 14.36 (Orthogonality proof)**.** Prove distinct cosine modes are orthogonal on a full period.

Hint. Use $\cos n \cos m = \frac{1}{2}[\cos(n-m) + \cos(n+m)]$. □

Solution. Both nonconstant integer-frequency cosines integrate to zero. □

Exercise 14.37 (Bessel inequality)**.** Prove the finite Bessel inequality for an orthonormal family.

Hint. Subtract the orthogonal projection and expand the nonnegative squared norm. □

Solution. $\|f - \sum_{k=1}^{N} \langle f, e_k \rangle e_k\|^2 = \|f\|^2 - \sum_{k=1}^{N} |\langle f, e_k \rangle|^2 \geq 0$. □

Exercise 14.38 (Parity coefficients)**.** Prove an odd function has zero cosine coefficients.

Hint. The product of odd f and even cosine is odd. □

Solution. The integral of an odd integrable function over $[-\pi, \pi]$ vanishes. □

Exercise 14.39 (Dirichlet midpoint mechanism)**.** Explain why a jump is represented by the average of one-sided limits under the standard theorem.

Hint. Symmetric kernel concentration samples both sides. □

Solution. The local symmetric contribution of the Dirichlet kernel averages the left and right limiting values; oscillatory far contributions cancel. □

Counterexamples and hypothesis testing

Exercise 14.40 (Formal coefficients insufficient)**.** Why does writing down Fourier coefficients not by itself prove pointwise convergence to f?

Hint. Coefficient definition and convergence of partial sums are separate questions. □

Solution. An expansion may converge in another sense, diverge at points, or converge to midpoint values at jumps. □

Exercise 14.41 (Uniform convergence warning)**.** Why can a Fourier series of a discontinuous periodic function not converge uniformly to that function if partial sums are continuous?

Hint. Uniform limits of continuous functions are continuous. □

Solution. A discontinuous target rules out uniform convergence of continuous trigonometric partial sums. ☐

Exercise 14.42 (Bessel not Parseval automatically)**.** Why does Bessel inequality alone not imply equality?

Hint. Equality requires completeness of the orthonormal system for the function space. ☐

Solution. Bessel gives only an upper bound on captured coefficient energy; uncaptured orthogonal energy may remain. ☐

Applications and investigations

Exercise 14.43 (Signal energy)**.** How does Parseval help verify a computed Fourier spectrum?

Hint. Compare time-domain squared norm with coefficient energy. ☐

Solution. A mismatch signals missing modes or normalization errors. ☐

Exercise 14.44 (Symmetry shortcut)**.** Why should parity be checked before computing coefficients numerically?

Hint. It predicts exact zeros. ☐

Solution. Using symmetry cuts work and provides a strong diagnostic for numerical integration error. ☐

Challenge problems

Exercise 14.45 (Best finite Fourier approximation)**.** Why is the N-mode Fourier partial sum the best L^2 approximation from the span of those modes?

Hint. It is the orthogonal projection. ☐

Solution. Projection minimizes the norm of the residual by Pythagoras. ☐

Exercise 14.46 (Coefficient decay from smoothness)**.** Use integration by parts heuristically to explain why smoother periodic functions have faster Fourier coefficient decay.

Hint. Move derivatives from the function onto e^{-inx}. ☐

Solution. Each integration by parts introduces a factor $1/n$ when boundary terms are controlled, so additional smoothness yields additional powers of decay. ☐

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.

Chapter 15

Pathological and Nowhere-Differentiable Functions

Pathological functions reveal the sharp limits of naive intuition. Continuity need not imply differentiability, pointwise limits can destroy regularity, and dense oscillation can coexist with precise global structure.

Learning goals

After this chapter, the reader should be able to:

- analyze continuity and differentiability of classical pathological examples such as Takagi- or Weierstrass-type functions;
- prove nowhere-differentiability or failure of regularity by isolating a scale-dependent oscillation;
- distinguish uniform convergence of a defining series from differentiability of its sum;
- estimate moduli of continuity for explicit pathological constructions;
- construct sequences of scales that expose incompatible difference quotients;
- explain which regularity theorem fails and which hypothesis is missing in a given counterexample;

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

15.1 Everywhere continuous, nowhere differentiable

Weierstrass-type series show continuity can coexist with complete failure of classical tangents.

15.2 Oscillation across scales

High-frequency components with slowly decaying amplitude create roughness at every scale.

Example 15.1 (Scale competition in a Weierstrass-type series)**.** For

$$W(x) = \sum_{n\geq 0} a^n \cos(b^n x), \qquad 0 < a < 1,$$

uniform convergence follows from $|a^n \cos(b^n x)| \leq a^n$. But the formal derivative has terms of size roughly

$$a^n b^n = (ab)^n.$$

When $ab > 1$, high-frequency derivative amplitudes grow rather than decay. This does not alone prove nowhere differentiability, but it identifies the scale conflict: continuity is controlled by a^n, while slopes are amplified by the frequency b^n.

15.3 Cantor-type functions

Monotone continuous functions can have derivative zero on a large set yet still increase globally.

Example 15.2 (The Cantor function separates derivative behavior from total change)**.** The Cantor function is constant on every removed middle-third interval, so its derivative is 0 at every point in the interiors of those intervals. Their union is dense and open. Yet the function increases from 0 to 1 on $[0, 1]$.

Thus knowing the derivative on a dense set is not enough to reconstruct global change when the derivative theorem's regularity hypotheses are absent. The increase is concentrated on the Cantor set.

15.4 Pointwise-limit pathologies

Limits can lose continuity when convergence is not uniform.

15.5 Dense discontinuities

Functions may be discontinuous on dense sets while retaining integrability properties.

15.6 Baire viewpoint

Category methods explain why rough functions are abundant rather than exceptional.

15.7 Moduli of continuity

Hölder estimates quantify regularity below differentiability.

Example 15.3 (A Hölder modulus predicts possible slope blow-up)**.** Suppose

$$|f(x) - f(y)| \leq C|x - y|^\alpha, \qquad 0 < \alpha < 1.$$

Then a difference quotient is bounded only by

$$\frac{|f(x) - f(y)|}{|x - y|} \leq C|x - y|^{\alpha - 1},$$

whose right side grows as $y \to x$. Hölder continuity therefore guarantees continuity but permits much rougher local behavior than Lipschitz continuity. The exponent explicitly measures the available regularity scale.

15.8 Lessons for hypotheses

Counterexamples identify which assumptions in major theorems are genuinely necessary.

15.9 Core structural results

Theorem 15.4 (Uniform convergence of a Weierstrass series)**.** *If* $\sum a^n \cos(b^n x)$ *has* $0 < a < 1$, *then it converges uniformly and defines a continuous function.*

Proof. Each term is bounded by a^n and the geometric majorant converges; the uniform limit of continuous functions is continuous. ☐

Theorem 15.5 (Cantor-function monotonicity)**.** *The Cantor function is continuous, nondecreasing, and constant on every removed middle-third interval.*

Proof. Its ternary/binary digit construction preserves order; uniform approximants are continuous monotone functions and stabilize on removed intervals. ☐

Theorem 15.6 (Hölder roughness principle)**.** *A Hölder estimate* $|f(x) - f(y)| \leq C|x - y|^\alpha$ *with* $0 < \alpha < 1$ *permits much rougher behavior than Lipschitz regularity.*

Proof. The quotient by $|x - y|$ may grow like $|x - y|^{\alpha-1}$, so differentiability is not forced. ☐

15.10 Worked examples

Example 15.7 (Uniform continuity of Weierstrass partial sums)**.** Why does the geometric amplitude guarantee a continuous limit? The M-test gives uniform convergence, and uniform limits preserve continuity.

Example 15.8 (Derivative heuristic)**.** Why can differentiating $a^n \cos(b^n x)$ termwise be dangerous when $ab > 1$? Derivative amplitudes behave like $(ab)^n$, which grow rather than decay, signaling loss of derivative control.

Example 15.9 (Cantor function and derivative zero)**.** How can a nonconstant monotone function have derivative zero on a dense open set? It is constant on each complementary middle-third interval; the union of these intervals is dense and open, yet the function increases through the Cantor set.

Example 15.10 (Pointwise discontinuous limit)**.** Use x^n on $[0, 1]$ to exhibit a discontinuous pointwise limit. The limit is 0 on $[0, 1)$ and 1 at 1.

15.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 15.11 (Uniform continuity of Weierstrass partial sums)**.** Why does the geometric amplitude guarantee a continuous limit?

Solution. The M-test gives uniform convergence, and uniform limits preserve continuity. ☐

Problem 15.12 (Derivative heuristic)**.** Why can differentiating $a^n \cos(b^n x)$ termwise be dangerous when $ab > 1$?

Solution. Derivative amplitudes behave like $(ab)^n$, which grow rather than decay, signaling loss of derivative control. ☐

Problem 15.13 (Cantor function and derivative zero)**.** How can a nonconstant monotone function have derivative zero on a dense open set?

Solution. It is constant on each complementary middle-third interval; the union of these intervals is dense and open, yet the function increases through the Cantor set. ☐

Problem 15.14 (Pointwise discontinuous limit)**.** Use x^n on $[0, 1]$ to exhibit a discontinuous pointwise limit.

Solution. The limit is 0 on $[0, 1)$ and 1 at 1. ☐

Problem 15.15 (Dirichlet function)**.** What is special about $1_{\mathbb{Q}}$?

Solution. It is discontinuous at every real point because every interval contains both rationals and irrationals. ☐

Problem 15.16 (Thomae contrast)**.** Describe why Thomae's function is discontinuous at rationals but continuous at irrationals.

Solution. Large values occur only at rationals with small denominators, of which there are finitely many in a bounded interval; near an irrational one can avoid those finitely many. ☐

Problem 15.17 (Hölder but not Lipschitz)**.** Show $f(x) = \sqrt{|x|}$ is Hölder-1/2 but not Lipschitz near zero.

Solution. $|\sqrt{|x|} - \sqrt{|y|}| \leq \sqrt{|x - y|}$, while $|f(x) - f(0)|/|x| = 1/\sqrt{|x|} \to \infty$. ☐

Problem 15.18 (Dense set of discontinuities)**.** Can a Riemann-integrable function have a dense set of discontinuities?

Solution. Yes; Thomae's function is Riemann integrable and discontinuous at every rational, a dense set of measure zero. ☐

Problem 15.19 (Nowhere differentiability lesson)**.** Why does continuity alone not control local slope?

Solution. Continuity bounds function increments absolutely, but differentiability requires increments to be asymptotically linear relative to distance. ☐

Problem 15.20 (Uniform-limit protection)**.** Which pathology does uniform convergence prevent?

Solution. It prevents continuous approximants from converging to a discontinuous limit. ☐

Problem 15.21 (Hypothesis testing)**.** How should a counterexample be used when reading a theorem?

Solution. Remove one hypothesis at a time and see whether the conclusion fails; a sharp counterexample shows that hypothesis controls a real mechanism. ☐

Problem 15.22 (Pathology synthesis)**.** Why are pathological examples central to analysis?

Solution. They separate intuition based on smooth finite-dimensional examples from the exact logical content of definitions and theorems. ☐

15.12 Exercises with complete solutions

Exercise 15.23. Is every continuous function differentiable somewhere?

Hint. Think of nowhere-differentiable examples. Isolate one scale of the construction where the oscillation is comparable to the increment under consideration. □

Solution. No. □

Exercise 15.24. Does uniform limit of continuous functions stay continuous?

Hint. For a nowhere-differentiability argument, choose increments adapted to the series frequencies so one term dominates the difference quotient. □

Solution. Yes. □

Exercise 15.25. Does pointwise limit?

Hint. Use x^n. Uniform convergence of the defining series proves continuity, not differentiability; analyze difference quotients separately. □

Solution. Not necessarily. □

Exercise 15.26. Can a monotone function be constant on dense open pieces yet nonconstant globally?

Hint. Cantor function. Split the series into low-frequency and high-frequency parts and estimate the two pieces differently. □

Solution. Yes. □

Exercise 15.27. Is $\sqrt{|x|}$ Lipschitz near zero?

Hint. Check quotient. Use the modulus of continuity to record how oscillation changes with scale. □

Solution. No. □

Exercise 15.28. Is it Hölder-1/2?

Hint. Use square-root inequality. To refute differentiability, produce two sequences of increments whose difference quotients cannot approach the same finite limit. □

Solution. Yes. □

Exercise 15.29. Is Dirichlet's function Riemann integrable?

Hint. Upper/lower sums. Check whether termwise differentiation would require a derivative series that actually converges. □

Solution. No. □

Exercise 15.30. Why study counterexamples?

Hint. Test hypotheses. Do not rely on a plot: convert visible oscillation into an explicit inequality at a chosen sequence of scales. □

Solution. They reveal necessity and boundaries of theorems. □

15.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 15.31 (M-test continuity)**.** Why does $\sum a^n \cos(b^n x)$ define a continuous function for $0 < a < 1$?

Hint. Compare each term with a^n. ☐

Solution. The M-test gives uniform convergence of continuous terms; the limit is continuous. ☐

Exercise 15.32 (Derivative amplitude)**.** What is the amplitude scale after formally differentiating $a^n \cos(b^n x)$?

Hint. Differentiate and inspect the coefficient. ☐

Solution. It is $a^n b^n = (ab)^n$. ☐

Exercise 15.33 (Cantor flat intervals)**.** What is the derivative of the Cantor function inside a removed middle-third interval?

Hint. The function is locally constant there. ☐

Solution. The derivative is 0. ☐

Exercise 15.34 (Hölder quotient)**.** If $|f(x) - f(y)| \leq C|x - y|^{1/2}$, what bound follows for a difference quotient?

Hint. Divide by $|x - y|$. ☐

Solution. It is at most $C|x - y|^{-1/2}$. ☐

Exercise 15.35 (Pointwise pathology)**.** What is the pointwise limit of x^n on $[0, 1]$?

Hint. Separate $x < 1$ and $x = 1$. ☐

Solution. It is 0 on $[0, 1)$ and 1 at 1. ☐

Proofs

Exercise 15.36 (Uniform continuity of Weierstrass sum)**.** Prove the uniformly convergent Weierstrass-type sum is continuous.

Hint. Use the uniform-limit theorem. ☐

Solution. The partial sums are continuous and converge uniformly by the M-test. ☐

Exercise 15.37 (Dense open flat set)**.** Explain why the union of removed middle-third intervals is dense in $[0, 1]$.

Hint. Every surviving interval is subdivided at later stages. ☐

Solution. Any nonempty interval eventually contains a removed middle-third interval. ☐

Exercise 15.38 (Hölder implies continuity)**.** Prove a Hölder estimate with exponent $\alpha > 0$ implies uniform continuity.

Hint. Choose $\delta = (\varepsilon/C)^{1/\alpha}$. □

Solution. Then $|x - y| < \delta$ gives $|f(x) - f(y)| < \varepsilon$. □

Exercise 15.39 (Uniform limit pathology restriction)**.** Prove a discontinuous function cannot be the uniform limit of continuous functions.

Hint. Use the uniform-limit theorem. □

Solution. Uniform limits preserve continuity. □

Counterexamples and hypothesis testing

Exercise 15.40 (Continuity not differentiability)**.** Give a simple continuous function not differentiable at one point.

Hint. Use absolute value. □

Solution. $|x|$ is continuous but not differentiable at 0. □

Exercise 15.41 (Derivative zero densely not constant)**.** Why does the Cantor function refute the naive claim that derivative zero on a dense set forces constancy?

Hint. Its derivative is zero on the dense open complement of the Cantor set but the function increases. □

Solution. The claim needs stronger derivative hypotheses; density alone is insufficient. □

Exercise 15.42 (Uniform convergence not derivative convergence)**.** Give a uniformly convergent sequence of differentiable functions whose derivatives do not converge uniformly.

Hint. Use $\sin(nx)/n$. □

Solution. $\sin(nx)/n \to 0$ uniformly, but derivatives $\cos(nx)$ fail to converge uniformly. □

Applications and investigations

Exercise 15.43 (Rough-data model)**.** Why are Hölder exponents useful for noisy or fractal signals?

Hint. They quantify how increments scale with distance. □

Solution. The exponent gives a measurable regularity rate even when classical derivatives do not exist. □

Exercise 15.44 (Hypothesis stress test)**.** What is the pedagogical role of pathological examples in analysis?

Hint. Identify the exact theorem hypothesis they violate. □

Solution. They distinguish genuinely necessary assumptions from convenient proof artifacts. □

Challenge problems

Exercise 15.45 (Takagi-scale heuristic)**.** For a dyadic tent-function sum with amplitudes 2^{-n}, explain why continuity can coexist with persistent slope changes.

Hint. Each finer term has smaller height but higher frequency. □

Solution. Uniformly summable heights ensure continuity, while slopes of individual scales can remain order one, creating unresolved oscillation at every scale. □

Exercise 15.46 (Baire viewpoint)**.** Why can nowhere-differentiable behavior be considered abundant in $C[0, 1]$?

Hint. Think of differentiability constraints as restrictive closed/nowhere-dense conditions in function space. □

Solution. Category arguments show the set of functions differentiable somewhere can be small in the Baire-category sense, so roughness is generic rather than exceptional. □

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.

Part IV

Fixed Points and Differential Equations

Chapter 16

Contraction Mappings

The contraction mapping theorem turns a quantitative shrinking estimate into existence, uniqueness, and an algorithm. It is one of the cleanest bridges between analysis and computation.

Learning goals

After this chapter, the reader should be able to:

- verify that a self-map is a contraction by finding a constant $q < 1$;
- prove existence and uniqueness of a fixed point by applying the contraction mapping theorem with all hypotheses checked;
- derive quantitative error bounds for Picard iteration;
- choose a closed invariant subset on which a proposed iteration is actually contractive;
- compare different fixed-point iterations by their contraction constants;
- construct examples showing why completeness or the strict contraction condition cannot simply be omitted;

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

16.1 Contractions

A contraction reduces all distances by a fixed factor $q < 1$.

16.2 Fixed points

A fixed point solves $Tx = x$.

16.3 Picard iteration

Repeated application $x_{n+1} = Tx_n$ is the natural algorithm.

Example 16.1 (Picard iteration with an exact fixed point check)**.** Let

$$T(x) = \frac{x+3}{4}.$$

Then

$$|T(x) - T(y)| = \frac{1}{4}|x - y|,$$

so $q = 1/4$. Starting from $x_0 = 0$,

$$x_1 = \frac{3}{4}, \qquad x_2 = \frac{15}{16}, \qquad x_3 = \frac{63}{64}.$$

The fixed-point equation gives $x_* = 1$, and indeed

$$|x_n - 1| = 4^{-n}.$$

The explicit formula checks the geometric convergence predicted by Banach's theorem.

16.4 Completeness

The iteration is Cauchy because geometric distance estimates telescope; completeness supplies the limit.

16.5 Uniqueness

Two fixed points would be strictly closer after applying T, impossible unless equal.

16.6 Error estimates

A priori and a posteriori geometric bounds quantify iteration error.

Example 16.2 (A posteriori error from the last step)**.** For a contraction with factor $q < 1$, Picard iterates satisfy

$$d(x_n, x_*) \le \frac{q}{1-q} d(x_n, x_{n-1}).$$

Thus if $q = 0.2$ and the most recent step has size 10^{-5}, then

$$d(x_n, x_*) \le \frac{0.2}{0.8} 10^{-5} = 2.5 \times 10^{-6}.$$

This is computationally useful because it estimates the unknown fixed-point error from quantities already produced by the iteration.

16.7 Parameter dependence

Contractive fixed points vary stably under perturbations.

Example 16.3 (Stable dependence of a fixed point on a parameter)**.** Consider

$$T_\lambda(x) = \frac{x+\lambda}{3}.$$

Every T_λ has contraction factor $1/3$ and fixed point $x_\lambda = \lambda/2$. If λ, μ are two parameters,

$$|x_\lambda - x_\mu| = \frac{1}{2}|\lambda - \mu|.$$

The general perturbation theorem gives the compatible bound

$$|x_\lambda - x_\mu| \leq \frac{1}{1 - 1/3} \sup_x |T_\lambda(x) - T_\mu(x)| = \frac{1}{2}|\lambda - \mu|.$$

Here the stability estimate is sharp.

16.8 Applications

Integral equations and ODEs can be reformulated as fixed-point problems.

16.9 Core structural results

Theorem 16.4 (Banach fixed-point theorem)**.** *A contraction on a nonempty complete metric space has a unique fixed point, and every Picard iteration converges to it.*

Proof. The successive differences decay geometrically, making iterates Cauchy. Completeness gives a limit; continuity of the contraction makes it fixed; the contraction inequality gives uniqueness. ☐

Theorem 16.5 (A priori error bound)**.** *If $x_{n+1} = Tx_n$ and contraction factor is q, then $d(x_n, x_*) \leq q^n(1-q)^{-1}d(x_1, x_0)$.*

Proof. Bound the tail by the geometric sum of successive differences. ☐

Theorem 16.6 (Stability under perturbation)**.** *If contractions T, S share factor $q < 1$ and fixed points x_T, x_S, then $d(x_T, x_S) \leq (1-q)^{-1} \sup_x d(Tx, Sx)$.*

Proof. Insert and subtract Tx_S and use the contraction bound before rearranging. ☐

16.10 Worked examples

Example 16.7 (Affine contraction)**.** Solve $x = (x+3)/4$. The map has factor $1/4$ and fixed point $x = 1$.

Example 16.8 (Iteration rate)**.** How many powers of q control the error? Error decays geometrically like q^n, up to the prefactor determined by the first step.

Example 16.9 (Uniqueness proof)**.** Show two fixed points must coincide. $d(x, y) = d(Tx, Ty) \leq qd(x, y)$; since $q < 1$, the distance is zero.

Example 16.10 (Why completeness matters)**.** Give a contraction on an incomplete space with no fixed point. On $(0, 1)$, $T(x) = (x+1)/2$ is a contraction whose fixed point 1 lies outside the space.

16.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 16.11 (Affine contraction)**.** Solve $x = (x + 3)/4$.

Solution. The map has factor $1/4$ and fixed point $x = 1$. □

Problem 16.12 (Iteration rate)**.** How many powers of q control the error?

Solution. Error decays geometrically like q^n, up to the prefactor determined by the first step. □

Problem 16.13 (Uniqueness proof)**.** Show two fixed points must coincide.

Solution. $d(x, y) = d(Tx, Ty) \leq qd(x, y)$; since $q < 1$, the distance is zero. □

Problem 16.14 (Why completeness matters)**.** Give a contraction on an incomplete space with no fixed point.

Solution. On $(0, 1)$, $T(x) = (x + 1)/2$ is a contraction whose fixed point 1 lies outside the space. □

Problem 16.15 (Matrix contraction)**.** When is $T(x) = Ax + b$ a Euclidean contraction?

Solution. Whenever the operator norm $\|A\|_2 < 1$. □

Problem 16.16 (Fixed-point series)**.** Solve $x = Ax + b$ with $\|A\| < 1$.

Solution. $x = (I - A)^{-1}b = \sum_{n\geq 0} A^n b$, with convergence from the geometric operator-norm bound. □

Problem 16.17 (A posteriori estimate)**.** Bound $d(x_n, x_*)$ using $d(x_n, x_{n-1})$.

Solution. The tail is at most $d(x_n, x_{n-1})/(1 - q)$. □

Problem 16.18 (Nested mapping)**.** Why must T map the complete set into itself?

Solution. Otherwise the iteration may leave the domain and the completeness argument no longer applies. □

Problem 16.19 (Parameter stability)**.** What happens if T is perturbed by at most epsilon uniformly?

Solution. The fixed point moves by at most $\varepsilon/(1 - q)$. □

Problem 16.20 (Integral-equation preview)**.** How does an integral operator become a contraction?

Solution. Bound the sup-norm difference of outputs by an interval-length times a Lipschitz constant times the input sup difference. □

Problem 16.21 (ODE preview)**.** How does Picard iteration arise for $y' = f(t, y)$?

Solution. Rewrite as $y(t) = y_0 + \int f(s, y(s))ds$ and iterate the integral operator. □

Problem 16.22 (Contraction synthesis)**.** Why is this theorem algorithmic?

Solution. Its proof constructs the solution by iteration and simultaneously provides computable geometric error bounds. □

16.12 Exercises with complete solutions

Exercise 16.23. Find fixed point of $T(x) = x/2 + 1$.

Hint. Solve algebraically. Compute a Lipschitz bound for T and verify the constant is strictly below one on the chosen domain. ☐

Solution. It is 2. ☐

Exercise 16.24. What is the contraction factor?

Hint. Coefficient of x. Check that the domain is complete and that T maps it into itself before invoking the contraction theorem. ☐

Solution. 1/2. ☐

Exercise 16.25. Why unique?

Hint. Use fixed-point inequality. Use $d(x_n, x_*) \leq q^n d(x_0, x_*)$ or the a posteriori geometric-tail estimate to quantify iteration error. ☐

Solution. Two fixed points have zero distance. ☐

Exercise 16.26. Does a contraction have to be linear?

Hint. If the global map is not contractive, restrict to a closed invariant subset where a sharper derivative or Lipschitz bound holds. ☐

Solution. No. ☐

Exercise 16.27. What property supplies the limit?

Hint. Completeness. Uniqueness follows by applying the contraction inequality to two hypothetical fixed points. ☐

Solution. Completeness of the space. ☐

Exercise 16.28. What sequence is used?

Hint. Picard iteration. To compare iterations, the smaller contraction factor gives the faster geometric asymptotic rate. ☐

Solution. $x_{n+1} = Tx_n$. ☐

Exercise 16.29. What is the qualitative error rate?

Hint. Geometric. If $q = 1$, the theorem no longer gives convergence or uniqueness; inspect a simple isometry as a counterexample. ☐

Solution. Order q^n. ☐

Exercise 16.30. Can $q = 1$ be used?

Hint. No strict contraction. Completeness is used to ensure the Cauchy iterate sequence has a limit inside the space. ☐

Solution. Not in Banach's theorem. ☐

16.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 16.31 (Affine fixed point)**.** Find the fixed point of $T(x) = (x + 5)/3$.

Hint. Solve $x = T(x)$. □

Solution. $x_* = 5/2$. □

Exercise 16.32 (Contraction constant)**.** Find a contraction constant for $T(x) = 0.7x + 1$ on $\mathbb{R}$.

Hint. Subtract outputs. □

Solution. $q = 0.7$. □

Exercise 16.33 (First Picard steps)**.** For $T(x) = (x + 3)/4$, compute x_1, x_2 from $x_0 = 0$.

Hint. Apply T twice. □

Solution. $x_1 = 3/4$, $x_2 = 15/16$. □

Exercise 16.34 (A priori error)**.** If $q = 1/2$ and $d(x_1, x_0) = 1$, bound $d(x_5, x_*)$.

Hint. Use $q^n(1 - q)^{-1}d(x_1, x_0)$. □

Solution. The bound is $2(1/2)^5 = 1/16$. □

Exercise 16.35 (A posteriori error)**.** If $q = 1/3$ and $d(x_n, x_{n-1}) = 0.006$, bound current error.

Hint. Use $q/(1 - q)$ times the last step. □

Solution. The error is at most 0.003. □

Proofs

Exercise 16.36 (Banach uniqueness)**.** Prove a contraction has at most one fixed point.

Hint. Apply the contraction inequality to two fixed points. □

Solution. $d(x, y) \le qd(x, y)$ with $q < 1$, so $d(x, y) = 0$. □

Exercise 16.37 (Picard Cauchy)**.** Prove Picard iterates form a Cauchy sequence.

Hint. Bound successive differences geometrically and sum a tail. □

Solution. $d(x_{n+k}, x_n) \le d(x_1, x_0) \sum_{j=n}^{n+k-1} q^j$, whose tail tends to zero. □

Exercise 16.38 (Fixed point at the limit)**.** Why is the Picard limit fixed?

Hint. Use continuity of a contraction. □

Solution. If $x_n \to x_*$, then $x_{n+1} = Tx_n \to Tx_*$, but also $x_{n+1} \to x_*$, so $Tx_* = x_*$. □

Exercise 16.39 (Perturbation bound)**.** Derive $d(x_T, x_S) \le (1 - q)^{-1} \sup d(Tx, Sx)$ for contractions with common factor q.

Hint. Insert Tx_S. □

Solution. $d(x_T, x_S) \le qd(x_T, x_S) + \sup d(Tx, Sx)$; rearrange. □

Counterexamples and hypothesis testing

Exercise 16.40 (Completeness required)**.** Give a contraction on an incomplete space with no fixed point.

Hint. Use $(0, 1)$ and a map whose fixed point is 1. □

Solution. $T(x) = (x + 1)/2$ is a contraction on $(0, 1)$ but has no fixed point there. □

Exercise 16.41 (Strict contraction required)**.** Give a nonexpansive self-map with no fixed point.

Hint. Use translation on $\mathbb{R}$. □

Solution. $T(x) = x + 1$ has Lipschitz constant 1 and no fixed point. □

Exercise 16.42 (Self-map required)**.** Why can a contractive formula fail Banach's theorem if it leaves the chosen domain?

Hint. The theorem requires $T : X \to X$. □

Solution. Without invariance, the iteration may exit the complete set on which the estimate was checked. □

Applications and investigations

Exercise 16.43 (Iteration stopping)**.** How can the a posteriori bound be used as a stopping rule?

Hint. Stop once the bound computed from the latest step is below tolerance. □

Solution. It certifies the unknown fixed-point error using observable iterate differences. □

Exercise 16.44 (Nonlinear equation)**.** How does solving $x = \cos x$ become a fixed-point problem?

Hint. Use $T(x) = \cos x$ on a suitable invariant interval. □

Solution. Choose an interval where $|\sin x| \le q < 1$; Banach then proves a unique local fixed point and convergence of iteration. □

Challenge problems

Exercise 16.45 (Choose an invariant interval)**.** Show $T(x) = \frac{1}{2}\cos x$ is a contraction mapping of $[-1, 1]$ into itself.

Hint. Bound both output size and derivative. □

Solution. $|T(x)| \le 1/2$, so the interval is invariant, and $|T'(x)| \le 1/2$, so it is a contraction. □

Exercise 16.46 (Optimal relaxation)**.** For $T_\alpha(x) = (1 - \alpha)x + \alpha c$, $0 < \alpha < 2$, find the contraction factor and discuss the best α.

Hint. Subtract outputs. □

Solution. The factor is $|1 - \alpha|$, minimized at $\alpha = 1$, which reaches the fixed point c in one step. □

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.

Chapter 17

Brouwer-Type Fixed-Point Ideas

Brouwer-type fixed-point ideas replace metric contraction by topology. Existence can survive without uniqueness or an iterative rate, revealing a distinct mechanism from Banach's theorem.

Learning goals

After this chapter, the reader should be able to:

- state and apply Brouwer-type fixed-point conclusions in finite-dimensional compact convex settings;
- verify compactness, convexity, and self-map hypotheses before invoking a fixed-point theorem;
- construct elementary one-dimensional or planar fixed-point arguments from continuity;
- distinguish Brouwer-type existence results from contraction-based uniqueness results;
- analyze examples where a fixed point exists but is not unique;
- identify which hypothesis fails in a fixed-point counterexample involving noncompact or nonconvex domains;

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

17.1 Fixed points without contraction

Continuous maps can have fixed points for topological rather than metric reasons.

17.2 Interval case

The intermediate-value theorem already yields a one-dimensional Brouwer theorem.

17.3 Convex compact sets

Brouwer's theorem applies to continuous self-maps of compact convex subsets of Euclidean space.

Example 17.1 (Checking Brouwer hypotheses on a disk)**.** Let

$$K = \{x \in \mathbb{R}^2 : \|x\| \le 1\}, \qquad T(x) = \frac{1}{2}x + \frac{1}{4}e_1.$$

The disk is nonempty, compact, and convex. For $\|x\| \le 1$,

$$\|T(x)\| \le \frac{1}{2} + \frac{1}{4} = \frac{3}{4} < 1,$$

so $T(K) \subset K$. Brouwer therefore gives a fixed point. Here we can also solve explicitly:

$$x = \frac{1}{2}x + \frac{1}{4}e_1 \quad \Rightarrow \quad x = \frac{1}{2}e_1.$$

The direct solution checks the theorem's conclusion after every hypothesis has been verified.

17.4 No-retraction intuition

A ball cannot continuously retract onto its boundary.

17.5 Degree intuition

Topological degree counts preimages with orientation and obstructs disappearance of fixed points.

17.6 Existence versus uniqueness

Brouwer gives existence but generally no uniqueness.

Example 17.2 (Existence does not imply uniqueness)**.** On the closed unit disk, the identity map is continuous and fixes every point. Brouwer correctly guarantees existence, but it supplies no uniqueness. More subtly, a continuous map can have several isolated fixed points without being contractive.

Thus fixed-point theorems answer different questions: topological hypotheses may force at least one equilibrium, while metric shrinking is what gives the strong uniqueness and iteration conclusions of Banach's theorem.

17.7 Applications

Equilibria in games, economics, and nonlinear systems often use Brouwer-type existence.

17.8 Comparison with Banach

Banach is quantitative and metric; Brouwer is qualitative and topological.

Example 17.3 (Banach and Brouwer certify different structures)**.** For $T(x) = x^2$ on $[0, 1]$, Brouwer applies because the interval is compact and convex and T is a continuous self-map. The fixed points are 0 and 1, so uniqueness fails. There is no global contraction factor $q < 1$ on the whole interval because

$$|T'(1)| = 2.$$

This single example separates the qualitative Brouwer mechanism from the quantitative Banach mechanism.

17.9 Core structural results

Theorem 17.4 (Interval fixed-point theorem)**.** *Every continuous $f : [a, b] \to [a, b]$ has a fixed point.*

Proof. Apply the intermediate-value theorem to $g(x) = f(x) - x$: $g(a) \geq 0$ and $g(b) \leq 0$. ☐

Theorem 17.5 (Brouwer fixed-point theorem)**.** *Every continuous map from a nonempty compact convex subset of $\mathbb{R}^n$ to itself has a fixed point.*

Proof. The full proof uses a topological obstruction such as no-retraction, Sperner's lemma, or degree; the key point is that a fixed-point-free map would induce impossible boundary data. ☐

Theorem 17.6 (No-retraction consequence)**.** *There is no continuous retraction from the closed ball onto its boundary sphere.*

Proof. A retraction would contradict Brouwer by composing radial constructions to obtain a fixed-point-free self-map of the ball. ☐

17.10 Worked examples

Example 17.7 (Interval proof)**.** Prove the fixed-point theorem on $[0, 1]$. For $g = f - \mathrm{id}$, $g(0) \geq 0$ and $g(1) \leq 0$; continuity gives a zero.

Example 17.8 (Failure without compactness)**.** Give a continuous self-map of $(0, 1)$ with no fixed point. $f(x) = (x + 1)/2$ maps $(0, 1)$ to itself and has only fixed point 1, outside the domain.

Example 17.9 (Failure without self-map condition)**.** Why does $f(x) = x + 1$ on $[0, 1]$ not contradict Brouwer? Its image is not contained in $[0, 1]$.

Example 17.10 (Nonuniqueness)**.** Give a Brouwer map with many fixed points. The identity map fixes every point.

17.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 17.11 (Interval proof)**.** Prove the fixed-point theorem on $[0, 1]$.

Solution. For $g = f - \mathrm{id}$, $g(0) \geq 0$ and $g(1) \leq 0$; continuity gives a zero. ☐

Problem 17.12 (Failure without compactness)**.** Give a continuous self-map of $(0, 1)$ with no fixed point.

Solution. $f(x) = (x + 1)/2$ maps $(0, 1)$ to itself and has only fixed point 1, outside the domain. ☐

Problem 17.13 (Failure without self-map condition)**.** Why does $f(x) = x + 1$ on $[0, 1]$ not contradict Brouwer?

Solution. Its image is not contained in $[0, 1]$. ☐

Problem 17.14 (Nonuniqueness)**.** Give a Brouwer map with many fixed points.

Solution. The identity map fixes every point. ☐

Problem 17.15 (Banach versus Brouwer)**.** Which theorem gives an iteration rate?

Solution. Banach does; Brouwer generally supplies existence only. ☐

Problem 17.16 (Convexity role)**.** Why is an annulus not covered directly by Brouwer's convex-set statement?

Solution. An annulus is not convex; straight segments between points may leave it. ☐

Problem 17.17 (A rotation of a disk)**.** Why must a disk rotation have a fixed point?

Solution. The center is fixed, consistent with Brouwer. ☐

Problem 17.18 (Sperner intuition)**.** How does Sperner's lemma relate to Brouwer?

Solution. A labeled triangulation forces a fully labeled simplex; shrinking mesh sizes produce approximate fixed points whose compactness limit is an exact one. ☐

Problem 17.19 (Equilibrium interpretation)**.** How can a fixed point encode equilibrium?

Solution. A response map sends a state to its updated best-response or normalized state; a fixed point is unchanged by the update and thus represents equilibrium. ☐

Problem 17.20 (Topological obstruction)**.** What does 'no retraction' mean intuitively?

Solution. One cannot continuously push every interior point to the boundary while leaving boundary points fixed. ☐

Problem 17.21 (Approximate fixed points)**.** Why are compactness arguments natural in Brouwer proofs?

Solution. Combinatorial or discretized arguments produce approximate fixed points; compactness extracts a convergent subsequence and continuity turns the limiting approximation error into zero. ☐

Problem 17.22 (Brouwer synthesis)**.** What fundamentally distinguishes Brouwer from Banach?

Solution. Brouwer relies on topology of compact convex sets and gives existence; Banach relies on quantitative shrinking in complete metric spaces and gives uniqueness plus convergence. ☐

17.12 Exercises with complete solutions

Exercise 17.23. Does every continuous $[0,1] \to [0,1]$ map have a fixed point?

Hint. Check compactness, convexity, and continuity/self-map conditions explicitly before invoking Brouwer. ☐

Solution. Yes. ☐

Exercise 17.24. Can it have several?

Hint. Identity. In one dimension, apply the intermediate value theorem to $f(x) - x$. ☐

Solution. Yes. ☐

Exercise 17.25. Does Brouwer guarantee uniqueness?

Hint. Brouwer yields existence, not uniqueness; look for multiple fixed points before asserting more. ☐

Solution. No. ☐

Exercise 17.26. Does compactness matter?

Hint. Open interval counterexample. To show convexity, verify that every segment $(1-t)x+ty$ stays in the set. ☐

Solution. Yes. ☐

Exercise 17.27. Does convexity appear in the standard theorem?

Hint. For a noncompact counterexample, try a translation that moves every point while still mapping the space into itself. ☐

Solution. Yes. ☐

Exercise 17.28. Is a disk convex?

Hint. Straight segments stay inside. For a nonconvex domain, construct a rotation or swap that avoids fixed points. ☐

Solution. Yes. ☐

Exercise 17.29. Is a circle convex?

Hint. Chord leaves circle. Compare with contraction mapping: a contraction supplies metric shrinkage and uniqueness, while Brouwer does not. ☐

Solution. No. ☐

Exercise 17.30. What one-dimensional theorem proves the interval case?

Hint. Intermediate value theorem. When reducing to a finite-dimensional ball, make sure the map really sends the ball back into itself. ☐

Solution. IVT. ☐

17.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 17.31 (Interval fixed point)**.** Why must $f(x) = 1 - x$ on $[0, 1]$ have a fixed point, and what is it?

Hint. Solve $f(x) = x$. □

Solution. The fixed point is $x = 1/2$. □

Exercise 17.32 (Self-map check)**.** Does $T(x) = x + 1$ satisfy Brouwer hypotheses on $[0, 1]$?

Hint. Check image containment. □

Solution. No; it is not a self-map of $[0, 1]$. □

Exercise 17.33 (Convexity)**.** Is the closed unit disk convex?

Hint. Use the triangle inequality on $tx + (1 - t)y$. □

Solution. Yes; its norm is at most $t\|x\| + (1 - t)\|y\| \leq 1$. □

Exercise 17.34 (Compactness)**.** Why is a closed bounded ball in $\mathbb{R}^n$ compact?

Hint. Use Heine–Borel. □

Solution. It is closed and bounded, hence compact. □

Exercise 17.35 (Nonunique fixed points)**.** How many fixed points does the identity map have?

Hint. Apply the definition. □

Solution. Every point is fixed. □

Proofs

Exercise 17.36 (Interval Brouwer)**.** Prove every continuous $f : [a, b] \to [a, b]$ has a fixed point.

Hint. Apply IVT to $g(x) = f(x) - x$. □

Solution. $g(a) \geq 0$ and $g(b) \leq 0$, so $g(c) = 0$ for some c. □

Exercise 17.37 (Compact convex self-map check)**.** Explain why each Brouwer hypothesis has a distinct role in the theorem statement.

Hint. Compare standard failures when compactness, convexity, or self-map containment is removed. □

Solution. Each prevents a different escape mechanism: points can run to infinity/boundary, domains can have holes, or the map can leave the domain. □

Exercise 17.38 (No-retraction consequence)**.** Explain how a retraction of a ball onto its sphere conflicts with Brouwer.

Hint. Use the retraction to construct a fixed-point-free self-map directed toward the opposite boundary. □

Solution. Such a construction would contradict that every continuous self-map of the ball has a fixed point. □

Exercise 17.39 (Continuous image relevance)**.** Why is continuity essential in fixed-point existence arguments based on IVT or topology?

Hint. Discontinuous maps can jump over the diagonal. □

Solution. Without continuity, the sign-change or topological obstruction need not force a zero/fixed point. □

Counterexamples and hypothesis testing

Exercise 17.40 (Noncompact failure)**.** Give a continuous self-map of $(0, 1)$ with no fixed point.

Hint. Use $(x + 1)/2$. □

Solution. Its only algebraic fixed point is 1, outside the domain. □

Exercise 17.41 (Nonconvex failure)**.** Give a compact nonconvex set with a continuous self-map having no fixed point.

Hint. Use the unit circle and a rotation. □

Solution. A nontrivial rotation of S^1 is continuous and fixed-point-free. □

Exercise 17.42 (Existence not uniqueness)**.** Give a Brouwer self-map with infinitely many fixed points.

Hint. Use the identity. □

Solution. The identity fixes the entire compact convex domain. □

Applications and investigations

Exercise 17.43 (Economic equilibrium)**.** Why are Brouwer-type theorems natural for equilibrium existence models?

Hint. An equilibrium is a fixed point of a continuous response map on a compact convex strategy/price set. □

Solution. The theorem can guarantee existence without requiring a contraction or a known iterative rate. □

Exercise 17.44 (Topological robustness)**.** What qualitative information can Brouwer provide when no useful Lipschitz constant exists?

Hint. It uses compactness, convexity and continuity instead of metric shrinking. □

Solution. It still certifies at least one fixed point. □

Challenge problems

Exercise 17.45 (Radial map)**.** Let $T(x) = \phi(\|x\|)x$ map the closed unit ball to itself continuously. Explain why 0 is always a fixed point.

Hint. Evaluate at the origin. ☐

Solution. $T(0) = \phi(0)0 = 0$. ☐

Exercise 17.46 (Banach implies Brouwer conclusion)**.** On a compact convex set, if a self-map is also a contraction, compare the two theorem conclusions.

Hint. State what each theorem adds. ☐

Solution. Both give existence; Banach additionally gives uniqueness, convergence of Picard iteration, and quantitative error bounds. ☐

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.

Chapter 18

Integral Equations

Integral equations encode unknown functions under an integral operator. Their analytic treatment combines norm estimates, compactness ideas, and fixed-point principles.

Learning goals

After this chapter, the reader should be able to:

- rewrite an integral equation as a fixed-point problem on a function space;
- estimate the associated integral operator in a sup or other relevant norm;
- verify conditions under which the integral operator is a contraction or maps a bounded set into itself;
- solve elementary separable or explicitly integrable integral equations;
- derive successive-approximation schemes and estimate their errors;
- distinguish existence, uniqueness, and stability conclusions for an integral equation;

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

18.1 Fredholm equations

The unknown appears under an integral over a fixed domain.

18.2 Volterra equations

The upper limit depends on the evaluation point, producing a triangular time structure.

18.3 Integral operators

Kernels define linear or nonlinear operators on function spaces.

18.4 Sup-norm estimates

Kernel bounds convert integral operators into norm inequalities.

Example 18.1 (A sharp sup-norm kernel estimate)**.** For

$$(Kf)(x) = \int_0^1 xt\, f(t)\, dt, \qquad 0 \le x \le 1,$$

we have

$$|Kf(x)| \le x\|f\|_\infty \int_0^1 t\, dt = \frac{x}{2}\|f\|_\infty.$$

Taking the supremum gives

$$\|K\|_\infty \le \frac{1}{2}.$$

Equality is attained for $f \equiv 1$ at $x = 1$, so the norm is exactly $1/2$. The sharp estimate immediately shows $I - \lambda K$ is invertible by a Neumann series whenever $|\lambda| < 2$.

18.5 Neumann series

Small operator norm permits inversion of $I - K$ by a geometric series.

Example 18.2 (Neumann series with an explicit operator error)**.** If $\|K\| = q < 1$, then

$$(I - K)^{-1} = I + K + K^2 + \cdots .$$

After truncating at N,

$$\left\|(I-K)^{-1} - \sum_{j=0}^{N} K^j\right\| \le \sum_{j=N+1}^{\infty} q^j = \frac{q^{N+1}}{1-q}.$$

Thus the same geometric series proving invertibility supplies a certified approximation of the inverse operator.

18.6 Contraction methods

Nonlinear integral equations can be solved by Banach's theorem under Lipschitz conditions.

18.7 Iterated kernels

Repeated substitution creates iterated kernel integrals.

18.8 Connection to ODEs

Initial-value ODEs are equivalent to Volterra integral equations.

Example 18.3 (An ODE becomes a causal Volterra equation)**.** The IVP

$$y'(t) = t + y(t), \qquad y(0) = 1$$

is equivalent to

$$y(t) = 1 + \int_0^t (s + y(s))\, ds.$$

A Picard scheme starts with $y_0(t) = 1$ and defines

$$y_{n+1}(t) = 1 + \int_0^t (s + y_n(s))\, ds.$$

The first iterate is $1 + t + t^2/2$. The upper integration limit t makes the operator causal: the value at time t depends only on earlier values.

18.9 Core structural results

Theorem 18.4 (Neumann-series inversion)**.** *If a bounded linear operator K satisfies $\|K\| < 1$, then $I - K$ is invertible with $(I - K)^{-1} = \sum_{n\geq 0} K^n$.*

Proof. The operator series converges absolutely in norm, and finite geometric identities pass to the limit. □

Theorem 18.5 (Kernel sup bound)**.** *For $(Kf)(x) = \int_a^b k(x,t)f(t)dt$, $\|K\|_\infty \leq (b-a)\|k\|_\infty$.*

Proof. Take absolute values, bound both kernel and function by sup norms, and integrate interval length. □

Theorem 18.6 (Volterra contraction on short intervals)**.** *If $F(t, y)$ is Lipschitz in y with constant L, the integral map $Ty = y_0 + \int_{t_0}^t F(s, y(s))ds$ is a contraction on an interval of length h when $Lh < 1$.*

Proof. The output difference is at most Lh times the input sup difference. □

18.10 Worked examples

Example 18.7 (Constant kernel)**.** Solve $f(x) = 1 + \lambda \int_0^1 f(t)dt$. Let $c = \int_0^1 f$. Then $f = 1 + \lambda c$ is constant and $c = 1 + \lambda c$, so for $\lambda \neq 1$, $f = 1/(1-\lambda)$.

Example 18.8 (Norm of a simple kernel operator)**.** For $Kf(x) = \int_0^1 xtf(t)dt$, bound $\|K\|_\infty$. Since $|xt| \leq 1$, the general bound gives $\|K\| \leq 1$; the sharper bound is $1/2$ because $\sup_x \int_0^1 xtdt = 1/2$.

Example 18.9 (Neumann approximation)**.** If $\|K\| = q < 1$, bound truncation after N terms. The inverse-series tail has norm at most $q^{N+1}/(1-q)$.

Example 18.10 (Volterra uniqueness)**.** Why are Volterra equations naturally compatible with iteration? The value at time t depends only on earlier times up to t, producing causal nested integrals and strong factorial-type bounds.

18.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 18.11 (Constant kernel)**.** Solve $f(x) = 1 + \lambda \int_0^1 f(t)dt$.

Solution. Let $c = \int_0^1 f$. Then $f = 1 + \lambda c$ is constant and $c = 1 + \lambda c$, so for $\lambda \neq 1$, $f = 1/(1-\lambda)$. □

Problem 18.12 (Norm of a simple kernel operator)**.** For $Kf(x) = \int_0^1 xtf(t)dt$, bound $\|K\|_\infty$.

Solution. Since $|xt| \leq 1$, the general bound gives $\|K\| \leq 1$; the sharper bound is $1/2$ because $\sup_x \int_0^1 xtdt = 1/2$. □

Problem 18.13 (Neumann approximation)**.** If $\|K\| = q < 1$, bound truncation after N terms.

Solution. The inverse-series tail has norm at most $q^{N+1}/(1-q)$. □

Problem 18.14 (Volterra uniqueness)**.** Why are Volterra equations naturally compatible with iteration?

Solution. The value at time t depends only on earlier times up to t, producing causal nested integrals and strong factorial-type bounds. □

Problem 18.15 (ODE equivalence)**.** Convert $y' = f(t, y)$, $y(t_0) = y_0$ to an integral equation.

Solution. Integrate to obtain $y(t) = y_0 + \int_{t_0}^t f(s, y(s))ds$. □

Problem 18.16 (Linear resolvent idea)**.** What does $(I - \lambda K)^{-1}$ represent?

Solution. It maps forcing terms to solutions of $f - \lambda K f = g$ when the inverse exists. □

Problem 18.17 (Short-interval contraction)**.** If Lipschitz constant is $L = 3$, how small should interval length be for direct Banach contraction?

Solution. Choose $h < 1/3$. □

Problem 18.18 (Iterated operator estimate)**.** If $\|K\| \leq q$, what is $\|K^n\|$?

Solution. At most q^n by submultiplicativity. □

Problem 18.19 (Kernel continuity)**.** Why does a continuous kernel on a compact square define a bounded sup-norm operator?

Solution. The kernel is bounded by compactness, and the kernel sup bound applies. □

Problem 18.20 (Nonlinear integral map)**.** How does a Lipschitz nonlinearity enter the estimate?

Solution. Take the difference inside the integral and use the pointwise Lipschitz bound before integration. □

Problem 18.21 (Fixed-point viewpoint)**.** What equation is actually solved by Banach?

Solution. The integral equation is rewritten as $f = Tf$ and the solution is the unique fixed point of T. □

Problem 18.22 (Integral-equation synthesis)**.** Why are integral equations often easier than differential equations?

Solution. They encode initial conditions automatically and replace derivatives by averaging operators, which are better behaved in norm estimates. □

18.12 Exercises with complete solutions

Exercise 18.23. What operator solves $f = g + Kf$?

Hint. Rearrange. Move every term except the unknown function to one side and define an operator Tu whose fixed points solve the equation. ☐

Solution. $(I - K)f = g$. ☐

Exercise 18.24. When does Neumann series converge?

Hint. Operator norm below one. Estimate $\|Tu - Tv\|_\infty$ by pulling $\|u - v\|_\infty$ outside the integral. ☐

Solution. If $\|K\| < 1$. ☐

Exercise 18.25. What norm is convenient on $C[a, b]$?

Hint. Sup norm. Bound the integral kernel uniformly to obtain a contraction constant. ☐

Solution. $\|f\|_\infty$. ☐

Exercise 18.26. What does a Volterra upper limit look like?

Hint. Variable upper endpoint. Before iterating, check that the operator maps the selected closed ball or function class into itself. ☐

Solution. $\int_a^x$. ☐

Exercise 18.27. What does a Fredholm equation use?

Hint. Fixed interval. Picard iteration starts from any convenient initial function and repeatedly applies the integral operator. ☐

Solution. $\int_a^b$. ☐

Exercise 18.28. How do ODEs connect?

Hint. Integrate once. For an explicit kernel, differentiate the integral equation only when regularity justifies doing so. ☐

Solution. They become Volterra equations. ☐

Exercise 18.29. Why does short interval help?

Hint. Multiplies Lipschitz constant. Existence and uniqueness are separate conclusions; identify exactly which estimate supplies each one. ☐

Solution. It makes $Lh < 1$. ☐

Exercise 18.30. Does $\|K\| < 1$ imply uniqueness for $(I - K)f = g$?

Hint. Use a geometric-series bound on successive differences to estimate the error after finitely many iterations. ☐

Solution. Yes, since $I - K$ is invertible. ☐

18.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 18.31 (Constant kernel)**.** Solve $f(x) = 2 + \lambda \int_0^1 f(t)\,dt$ for $\lambda \neq 1$.

Hint. The solution must be constant; call its integral c. ☐

Solution. $c = 2 + \lambda c$, so $f \equiv 2/(1-\lambda)$. ☐

Exercise 18.32 (Kernel bound)**.** For $Kf(x) = \int_0^1 x^2 t\, f(t)\,dt$, bound $\|K\|_\infty$.

Hint. Integrate t and maximize x^2. ☐

Solution. $\|K\| \leq 1/2$, and equality occurs for $f \equiv 1, x = 1$. ☐

Exercise 18.33 (Neumann tail)**.** If $\|K\| = 1/4$, bound the inverse-series tail after $N = 3$.

Hint. Use $q^{N+1}/(1-q)$. ☐

Solution. The bound is $(1/4)^4/(3/4) = 1/192$. ☐

Exercise 18.34 (Volterra first iterate)**.** For $y = 1 + \int_0^t y(s)\,ds$, compute the first two Picard iterates from $y_0 = 1$.

Hint. Substitute successively. ☐

Solution. $y_1 = 1 + t$, $y_2 = 1 + t + t^2/2$. ☐

Exercise 18.35 (Contraction interval)**.** If the state Lipschitz constant is $L = 5$, how short must an interval be to make the Volterra map contractive?

Hint. Require $Lh < 1$. ☐

Solution. Take $h < 1/5$. ☐

Proofs

Exercise 18.36 (Kernel operator bound)**.** Prove $\|K\|_\infty \leq (b-a)\|k\|_\infty$.

Hint. Take absolute values inside the integral. ☐

Solution. $|Kf(x)| \leq (b-a)\|k\|_\infty \|f\|_\infty$; take the sup. ☐

Exercise 18.37 (Neumann inversion)**.** Prove $(I - K)\sum_{j=0}^{N} K^j = I - K^{N+1}$.

Hint. Use the finite geometric identity. ☐

Solution. All intermediate powers cancel; norm convergence lets $N \to \infty$ when $\|K\| < 1$. ☐

Exercise 18.38 (Volterra contraction)**.** Prove the Picard integral operator has contraction factor at most Lh on an interval of length h.

Hint. Estimate the integral of the Lipschitz difference. ☐

Solution. $\|Ty - Tz\|_\infty \leq Lh\|y - z\|_\infty$. ☐

Exercise 18.39 (ODE equivalence)**.** Prove a C^1 solution of the IVP is equivalent to its Volterra integral equation.

Hint. Integrate the ODE and differentiate the integral equation. □

Solution. The fundamental theorem of calculus gives both directions. □

Counterexamples and hypothesis testing

Exercise 18.40 (Neumann condition sufficient not necessary)**.** Why does $\|K\| < 1$ give a sufficient but not necessary invertibility condition for $I - K$?

Hint. An operator may have norm above one while 1 is still outside its spectrum. □

Solution. The geometric-series proof fails when the norm is large, but invertibility can still hold by other arguments. □

Exercise 18.41 (Kernel bound not sharp)**.** Why can $(b - a)\|k\|_\infty$ overestimate the true operator norm?

Hint. It ignores cancellation and the integral profile in t. □

Solution. A sharper bound often uses $\sup_x \int |k(x,t)|dt$. □

Exercise 18.42 (No uniqueness without Lipschitz)**.** Why may a nonlinear integral equation have multiple solutions without a contraction condition?

Hint. Banach's uniqueness mechanism is lost. □

Solution. Continuity or self-map properties alone can imply existence without uniqueness. □

Applications and investigations

Exercise 18.43 (Integral-equation solver)**.** How does successive approximation turn an integral equation into an algorithm?

Hint. Iterate the associated operator. □

Solution. Each step evaluates known integrals from the previous approximation; contraction estimates certify convergence and error. □

Exercise 18.44 (Resolvent stability)**.** Why does small $\|K\|$ make $f - Kf = g$ stable?

Hint. Bound $\|(I - K)^{-1}\|$. □

Solution. The Neumann series gives $\|(I - K)^{-1}\| \leq 1/(1 - \|K\|)$, controlling amplification of data errors. □

Challenge problems

Exercise 18.45 (Factorial Volterra bound)**.** Explain why iterated Volterra operators can have $1/n!$-type bounds even when the one-step contraction estimate is not small globally.

Hint. Nested time integrals occupy an ordered simplex. □

Solution. The n-fold integral region has volume $h^n/n!$, producing stronger decay for iterates. □

Exercise 18.46 (Rank-one Fredholm operator)**.** For $Kf = \phi(x)\int \psi(t)f(t)dt$, reduce $f - \lambda Kf = g$ to a scalar equation.

Hint. Let $c = \int \psi f$. □

Solution. Then $f = g + \lambda c\phi$, and $c = \int \psi g + \lambda c \int \psi\phi$; solve this scalar equation when the denominator is nonzero. □

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.

Chapter 19

Elementary Existence Theory for ODEs

Elementary ODE existence theory is most transparent when an initial-value problem is rewritten as an integral fixed-point equation. Continuity gives existence mechanisms; Lipschitz control supplies uniqueness.

Learning goals

After this chapter, the reader should be able to:

- convert a first-order ODE into an equivalent integral equation;
- verify local Lipschitz conditions in the dependent variable for concrete right-hand sides;
- apply Picard iteration to obtain successive approximations;
- derive local existence and uniqueness from contraction estimates on a suitable rectangle or ball;
- identify blow-up, nonuniqueness, or loss-of-Lipschitz examples where the standard theorem does not apply;
- estimate how the solution changes under perturbations of initial data or the right-hand side in elementary settings;

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

19.1 Initial-value problems

An ODE with initial data asks for a curve satisfying a differential rule and a point condition.

19.2 Integral formulation

Integrating converts $y' = f(t, y)$ into a Volterra equation.

19.3 Picard iteration

Successive substitution generates approximate solutions.

Example 19.1 (Picard iterates reveal the exponential series)**.** For

$$y' = y, \qquad y(0) = 1,$$

the integral equation is

$$y(t) = 1 + \int_0^t y(s)\,ds.$$

Starting from $y_0 = 1$,

$$y_1 = 1 + t, \qquad y_2 = 1 + t + \frac{t^2}{2}, \qquad y_3 = 1 + t + \frac{t^2}{2} + \frac{t^3}{6}.$$

The Picard iterates are successive Taylor polynomials for e^t. The fixed point is therefore e^t, and the iteration exhibits existence constructively.

19.4 Local existence

Continuity and compactness can yield existence; Lipschitz hypotheses enable Banach's theorem.

19.5 Uniqueness

Lipschitz dependence on the state variable prevents branching solutions.

19.6 Continuation

Solutions extend while they remain in regions where the hypotheses stay controlled.

19.7 Continuous dependence

Small changes in initial data or vector field produce controlled changes in solutions.

Example 19.2 (Gronwall quantifies dependence on initial data)**.** Suppose two solutions of $y' = f(t, y)$ satisfy a common state Lipschitz bound L and initial values y_0, z_0. Subtracting the integral equations gives

$$|y(t) - z(t)| \le |y_0 - z_0| + L \int_0^t |y(s) - z(s)|\,ds.$$

Gronwall yields

$$|y(t) - z(t)| \le |y_0 - z_0| e^{Lt}.$$

Thus uniqueness is the zero-initial-separation case, while continuous dependence is the quantitative version for nonzero perturbations.

19.8 Autonomous systems

Phase-space geometry reframes $y' = F(y)$ as a flow problem.

Example 19.3 (Phase-line geometry for an autonomous equation)**.** For

$$y' = y(1 - y),$$

equilibria occur at $y = 0, 1$. On $0 < y < 1$, the derivative is positive, so solutions move upward; for $y > 1$, the derivative is negative, so solutions move downward. Hence $y = 1$ is attracting from both nearby sides, while $y = 0$ is unstable to positive perturbations.

The autonomous formulation converts the differential equation into geometry of the vector field on phase space.

19.9 Core structural results

Theorem 19.4 (Picard–Lindelöf)**.** *If f is continuous and locally Lipschitz in y near (t_0, y_0), then the IVP $y' = f(t, y)$, $y(t_0) = y_0$ has a unique local solution.*

Proof. On a sufficiently small rectangle, the integral operator maps a closed sup-norm ball into itself and is a contraction; Banach gives a unique fixed point. □

Theorem 19.5 (Gronwall inequality)**.** *If $u(t) \le A + B\int_0^t u(s)ds$ with $u \ge 0$, then $u(t) \le Ae^{Bt}$.*

Proof. Define the right-hand side majorant and differentiate it; multiplying by e^{-Bt} yields a monotone quantity that bounds u. □

Theorem 19.6 (Continuous dependence)**.** *Under a Lipschitz bound L, two solutions satisfy an exponential stability estimate controlled by initial separation.*

Proof. Subtract the integral equations, estimate by the initial difference plus L times the integral of the difference, then apply Gronwall. □

19.10 Worked examples

Example 19.7 (Linear scalar IVP)**.** Solve $y' = ay$, $y(0) = y_0$. The solution is $y(t) = y_0e^{at}$.

Example 19.8 (Integral reformulation)**.** Rewrite $y' = t+y$, $y(0) = 1$. $y(t) = 1+\int_0^t(s+y(s))ds$.

Example 19.9 (Picard first iterate)**.** Starting from $y_0(t) = 1$, compute one Picard iterate for the previous equation. $y_1(t) = 1 + \int_0^t(s + 1)ds = 1 + t + t^2/2$.

Example 19.10 (Uniqueness failure without Lipschitz)**.** Why can $y' = \sqrt{|y|}$, $y(0) = 0$ have nonunique solutions? The square-root field is not Lipschitz at zero; solutions may remain zero for a while and then depart along a translated quadratic branch.

19.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 19.11 (Linear scalar IVP)**.** Solve $y' = ay$, $y(0) = y_0$.

Solution. The solution is $y(t) = y_0e^{at}$. □

Problem 19.12 (Integral reformulation). Rewrite $y' = t + y$, $y(0) = 1$.

Solution. $y(t) = 1 + \int_0^t (s + y(s))ds$. □

Problem 19.13 (Picard first iterate). Starting from $y_0(t) = 1$, compute one Picard iterate for the previous equation.

Solution. $y_1(t) = 1 + \int_0^t (s + 1)ds = 1 + t + t^2/2$. □

Problem 19.14 (Uniqueness failure without Lipschitz). Why can $y' = \sqrt{|y|}$, $y(0) = 0$ have nonunique solutions?

Solution. The square-root field is not Lipschitz at zero; solutions may remain zero for a while and then depart along a translated quadratic branch. □

Problem 19.15 (Local contraction scale). If $|f(t, y) - f(t, z)| \leq L|y - z|$, what interval condition gives direct contraction?

Solution. Choose interval half-width or length h with $Lh < 1$. □

Problem 19.16 (Gronwall stability). If two solutions have initial gap δ, bound their later gap.

Solution. Under Lipschitz constant L, the gap is at most $\delta e^{L|t-t_0|}$. □

Problem 19.17 (Blow-up and continuation). Why does $y' = y^2$, $y(0) = 1$ not extend for all positive time?

Solution. The solution $y = 1/(1 - t)$ blows up at $t = 1$, leaving every bounded region where local continuation could proceed. □

Problem 19.18 (Autonomous equilibrium). What is an equilibrium of $y' = F(y)$?

Solution. A point y_* with $F(y_*) = 0$, giving the constant solution $y(t) = y_*$. □

Problem 19.19 (Separable example). Solve $y' = -2y$, $y(0) = 3$.

Solution. $y(t) = 3e^{-2t}$. □

Problem 19.20 (Parameter dependence). Why does a Lipschitz theorem imply robustness?

Solution. The same Gronwall estimate controlling uniqueness also bounds changes caused by perturbed initial data or forcing. □

Problem 19.21 (Picard convergence). What makes successive approximations converge?

Solution. The integral operator is a contraction on a sufficiently small complete function-space ball. □

Problem 19.22 (ODE synthesis). Why is integration the natural foundation for existence theory?

Solution. Differential equations ask for unknown derivatives, whereas the integral equation defines a self-map on a space of continuous functions where completeness and norm estimates can be applied. □

19.12 Exercises with complete solutions

Exercise 19.23. Solve $y' = y$, $y(0) = 2$.

Hint. Exponential. Integrate the ODE from the initial time to rewrite it as $y(t) = y_0 + \int f(s, y(s))\, ds$. □

Solution. $2e^t$. □

Exercise 19.24. What theorem gives local uniqueness?

Hint. Picard–Lindelöf. Estimate the derivative of f with respect to y on the chosen rectangle to obtain a local Lipschitz constant. □

Solution. Picard–Lindelöf. □

Exercise 19.25. What hypothesis controls uniqueness?

Hint. Lipschitz in state. Choose the time interval short enough that the Picard operator maps the function ball into itself and becomes a contraction. □

Solution. Local Lipschitz continuity in y. □

Exercise 19.26. What does Gronwall control?

Hint. Growth of differences. Compute the first few Picard iterates directly from the integral equation rather than differentiating them afterward. □

Solution. Exponential stability bounds. □

Exercise 19.27. Can solutions blow up in finite time?

Hint. If uniqueness fails, look for a right-hand side that is continuous but not locally Lipschitz in y. □

Solution. Yes, e.g. $y' = y^2$. □

Exercise 19.28. What is Picard iteration applied to?

Hint. Integral map. To detect finite-time blow-up, solve or compare with a scalar equation whose growth can be integrated explicitly. □

Solution. The Volterra integral formulation. □

Exercise 19.29. What is an equilibrium?

Hint. Zero of vector field. For stability, subtract the two integral equations and estimate the difference before applying a Grönwall-type bound if available. □

Solution. A point $F(y_*) = 0$. □

Exercise 19.30. Does continuity alone always imply uniqueness?

Hint. Keep the existence interval local unless you have an a priori bound preventing escape from the rectangle. □

Solution. No. □

19.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 19.31 (Integral form). Rewrite $y' = t - y,\ y(0) = 2$ as an integral equation.

Hint. Integrate from 0 to t. □

Solution. $y(t) = 2 + \int_0^t (s - y(s))ds$. □

Exercise 19.32 (Picard iterate). Starting from $y_0 = 2$, compute one Picard iterate for the previous IVP.

Hint. Insert the constant guess. □

Solution. $y_1(t) = 2 + t^2/2 - 2t$. □

Exercise 19.33 (Lipschitz check). Is $f(t, y) = t + y^3$ locally Lipschitz in y?

Hint. Bound $f_y = 3y^2$ on a bounded rectangle. □

Solution. Yes, locally; on $|y| \le M$ a Lipschitz constant is $3M^2$. □

Exercise 19.34 (Gronwall bound). If initial separation is 0.01 and $L = 2$, bound separation at $t = 1$.

Hint. Use e^{Lt}. □

Solution. At most $0.01e^2$. □

Exercise 19.35 (Equilibria). Find equilibria of $y' = y(y - 2)$.

Hint. Set the right side to zero. □

Solution. $y = 0$ and $y = 2$. □

Proofs

Exercise 19.36 (Picard–Lindelöf mechanism). Explain how local Lipschitz control yields a contraction on a short time interval.

Hint. Use the Volterra integral operator and require $Lh < 1$. □

Solution. The sup difference of two outputs is bounded by Lh times the input sup difference. □

Exercise 19.37 (Uniqueness via Gronwall). Prove two solutions with the same initial value coincide under a Lipschitz bound.

Hint. Apply Gronwall to their difference. □

Solution. The initial separation is zero, so the exponential bound is identically zero. □

Exercise 19.38 (Integral/differential equivalence). Prove a continuous f makes the integral formulation differentiable and equivalent to the ODE.

Hint. Use the fundamental theorem of calculus. □

Solution. Differentiating the integral gives $f(t, y(t))$; integrating the ODE recovers the integral equation. ☐

Exercise 19.39 (Continuation principle)**.** Explain why a local solution can often be extended while it remains inside a compact region where f and its Lipschitz constants stay controlled.

Hint. Restart the local theorem at a later point. ☐

Solution. Uniform local bounds allow repeated continuation steps; obstruction occurs only when the solution leaves every controlled region or time reaches a boundary. ☐

Counterexamples and hypothesis testing

Exercise 19.40 (Nonuniqueness)**.** Give an IVP with continuous right side but nonunique solutions.

Hint. Use $y' = \sqrt{|y|}, y(0) = 0$. ☐

Solution. The zero solution and delayed quadratic departures show nonuniqueness; the field is not Lipschitz at 0. ☐

Exercise 19.41 (Finite-time blow-up)**.** Show $y' = y^2,\ y(0) = 1$ cannot be continued for all positive time.

Hint. Solve explicitly. ☐

Solution. $y(t) = 1/(1 - t)$, which blows up at $t = 1$. ☐

Exercise 19.42 (Continuity alone insufficient for uniqueness)**.** Why does Peano-type existence not imply uniqueness?

Hint. Continuity lacks a state-difference estimate. ☐

Solution. Without Lipschitz or related control, solution branches can separate from the same initial point. ☐

Applications and investigations

Exercise 19.43 (Error propagation)**.** Why is the Gronwall estimate useful in numerical ODE analysis?

Hint. It converts local perturbation estimates into global-in-time stability bounds. ☐

Solution. It shows how discretization or data errors can grow at most exponentially under a Lipschitz field. ☐

Exercise 19.44 (Population model)**.** Interpret equilibria and stability for the logistic equation $y' = ry(1 - y/K)$.

Hint. Check the sign of the vector field around 0 and K. ☐

Solution. 0 and K are equilibria; for $r > 0$, K is attracting for positive states and 0 is unstable. ☐

Challenge problems

Exercise 19.45 (Comparison estimate)**.** If $|f(t,y) - g(t,y)| \le \eta$ and both are L-Lipschitz in y, derive a bound between solutions with the same initial value.

Hint. Subtract integral equations and apply Gronwall with forcing ηt. ☐

Solution. A standard bound is $\frac{\eta}{L}(e^{Lt} - 1)$ for $L > 0$, and ηt for $L = 0$. ☐

Exercise 19.46 (Invariant interval)**.** For $y' = y(1-y)$, explain why a solution starting in $[0,1]$ stays there.

Hint. Use uniqueness at the equilibrium boundary solutions. ☐

Solution. A solution cannot cross 0 or 1 without first meeting an equilibrium solution; uniqueness would then force it to coincide, preventing crossing. ☐

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.

Part V

Approximation

Chapter 20

Polynomial Interpolation

Polynomial interpolation reconstructs a polynomial from finite data. Lagrange and Newton forms expose uniqueness, computation, divided differences, and the remainder term.

Learning goals

After this chapter, the reader should be able to:

- construct Lagrange and Newton interpolating polynomials from given data;
- prove uniqueness of the degree-at-most-n interpolant from the root-counting argument;
- compute divided differences and update a Newton interpolant when a new node is added;
- derive or apply the interpolation remainder formula to bound the error;
- compare equally spaced and Chebyshev-type node choices through the node polynomial;
- translate interpolation into a Vandermonde linear system and explain why distinct nodes are essential;

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

20.1 Interpolation problem

Given distinct nodes and values, seek a polynomial matching all data.

20.2 Lagrange basis

Cardinal polynomials isolate one node at a time.

20.3 Uniqueness

A polynomial of degree at most n matching $n + 1$ distinct data points is unique.

20.4 Newton form

Divided differences produce a nested incremental representation.

Example 20.1 (Newton form updates without rebuilding the interpolant)**.** For data

$$(0,1), \quad (1,2), \quad (2,5),$$

the divided differences are

$$f[0]=1, \qquad f[0,1]=1, \qquad f[1,2]=3, \qquad f[0,1,2]=1.$$

Hence

$$p_2(x)=1+x+x(x-1)=1+x^2.$$

If a new node is added, Newton form appends one new divided-difference term rather than rebuilding the whole polynomial. Evaluating at the three old nodes checks the interpolation conditions.

20.5 Divided differences

Recursive coefficients encode discrete derivatives.

20.6 Error formula

The interpolation remainder involves the next derivative and the node polynomial.

Example 20.2 (Interpolation remainder becomes an error bound)**.** Interpolate $f(x)=e^x$ at 0 and 1 by a line p_1. For $x\in[0,1]$,

$$f(x)-p_1(x)=\frac{e^{\xi}}{2}x(x-1)$$

for some $\xi\in(0,1)$. Since $e^{\xi}\le e$ and

$$|x(x-1)|\le\frac{1}{4},$$

we get

$$|f(x)-p_1(x)|\le\frac{e}{8}.$$

The remainder formula separates smoothness control from node geometry.

20.7 Node choice

Equally spaced nodes can be unstable at high degree; special nodes reduce worst-case error.

Example 20.3 (Node choice controls the node polynomial)**.** The interpolation error contains

$$\omega_{n+1}(x)=\prod_{j=0}^{n}(x-x_j).$$

For a fixed derivative bound, reducing $\|\omega_{n+1}\|_\infty$ reduces worst-case interpolation error. Chebyshev nodes are designed precisely to control this factor on an interval. This explains why node placement is not a cosmetic choice: equally spaced nodes can produce a much larger oscillatory node polynomial near endpoints.

20.8 Hermite preview

Derivative data can be incorporated by repeated-node ideas.

20.9 Core structural results

Theorem 20.4 (Interpolation existence and uniqueness). *For $n+1$ distinct nodes, there is a unique polynomial of degree at most n taking prescribed values.*

Proof. Lagrange basis polynomials give existence. The difference of two interpolants has $n+1$ roots but degree at most n, so it is zero. ☐

Theorem 20.5 (Lagrange formula). *The interpolant is $p(x) = \sum_{j=0}^{n} y_j L_j(x)$ with $L_j(x_i) = \delta_{ij}$.*

Proof. By construction each basis polynomial is one at its own node and zero at the others, so the sum matches all data. ☐

Theorem 20.6 (Interpolation remainder). *If $f \in C^{n+1}$, then $f(x) - p_n(x) = f^{(n+1)}(\xi)(\prod_{j=0}^{n}(x - x_j))/(n+1)!$ for a suitable ξ.*

Proof. Subtract a multiple of the node polynomial chosen to vanish also at x, then apply Rolle's theorem repeatedly to the resulting function. ☐

20.10 Worked examples

Example 20.7 (Two-point interpolation). Interpolate $(0,1)$ and $(1,3)$. The unique linear interpolant is $p(x) = 1 + 2x$.

Example 20.8 (Three-point Lagrange). Interpolate $f(0) = 0, f(1) = 1, f(2) = 4$. The data lie on x^2, so uniqueness gives $p(x) = x^2$.

Example 20.9 (Uniqueness by roots). Why cannot two degree-n interpolants agree at $n+1$ distinct nodes and differ elsewhere? Their difference would be a nonzero degree-at-most-n polynomial with too many roots.

Example 20.10 (Lagrange basis property). Verify $L_j(x_i) = \delta_{ij}$. At $i = j$ all factors equal one; for $i \neq j$ one numerator factor vanishes.

20.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 20.11 (Two-point interpolation). Interpolate $(0,1)$ and $(1,3)$.

Solution. The unique linear interpolant is $p(x) = 1 + 2x$. ☐

Problem 20.12 (Three-point Lagrange). Interpolate $f(0) = 0, f(1) = 1, f(2) = 4$.

Solution. The data lie on x^2, so uniqueness gives $p(x) = x^2$. ☐

Problem 20.13 (Uniqueness by roots). Why cannot two degree-n interpolants agree at $n+1$ distinct nodes and differ elsewhere?

Solution. Their difference would be a nonzero degree-at-most-n polynomial with too many roots. □

Problem 20.14 (Lagrange basis property)**.** Verify $L_j(x_i) = \delta_{ij}$.

Solution. At $i = j$ all factors equal one; for $i \neq j$ one numerator factor vanishes. □

Problem 20.15 (Newton divided difference)**.** Compute the first divided difference for data $(x_0, y_0), (x_1, y_1)$.

Solution. It is $(y_1 - y_0)/(x_1 - x_0)$. □

Problem 20.16 (Incremental interpolation)**.** Why is Newton form convenient when adding one new node?

Solution. The previous polynomial remains intact and one new product term is appended. □

Problem 20.17 (Error node product)**.** Why do roots of the interpolation error include all nodes?

Solution. The interpolant matches the function exactly at every node. □

Problem 20.18 (Linear interpolation error)**.** State the error form for two nodes.

Solution. $f(x) - p_1(x) = f''(\xi)(x - x_0)(x - x_1)/2$. □

Problem 20.19 (Runge warning)**.** What can go wrong with high-degree equally spaced interpolation?

Solution. Endpoint oscillations can grow despite exact matching at nodes. □

Problem 20.20 (Chebyshev motivation)**.** Why choose nodes near Chebyshev points?

Solution. They reduce the maximum magnitude of the node polynomial and thereby control worst-case interpolation error. □

Problem 20.21 (Vandermonde formulation)**.** How can interpolation be written as a linear system?

Solution. Writing $p(x) = \sum a_k x^k$ gives a Vandermonde matrix whose invertibility follows from distinct nodes. □

Problem 20.22 (Interpolation synthesis)**.** What is being approximated and what is exact?

Solution. The polynomial matches finitely many data exactly, while the remainder formula quantifies error between those nodes in terms of higher derivatives and node geometry. □

20.12 Exercises with complete solutions

Exercise 20.23. Interpolate two points $(0, 0), (1, 1)$.

Hint. Line through them. For two nodes, write the unique affine function through them before invoking any general interpolation machinery. □

Solution. $p = x$. □

Exercise 20.24. How many data points determine degree at most n?

Hint. Distinct nodes. A degree-at-most-n polynomial is determined by $n+1$ distinct values because the difference of two candidates would have too many roots. □

Solution. $n+1$. □

Exercise 20.25. Why distinct nodes?

Hint. Vandermonde determinant. Distinct nodes make the Vandermonde determinant nonzero; repeated nodes require extra derivative data instead. □

Solution. Otherwise uniqueness fails or data may conflict. □

Exercise 20.26. What are Lagrange basis values at nodes?

Hint. Kronecker delta. For a Lagrange basis polynomial, inspect its factors at each node to obtain the Kronecker-delta values. □

Solution. 0 or 1. □

Exercise 20.27. What degree is the node polynomial?

Hint. One factor per node. The node polynomial has one linear factor for each interpolation node; count those factors to get its degree. □

Solution. $n+1$. □

Exercise 20.28. What theorem proves uniqueness?

Hint. Root count. Uniqueness is a root-counting argument: a nonzero polynomial cannot have more roots than its degree. □

Solution. A nonzero degree-n polynomial has at most n roots. □

Exercise 20.29. Does interpolation guarantee small error?

Hint. Interpolation is exact at the nodes but may oscillate badly between them; inspect the remainder and the node placement. □

Solution. Node placement and smoothness matter. □

Exercise 20.30. What form supports incremental data?

Hint. Newton form. Newton form is incremental because adding one node appends one new product term without changing the previous coefficients. □

Solution. Newton divided-difference form. □

20.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 20.31 (Linear interpolation)**.** Interpolate $(0,2)$ and $(2,6)$.

Hint. Find the line through the points. ☐

Solution. $p(x) = 2 + 2x$. ☐

Exercise 20.32 (Lagrange basis)**.** For nodes $0, 1, 2$, write $L_1(x)$.

Hint. Use products over the other nodes. ☐

Solution. $L_1(x) = \frac{x(x-2)}{(1-0)(1-2)} = -x(x-2)$. ☐

Exercise 20.33 (Divided difference)**.** Compute $f[0,1]$ for values $f(0) = 1, f(1) = 4$.

Hint. Use the first divided-difference formula. ☐

Solution. $f[0,1] = 3$. ☐

Exercise 20.34 (Vandermonde determinant)**.** What condition on nodes makes the Vandermonde matrix invertible?

Hint. Recall its determinant product. ☐

Solution. The nodes must be pairwise distinct. ☐

Exercise 20.35 (Remainder factor)**.** For quadratic interpolation at nodes $0, 1, 2$, what node product appears in the remainder?

Hint. Multiply $x - x_j$. ☐

Solution. $x(x-1)(x-2)$. ☐

Proofs

Exercise 20.36 (Uniqueness)**.** Prove a degree-at-most-n interpolant at $n+1$ distinct nodes is unique.

Hint. Subtract two interpolants. ☐

Solution. The difference has $n+1$ distinct roots but degree at most n, so it is zero. ☐

Exercise 20.37 (Lagrange existence)**.** Prove the Lagrange formula matches prescribed node values.

Hint. Use $L_j(x_i) = \delta_{ij}$. ☐

Solution. At node x_i, every term vanishes except $y_i L_i(x_i) = y_i$. ☐

Exercise 20.38 (Newton update)**.** Explain why adding one node adds only one term to Newton form.

Hint. The existing interpolant already matches old nodes; multiply the correction by the old node polynomial. ☐

Solution. The correction $c\prod_{j=0}^{n}(x - x_j)$ vanishes at all old nodes, and c is chosen to fit the new one. ☐

Exercise 20.39 (Remainder theorem mechanism)**.** Outline the repeated Rolle proof of the interpolation remainder.

Hint. Subtract a multiple of the node polynomial so the error also vanishes at the evaluation point. ☐

Solution. The constructed function has $n+2$ zeros; repeated Rolle gives a zero of its $(n+1)$-st derivative and yields the formula. ☐

Counterexamples and hypothesis testing

Exercise 20.40 (Repeated nodes)**.** Why do repeated nodes break ordinary Lagrange interpolation formulas?

Hint. Denominators contain node differences. ☐

Solution. Distinctness fails and denominators vanish; derivative data require Hermite-type interpolation instead. ☐

Exercise 20.41 (High-degree instability)**.** Why can equally spaced high-degree interpolation behave poorly near interval endpoints?

Hint. Node-product and Lebesgue factors can grow strongly. ☐

Solution. Runge-type oscillations can amplify data/approximation error despite exact interpolation at nodes. ☐

Exercise 20.42 (Interpolation not approximation optimum)**.** Why need an interpolating polynomial not be the best uniform approximant of the same degree?

Hint. Interpolation enforces exact values at chosen nodes, while minimax optimizes worst-case error globally. ☐

Solution. The two optimization criteria are different. ☐

Applications and investigations

Exercise 20.43 (Table update)**.** Why is Newton form attractive for streaming data?

Hint. New data append divided differences without changing old coefficients. ☐

Solution. It supports incremental interpolation with less recomputation. ☐

Exercise 20.44 (Error budgeting)**.** How can derivative and node-product bounds guide node count?

Hint. Use the remainder inequality. ☐

Solution. Choose degree/nodes until $M\|\omega\|_\infty/(n+1)!$ falls below the desired tolerance. ☐

Challenge problems

Exercise 20.45 (Chebyshev node motivation)**.** Explain why roots of a Chebyshev polynomial are natural interpolation nodes.

Hint. They minimize the size of an associated monic node polynomial up to scaling. ☐

Solution. This reduces the worst-case factor in the interpolation remainder. ☐

Exercise 20.46 (Hermite dimension count)**.** Why do values and first derivatives at m distinct nodes determine a polynomial of degree at most $2m - 1$ uniquely?

Hint. The difference of two candidates has a double zero at each node. ☐

Solution. It would have at least $2m$ zeros counting multiplicity but degree at most $2m - 1$, so it must vanish. ☐

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.

Chapter 21

Polynomial Approximation

Polynomial approximation asks not for exact matching at finitely many points but for small error on an entire compact interval. The Weierstrass theorem says continuous functions admit arbitrarily accurate uniform polynomial approximations.

Learning goals

After this chapter, the reader should be able to:

- construct explicit polynomial approximants to continuous functions in the settings developed in the chapter;
- apply a polynomial approximation theorem only after checking compactness and continuity hypotheses;
- estimate approximation error using modulus-of-continuity or constructive bounds when available;
- distinguish interpolation from uniform approximation;
- prove density statements by reducing them to the appropriate approximation theorem;
- construct examples showing why smoothness can improve approximation rates without being necessary for mere density;

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

21.1 Uniform approximation

Approximation quality is measured by the sup norm.

21.2 Weierstrass theorem

Polynomials are dense in $C[a, b]$.

21.3 Bernstein polynomials

On $[0, 1]$, positive averaging polynomials give a constructive proof.

Example 21.1 (Bernstein polynomials as probabilistic averaging)**.** For $f \in C[0,1]$,

$$B_n f(x) = \sum_{k=0}^{n} f(k/n) \binom{n}{k} x^k (1-x)^{n-k}.$$

The coefficients form a binomial probability distribution, so

$$B_n f(x) = \mathbb{E}\, f(X_n/n), \qquad X_n \sim \text{Binomial}(n, x).$$

Constants and the identity are reproduced because probabilities sum to one and $\mathbb{E}(X_n/n) = x$. This averaging interpretation explains positivity and why concentration of X_n/n near x drives convergence.

21.4 Modulus of continuity

Uniform continuity controls approximation error.

Example 21.2 (Modulus of continuity separates near and far samples)**.** Let

$$\omega_f(\delta) = \sup_{|x-y|\le\delta} |f(x) - f(y)|.$$

In Bernstein's proof, sample points k/n near x contribute at most $\omega_f(\delta)$, while far sample points are controlled by a probability tail. Uniform continuity makes the near contribution small, and binomial concentration makes the far contribution small uniformly in x.

This decomposition turns qualitative continuity into a quantitative approximation argument.

21.5 Shape preservation

Positive approximation operators can preserve positivity and convexity features.

21.6 Density viewpoint

Approximation is a statement about closure of a subspace in a normed function space.

21.7 Approximation rates

Extra smoothness improves quantitative convergence rates.

Example 21.3 (Extra smoothness improves an approximation rate)**.** For a Lipschitz function with constant L,

$$|f(u) - f(v)| \le L|u - v|.$$

Using the Bernstein probabilistic form and Cauchy–Schwarz,

$$|B_n f(x) - f(x)| \le L\,\mathbb{E}|X_n/n - x| \le L\sqrt{\frac{x(1-x)}{n}} \le \frac{L}{2\sqrt{n}}.$$

Thus smoothness stronger than continuity yields an explicit convergence rate.

21.8 Beyond polynomials

The same density philosophy leads to trigonometric and rational approximation.

21.9 Core structural results

Theorem 21.4 (Weierstrass approximation theorem). *Every continuous real function on $[a, b]$ can be uniformly approximated by polynomials.*

Proof. After rescaling to $[0, 1]$, Bernstein polynomials converge uniformly; uniform continuity controls nearby samples and binomial concentration controls distant samples. ☐

Theorem 21.5 (Bernstein reproduction). *Bernstein operators reproduce constants and the identity function.*

Proof. Binomial identities give $B_n 1 = 1$ and $B_n x = x$. ☐

Theorem 21.6 (Density consequence). *For every $f \in C[a, b]$ and $\varepsilon > 0$, some polynomial p satisfies $\|f - p\|_\infty < \varepsilon$.*

Proof. This is exactly the statement that the polynomial subspace is dense in the sup-norm space. ☐

21.10 Worked examples

Example 21.7 (Approximate absolute value). Why can $|x|$ on $[-1, 1]$ be uniformly approximated by polynomials despite not being differentiable at zero? Weierstrass requires continuity only, not differentiability.

Example 21.8 (Bernstein constant). Show $B_n 1 = 1$. The Bernstein basis sums to $(x + (1 - x))^n = 1$.

Example 21.9 (Bernstein identity). Show $B_n x = x$. The weighted sum of k/n against binomial probabilities equals the mean of a Binomial(n, x) variable divided by n, namely x.

Example 21.10 (Uniform versus pointwise approximation). Why is sup norm stronger? It bounds the worst error at every point simultaneously rather than allowing the bad point to vary with the approximant.

21.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 21.11 (Approximate absolute value). Why can $|x|$ on $[-1, 1]$ be uniformly approximated by polynomials despite not being differentiable at zero?

Solution. Weierstrass requires continuity only, not differentiability. ☐

Problem 21.12 (Bernstein constant). Show $B_n 1 = 1$.

Solution. The Bernstein basis sums to $(x + (1 - x))^n = 1$. ☐

Problem 21.13 (Bernstein identity). Show $B_n x = x$.

Solution. The weighted sum of k/n against binomial probabilities equals the mean of a Binomial(n, x) variable divided by n, namely x. ☐

Problem 21.14 (Uniform versus pointwise approximation). Why is sup norm stronger?

Solution. It bounds the worst error at every point simultaneously rather than allowing the bad point to vary with the approximant. □

Problem 21.15 (Density interpretation)**.** What does polynomial density mean geometrically?

Solution. Every open sup-norm ball around every continuous function contains a polynomial. □

Problem 21.16 (No exact representation required)**.** Does Weierstrass say every continuous function is a polynomial?

Solution. No; it says polynomials can approach arbitrarily closely in sup norm. □

Problem 21.17 (Approximation of e^x)**.** Give one natural polynomial approximant.

Solution. Taylor polynomials provide explicit approximants; on a compact interval their remainders can be bounded uniformly. □

Problem 21.18 (Approximation and integration)**.** If $p_n \to f$ uniformly, what happens to integrals?

Solution. The integrals converge because integration is continuous in the sup norm. □

Problem 21.19 (Approximation and roots)**.** Does a uniformly close polynomial necessarily preserve every zero?

Solution. No; zeros can move or disappear unless extra sign or transversality information is available. □

Problem 21.20 (Smoothness and rate)**.** Why should smoother functions be easier to approximate rapidly?

Solution. Higher regularity controls oscillation and derivative growth, enabling sharper remainder and constructive approximation estimates. □

Problem 21.21 (Positive operators)**.** Why are Bernstein polynomials stable?

Solution. They are convex combinations of sampled values, so they do not amplify the sup norm and preserve positivity. □

Problem 21.22 (Approximation synthesis)**.** How does this differ from interpolation?

Solution. Interpolation enforces exact agreement at selected points; approximation minimizes or controls global error across the whole interval. □

21.12 Exercises with complete solutions

Exercise 21.23. What norm does Weierstrass use?

Hint. Sup norm. Separate existence of a polynomial approximant from construction of a specific interpolant; these are different problems. □

Solution. Uniform norm. □

Exercise 21.24. Must f be differentiable?

Hint. Check continuity on a compact interval before invoking the uniform polynomial approximation theorem. □

Solution. Continuity suffices. ☐

Exercise 21.25. What interval is standard for Bernstein polynomials?

Hint. $[0, 1]$. Use the modulus of continuity to convert closeness of nearby arguments into a uniform function-value estimate. ☐

Solution. $[0, 1]$. ☐

Exercise 21.26. Do Bernstein polynomials reproduce constants?

Hint. If a constructive approximant is given, estimate its error directly in the sup norm. ☐

Solution. Yes. ☐

Exercise 21.27. Do they reproduce x?

Hint. Density means every $\varepsilon > 0$ admits some polynomial approximant; do not confuse this with convergence of one fixed sequence unless specified. ☐

Solution. Yes. ☐

Exercise 21.28. Does density mean equality?

Hint. Compare the approximating polynomial with the target on the whole interval, not only at selected nodes. ☐

Solution. Arbitrarily close approximation. ☐

Exercise 21.29. Does uniform approximation imply integral approximation?

Hint. Additional smoothness can improve rates, but uniform approximability of continuous functions does not require differentiability. ☐

Solution. Yes on compact intervals. ☐

Exercise 21.30. Can $|x|$ be polynomially approximated uniformly?

Hint. To disprove an overstrong claim, choose a continuous function with limited smoothness and test the asserted rate rather than the existence theorem. ☐

Solution. Yes. ☐

21.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 21.31 (Bernstein constant)**.** Compute $B_n 1$.

Hint. Use the binomial theorem. □

Solution. $B_n 1 = 1$. □

Exercise 21.32 (Bernstein identity)**.** Compute $B_n x$.

Hint. Use the mean of a binomial variable. □

Solution. $B_n x = x$. □

Exercise 21.33 (Sup error meaning)**.** What does $\|f - p\|_\infty < 0.01$ mean?

Hint. Unpack the supremum norm. □

Solution. $|f(x) - p(x)| < 0.01$ for every x in the interval. □

Exercise 21.34 (Weierstrass hypothesis)**.** Can $|x|$ on $[-1, 1]$ be uniformly approximated by polynomials?

Hint. Check continuity, not differentiability. □

Solution. Yes, by Weierstrass. □

Exercise 21.35 (Density statement)**.** What does it mean that polynomials are dense in $C[a, b]$?

Hint. Use epsilon and sup norm. □

Solution. For every continuous f and every $\varepsilon > 0$, some polynomial p satisfies $\|f - p\|_\infty < \varepsilon$. □

Proofs

Exercise 21.36 (Bernstein positivity)**.** Prove $f \geq 0$ implies $B_n f \geq 0$.

Hint. Each Bernstein basis weight is nonnegative. □

Solution. The polynomial is a nonnegative weighted sum of nonnegative sample values. □

Exercise 21.37 (Constant reproduction)**.** Prove Bernstein polynomials reproduce constants.

Hint. Sum the basis weights. □

Solution. $\sum_k \binom{n}{k} x^k (1 - x)^{n-k} = 1$. □

Exercise 21.38 (Identity reproduction)**.** Prove $B_n x = x$.

Hint. Use $k\binom{n}{k} = n\binom{n-1}{k-1}$. □

Solution. The weighted sum becomes $x(x + 1 - x)^{n-1} = x$. □

Exercise 21.39 (Density consequence)**.** Explain why uniform Bernstein convergence proves Weierstrass on $[0, 1]$.

Hint. Each $B_n f$ is a polynomial. □

Solution. For every epsilon, some Bernstein polynomial lies within epsilon in sup norm. □

Counterexamples and hypothesis testing

Exercise 21.40 (Interpolation vs approximation)**.** Why is a best or good uniform approximant not required to match f at any selected nodes?

Hint. Uniform approximation optimizes global error rather than exact data constraints. ☐

Solution. Matching nodes is an interpolation requirement, not an approximation requirement. ☐

Exercise 21.41 (Continuity necessary for uniform polynomial limit)**.** Why can a discontinuous function not be uniformly approximated arbitrarily well by polynomials?

Hint. A uniform limit of continuous functions is continuous. ☐

Solution. If errors tended to zero uniformly, the polynomial sequence would converge uniformly to the discontinuous target, impossible. ☐

Exercise 21.42 (Smoothness not necessary)**.** Why is differentiability not required for Weierstrass approximation?

Hint. The theorem assumes continuity only. ☐

Solution. Nondifferentiable continuous functions such as $|x|$ still admit arbitrary uniform polynomial approximation. ☐

Applications and investigations

Exercise 21.43 (Function compression)**.** Why are polynomial approximants useful computationally?

Hint. They replace a function by finitely many coefficients and support fast evaluation/differentiation/integration. ☐

Solution. A certified sup error makes that finite representation globally reliable. ☐

Exercise 21.44 (Shape preservation)**.** What shape property does positivity of Bernstein weights help preserve?

Hint. Positive averaging cannot create negative values from nonnegative data. ☐

Solution. At minimum it preserves nonnegativity; related averaging arguments can preserve monotonicity/convexity under suitable conditions. ☐

Challenge problems

Exercise 21.45 (Lipschitz Bernstein rate)**.** Derive $\|B_n f - f\|_\infty \le L/(2\sqrt{n})$ for L-Lipschitz f.

Hint. Use the binomial variance. ☐

Solution. $\mathbb{E}|X_n/n - x| \le \sqrt{x(1-x)/n} \le 1/(2\sqrt{n})$; multiply by L. ☐

Exercise 21.46 (Rescale interval)**.** Explain how Weierstrass on $[0,1]$ implies it on any $[a,b]$.

Hint. Use the affine change $x = a + (b-a)t$. ☐

Solution. Compose f with the affine map, approximate in t, then substitute $t = (x-a)/(b-a)$; a polynomial remains a polynomial. ☐

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.

Chapter 22

Chebyshev and Minimax Approximation

Chebyshev and minimax approximation focus on worst-case uniform error. Equioscillation and Chebyshev polynomials reveal why optimal approximants spread their error rather than concentrating it.

Learning goals

After this chapter, the reader should be able to:

- formulate the minimax approximation problem in the sup norm;
- compute simple best approximants by balancing extreme errors;
- use Chebyshev polynomials to control the maximum size of monic polynomials on an interval;
- compare interpolation error with best-uniform-approximation error;
- identify equioscillation patterns in explicit low-degree examples;
- derive error lower bounds that certify optimality of a candidate minimax approximant;

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

22.1 Minimax problem

Choose an approximant minimizing the maximum absolute error.

22.2 Chebyshev polynomials

$T_n(\cos\theta) = \cos(n\theta)$ oscillates between ± 1 with controlled leading coefficient.

Example 22.1 (Chebyshev recurrence and bounded oscillation)**.** From

$$T_n(\cos\theta) = \cos(n\theta)$$

and the cosine addition identity,

$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x).$$

Starting with $T_0 = 1, T_1 = x$, this gives

$$T_2 = 2x^2 - 1, \qquad T_3 = 4x^3 - 3x.$$

On $[-1, 1]$, $|T_n(x)| \leq 1$, and the extrema alternate between $+1$ and -1 at $x_k = \cos(k\pi/n)$. This controlled oscillation is exactly the pattern used in minimax optimality.

22.3 Extremal property

Among monic degree-n polynomials, a scaled Chebyshev polynomial minimizes sup norm on $[-1, 1]$.

22.4 Equioscillation

Optimal errors alternate in sign at enough extremal points.

Example 22.2 (Balancing error gives the best constant approximant)**.** For a continuous function f with minimum m and maximum M, approximate it by a constant c. The worst error is

$$E(c) = \max(M - c, c - m).$$

This is minimized when the two active errors are equal:

$$M - c = c - m,$$

so

$$c_* = \frac{M + m}{2}, \qquad E_* = \frac{M - m}{2}.$$

The optimal error alternates with equal magnitude at points where $f = M$ and $f = m$, the degree-zero case of equioscillation.

22.5 Uniqueness

Alternation forces uniqueness of the best approximant in classical polynomial spaces.

22.6 Node placement

Chebyshev extrema and zeros guide stable interpolation and quadrature.

Example 22.3 (Chebyshev nodes reduce the dangerous interpolation factor)**.** For degree-n interpolation, the remainder contains the node polynomial

$$\omega_{n+1}(x) = \prod_{j=0}^{n} (x - x_j).$$

Choosing nodes related to zeros of a Chebyshev polynomial makes the associated monic polynomial have near-minimal sup norm on $[-1, 1]$. The nodes cluster near the endpoints, precisely where equally spaced high-degree interpolation often develops large oscillations.

Thus Chebyshev placement is derived from a minimax property, not an arbitrary nonuniform spacing rule.

22.7 Error scaling

Affine changes transfer results from $[-1, 1]$ to any compact interval.

22.8 Numerical viewpoint

Minimax approximation optimizes worst-case performance rather than average error.

22.9 Core structural results

Theorem 22.4 (Chebyshev extremal property)**.** *The monic polynomial* $2^{1-n}T_n$ *has minimal sup norm among all monic degree-n polynomials on* $[-1, 1]$*.*

Proof. If another monic polynomial had smaller norm, their difference would have degree at most $n - 1$ yet would change sign between the $n + 1$ alternating extrema of the Chebyshev polynomial, forcing at least n roots, contradiction. □

Theorem 22.5 (Alternation criterion)**.** *A best degree-at-most-n uniform approximant to a continuous function is characterized by at least* $n + 2$ *alternating extremal error points.*

Proof. If alternation fails, one can perturb within the approximation space to reduce the maximum error; if alternation holds, any competitor difference has too many sign changes for its degree. □

Theorem 22.6 (Uniqueness of minimax polynomial)**.** *The best uniform polynomial approximant of prescribed degree is unique.*

Proof. Two distinct best approximants averaged together would force a smaller maximum error unless their error extrema align; alternation then forces their difference to have too many zeros. □

22.10 Worked examples

Example 22.7 (Compute low Chebyshev polynomials)**.** Find T_0, T_1, T_2, T_3. From $T_n(\cos\theta) = \cos n\theta$: $1, x, 2x^2 - 1, 4x^3 - 3x$.

Example 22.8 (Recurrence)**.** Derive $T_{n+1} = 2xT_n - T_{n-1}$. Use $\cos((n+1)\theta) = 2\cos\theta\cos n\theta - \cos((n-1)\theta)$.

Example 22.9 (Extrema)**.** Where does T_n attain ± 1? At $x_k = \cos(k\pi/n)$, $k = 0, \dots, n$.

Example 22.10 (Zeros)**.** Where are the zeros of T_n? At $\cos((2k - 1)\pi/(2n))$, $k = 1, \dots, n$.

22.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 22.11 (Compute low Chebyshev polynomials)**.** Find T_0, T_1, T_2, T_3.

Solution. From $T_n(\cos\theta) = \cos n\theta$: $1, x, 2x^2 - 1, 4x^3 - 3x$. □

Problem 22.12 (Recurrence)**.** Derive $T_{n+1} = 2xT_n - T_{n-1}$.

Solution. Use $\cos((n+1)\theta) = 2\cos\theta\cos n\theta - \cos((n-1)\theta)$. ☐

Problem 22.13 (Extrema)**.** Where does T_n attain ± 1?

Solution. At $x_k = \cos(k\pi/n)$, $k = 0, \ldots, n$. ☐

Problem 22.14 (Zeros)**.** Where are the zeros of T_n?

Solution. At $\cos((2k-1)\pi/(2n))$, $k = 1, \ldots, n$. ☐

Problem 22.15 (Monic scaling)**.** Why multiply by 2^{1-n}?

Solution. For $n \geq 1$, the leading coefficient of T_n is 2^{n-1}, so scaling makes it monic. ☐

Problem 22.16 (Worst-case interpretation)**.** What does minimax optimize?

Solution. The largest absolute error over the entire interval. ☐

Problem 22.17 (Alternating errors)**.** Why should an optimum touch both positive and negative error extremes?

Solution. If all extreme errors had the same sign, a small vertical shift would reduce the maximum magnitude. ☐

Problem 22.18 (Linear approximation intuition)**.** For degree one, how many alternating extremal points characterize the minimax fit?

Solution. At least three. ☐

Problem 22.19 (Affine rescaling)**.** How do Chebyshev points transfer to $[a, b]$?

Solution. Map $x \in [-1, 1]$ by $(a+b)/2 + (b-a)x/2$. ☐

Problem 22.20 (Interpolation connection)**.** Why do Chebyshev nodes reduce endpoint problems?

Solution. They cluster near endpoints, reducing the maximum node-polynomial magnitude. ☐

Problem 22.21 (Uniqueness mechanism)**.** How does root counting enter minimax uniqueness?

Solution. Alternating error inequalities force the difference of two candidates to change sign too many times for its degree. ☐

Problem 22.22 (Minimax synthesis)**.** How is minimax different from least squares?

Solution. Minimax controls the maximum error; least squares minimizes an integrated squared error and may tolerate larger localized peaks. ☐

22.12 Exercises with complete solutions

Exercise 22.23. What is T_2?

Hint. Use recurrence. Write the error function $e = f - p$ and locate its largest positive and negative values before adjusting the approximant. □

Solution. $2x^2 - 1$. □

Exercise 22.24. What is T_3?

Hint. Use recurrence. For a constant best approximant, balance the maximum positive and negative errors. □

Solution. $4x^3 - 3x$. □

Exercise 22.25. What is $\|T_n\|_\infty$ on $[-1, 1]$?

Hint. Cosine definition. Chebyshev polynomials are extremal because they oscillate with equal magnitude; scale them to the required leading coefficient. □

Solution. 1. □

Exercise 22.26. How many extrema are used in alternation for degree n?

Hint. One more than dimension. To certify optimality, derive a lower bound on every competitor's maximum error and show the candidate attains it. □

Solution. At least $n + 2$. □

Exercise 22.27. Are Chebyshev zeros equally spaced in x?

Hint. They are equally spaced in angle. Interpolation need not be minimax; compare its sup error with an equioscillating competitor. □

Solution. No. □

Exercise 22.28. Why scale T_n to monic?

Hint. Compare monic polynomials. Normalize the interval to $[-1, 1]$ before applying standard Chebyshev formulas. □

Solution. To formulate the extremal property. □

Exercise 22.29. What norm is minimized?

Hint. Sup norm. Count alternating extreme points in the error curve; insufficient alternation suggests the approximation can still be improved. □

Solution. Uniform norm. □

Exercise 22.30. Does minimax equal least squares?

Hint. Generally no. Use symmetry of the target function to reduce the possible form of a best approximating polynomial. □

Solution. No. □

22.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 22.31 (Compute T_4)**.** Use the recurrence to compute T_4.

Hint. Use $T_4 = 2xT_3 - T_2$. □

Solution. $T_4 = 8x^4 - 8x^2 + 1$. □

Exercise 22.32 (Extrema)**.** Where does T_n attain ± 1?

Hint. Set $x = \cos\theta$. □

Solution. At $x_k = \cos(k\pi/n)$, $k = 0, \dots, n$. □

Exercise 22.33 (Zeros)**.** Where are the zeros of T_n?

Hint. Solve $\cos(n\theta) = 0$. □

Solution. $\cos((2k-1)\pi/(2n))$, $k = 1, \dots, n$. □

Exercise 22.34 (Best constant)**.** Find the best constant approximation to $f(x) = x^2$ on $[-1, 1]$.

Hint. The range is $[0, 1]$. □

Solution. $c = 1/2$ with sup error $1/2$. □

Exercise 22.35 (Scaled monic Chebyshev)**.** What scaling makes T_n monic for $n \geq 1$?

Hint. Its leading coefficient is 2^{n-1}. □

Solution. $2^{1-n}T_n$. □

Proofs

Exercise 22.36 (Recurrence proof)**.** Derive the Chebyshev recurrence.

Hint. Use $\cos((n+1)\theta) + \cos((n-1)\theta) = 2\cos\theta\cos(n\theta)$. □

Solution. Substitute $x = \cos\theta$ to obtain $T_{n+1} = 2xT_n - T_{n-1}$. □

Exercise 22.37 (Extremal property idea)**.** Explain the root-counting contradiction behind the monic Chebyshev extremal theorem.

Hint. Subtract a supposedly smaller-norm monic polynomial from the scaled Chebyshev polynomial. □

Solution. The leading terms cancel, leaving degree at most $n - 1$, but alternating extrema force at least n sign-change roots. □

Exercise 22.38 (Uniqueness minimax)**.** Why is the best polynomial approximant of fixed degree unique?

Hint. Use alternation or average two candidates. □

Solution. Alternation forces the difference of two best candidates to have too many zeros unless it is identically zero. ☐

Exercise 22.39 (Affine scaling)**.** Explain how Chebyshev minimax results on $[-1, 1]$ transfer to $[a, b]$.

Hint. Use an affine bijection between intervals. ☐

Solution. Compose with $t = (2x - a - b)/(b - a)$; polynomial degree is preserved and sup norms correspond. ☐

Counterexamples and hypothesis testing

Exercise 22.40 (Interpolation not minimax)**.** Why need a Chebyshev-node interpolant not equal the best minimax polynomial?

Hint. Node interpolation imposes exact values, while minimax enforces optimal global error. ☐

Solution. Good nodes improve interpolation but do not make the optimization problems identical. ☐

Exercise 22.41 (Average error not sup error)**.** Why can a least-squares best fit differ from a minimax fit?

Hint. The objectives weight errors differently. ☐

Solution. Least squares minimizes integrated/summed squared error, whereas minimax minimizes the largest absolute error. ☐

Exercise 22.42 (Too few alternating points)**.** Why does an error pattern with fewer than $n + 2$ alternating extrema fail to certify degree-n optimality?

Hint. The alternation theorem requires enough sign constraints to block lower-degree perturbations. ☐

Solution. Without them, a perturbation may reduce the maximum error. ☐

Applications and investigations

Exercise 22.43 (Worst-case approximation)**.** When is minimax approximation preferable to least squares?

Hint. When a guaranteed maximum error matters. ☐

Solution. It optimizes worst-case deviation rather than average performance. ☐

Exercise 22.44 (Stable node placement)**.** Why do endpoint-clustered Chebyshev nodes help high-degree interpolation?

Hint. They reduce the sup size of the node polynomial and moderate endpoint amplification. ☐

Solution. This lowers a central worst-case factor in the interpolation error. ☐

Challenge problems

Exercise 22.45 (Best monic quadratic)**.** Find the monic quadratic of smallest sup norm on $[-1, 1]$.

Hint. Scale $T_2 = 2x^2 - 1$ to be monic. ☐

Solution. $x^2 - \frac{1}{2}$, with sup norm 1/2. ☐

Exercise 22.46 (Alternation lower bound)**.** If a degree-n candidate error alternates at $n + 2$ points with magnitude E, explain why no competitor can have sup error below E.

Hint. Compare the candidate with a hypothetical better competitor and track signs of their polynomial difference. ☐

Solution. The difference must change sign in each adjacent interval, giving at least $n + 1$ roots despite degree at most n, contradiction. ☐

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.

Chapter 23

The Alternation Principle

The alternation principle is the structural certificate for best uniform approximation. It converts an optimization problem into a finite pattern of equal-magnitude alternating errors.

Learning goals

After this chapter, the reader should be able to:

- state the alternation criterion with the correct number and ordering of alternating extrema;
- use an alternation pattern to certify that a polynomial approximant is minimax;
- construct a perturbation argument showing that insufficient alternation cannot be optimal;
- identify the error function and its extreme points in explicit approximation problems;
- distinguish necessary from sufficient parts of the alternation principle under the chapter's hypotheses;
- apply the principle to low-degree examples without solving a larger optimization problem directly;

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

23.1 Error function

For approximant p, study $e = f - p$.

23.2 Extreme points

Minimax optimality is witnessed where $|e|$ reaches its sup norm.

23.3 Alternation pattern

Signs alternate at successive extremal points.

Example 23.1 (Degree-one minimax certification by three alternating errors)**.** To certify that a line p is the best uniform degree-one approximation to f, it is enough to find three ordered points

$$x_0 < x_1 < x_2$$

such that

$$f(x_j) - p(x_j) = (-1)^j E$$

for one $E > 0$, and $\|f - p\|_\infty = E$. The equal magnitude says all three points are active constraints; alternating signs are what force a competitor difference to cross zero between successive points.

23.4 Necessity

Without enough alternation one can construct a perturbation reducing the maximum error.

23.5 Sufficiency

Enough alternating extrema block every lower-degree perturbation by root counting.

Example 23.2 (Why alternation blocks every better competitor)**.** Suppose p has degree at most n and its error alternates with magnitude E at $n+2$ ordered points. If q had smaller sup error, then at each alternation point the sign of

$$q - p = (f - p) - (f - q)$$

would match the sign of $f - p$. Hence $q - p$ would change sign between every adjacent pair, producing at least $n + 1$ distinct zeros. But $q - p$ has degree at most n, contradiction.

23.6 Uniqueness

Two best approximants would force their difference to have too many zeros.

23.7 Remez idea

Algorithms update candidate alternation points until equal error levels stabilize.

Example 23.3 (Remez iteration seeks a balanced finite certificate)**.** A Remez-style iteration begins with $n + 2$ candidate extremal points, solves for a degree-n polynomial p and level E satisfying alternating equations

$$f(x_j) - p(x_j) = (-1)^j E,$$

then searches for where the actual error is larger. Replacing weak points by stronger extrema repeats the process until the error peaks are nearly equal.

The algorithm is therefore a numerical search for the finite alternation certificate guaranteed by the theorem.

23.8 Approximation certificates

Alternation gives a verifiable finite certificate of global optimality.

23.9 Core structural results

Theorem 23.4 (Chebyshev alternation theorem). *A polynomial p of degree at most n is best uniform approximation to continuous f on $[a, b]$ iff the error attains its maximum magnitude at at least $n + 2$ ordered points with alternating signs.*

Proof. Sufficiency follows because any better competitor would force the difference polynomial to change sign between each adjacent pair, producing more roots than its degree. Necessity follows from a separation/perturbation argument when no such alternating set exists. □

Theorem 23.5 (Uniqueness). *The best degree-n uniform polynomial approximant is unique.*

Proof. If two best approximants existed, alternation for one and the competitor inequalities would force their difference to have at least $n + 1$ zeros despite degree at most n. □

Theorem 23.6 (Finite certificate). *Once $n + 2$ alternating equal-magnitude error points are verified, global minimax optimality follows.*

Proof. This is the sufficiency direction of the alternation theorem. □

23.10 Worked examples

Example 23.7 (Constant approximation). How many alternating points certify a best constant approximant? Two points with errors $+E$ and $-E$; the best constant centers the range between maximum and minimum values.

Example 23.8 (Best constant formula). For continuous f, find the best constant in sup norm. It is $(M + m)/2$, where $M = \max f$ and $m = \min f$, with error $(M - m)/2$.

Example 23.9 (Degree-one certificate). How many alternating extrema certify a best line? Three.

Example 23.10 (Why equal magnitude?). Why must active extrema in the alternation pattern have the same magnitude? If one were strictly smaller, the true sup norm would be determined elsewhere and the finite optimality certificate would not be balanced.

23.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 23.11 (Constant approximation). How many alternating points certify a best constant approximant?

Solution. Two points with errors $+E$ and $-E$; the best constant centers the range between maximum and minimum values. □

Problem 23.12 (Best constant formula). For continuous f, find the best constant in sup norm.

Solution. It is $(M + m)/2$, where $M = \max f$ and $m = \min f$, with error $(M - m)/2$. □

Problem 23.13 (Degree-one certificate). How many alternating extrema certify a best line?

Solution. Three. □

Problem 23.14 (Why equal magnitude?). Why must active extrema in the alternation pattern have the same magnitude?

Solution. If one were strictly smaller, the true sup norm would be determined elsewhere and the finite optimality certificate would not be balanced. □

Problem 23.15 (Root-count sufficiency). Explain the sign-change argument.

Solution. A uniformly better competitor must lie closer to f at each alternating extremum, forcing the difference of the two approximants to alternate sign between consecutive extremal points and thus have too many roots. □

Problem 23.16 (Uniqueness). Why can two different minimax polynomials not share the same alternation certificate?

Solution. Their difference would need to vanish or change sign in too many intervals for its degree. □

Problem 23.17 (Remez iteration). What does a Remez step update?

Solution. It solves for an approximant and common error level on a chosen alternating set, then replaces points using extrema of the new error. □

Problem 23.18 (Error centering). For constant approximation, why center the maximum and minimum?

Solution. Any shift away from the midpoint increases the larger of the two endpoint range deviations. □

Problem 23.19 (Monotone target). If f is monotone, where are extreme errors for the best constant?

Solution. At points attaining the minimum and maximum of f. □

Problem 23.20 (Certificate versus search). Why is alternation valuable computationally?

Solution. Searching for the optimum is global, but once a candidate displays the required alternation, optimality can be certified finitely. □

Problem 23.21 (Generalized spaces). What property of polynomials enables alternation arguments?

Solution. A nonzero degree-n polynomial cannot have more than n roots; more generally Chebyshev systems have controlled zero counts. □

Problem 23.22 (Alternation synthesis). What does the theorem reveal conceptually?

Solution. Best worst-case approximations distribute their unavoidable error evenly with alternating sign rather than allowing one-sided or localized excess. □

23.12 Exercises with complete solutions

Exercise 23.23. Best constant to f uses which values?

Hint. Maximum and minimum. Form the error $e = f - p$ and list its ordered extreme points with their signs. ☐

Solution. Their midpoint. ☐

Exercise 23.24. How many points for degree n?

Hint. $n + 2$. For degree n, look for at least $n + 2$ alternating extreme errors under the theorem's hypotheses. ☐

Solution. At least $n + 2$. ☐

Exercise 23.25. What alternates?

Hint. Error sign. To prove optimality, suppose a better approximant exists and examine the sign changes of the difference of the two approximants. ☐

Solution. Signs of extremal error. ☐

Exercise 23.26. What is equal?

Hint. Error magnitude. Too many sign changes force a low-degree polynomial difference to have too many zeros. ☐

Solution. The sup error magnitude. ☐

Exercise 23.27. What proves sufficiency?

Hint. Root count. If alternation is missing, perturb the candidate in a direction that lowers the largest errors without enlarging the others. ☐

Solution. Sign changes and root counting. ☐

Exercise 23.28. What algorithm uses alternation?

Hint. Remez. Keep the extrema ordered along the interval; alternation is not merely a count of positive and negative values. ☐

Solution. Remez algorithm. ☐

Exercise 23.29. Is minimax best approximant unique?

Hint. For polynomials, yes. Use symmetry to predict where alternating extrema should occur in low-degree examples. ☐

Solution. Yes. ☐

Exercise 23.30. What norm is involved?

Hint. Sup norm. Check that the error magnitudes at the alternating points are equal before invoking the certification theorem. ☐

Solution. Uniform norm. ☐

23.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 23.31 (Degree count)**.** How many alternating extrema certify a degree-2 minimax polynomial?

Hint. Use $n + 2$. ☐

Solution. Four. ☐

Exercise 23.32 (Best constant)**.** For a continuous function with range $[-3, 5]$, find the best constant and error.

Hint. Center the range. ☐

Solution. The best constant is 1, with error 4. ☐

Exercise 23.33 (Error function)**.** For candidate p, what function should be inspected for alternation?

Hint. Subtract approximant from target. ☐

Solution. $e = f - p$. ☐

Exercise 23.34 (Active extremum)**.** What does it mean for x_0 to be an active error point?

Hint. Compare $|e(x_0)|$ with $\|e\|_\infty$. ☐

Solution. $|e(x_0)| = \|e\|_\infty$. ☐

Exercise 23.35 (Line certificate)**.** How many alternating points certify a best line?

Hint. Degree one means $1 + 2$. ☐

Solution. Three. ☐

Proofs

Exercise 23.36 (Sufficiency)**.** Prove the sufficiency direction of alternation.

Hint. Assume a better competitor and use sign changes of the difference. ☐

Solution. The competitor difference has at least $n + 1$ roots despite degree at most n, impossible. ☐

Exercise 23.37 (Uniqueness)**.** Use alternation to prove uniqueness of a minimax degree-n polynomial.

Hint. Compare two best candidates at an alternating set. ☐

Solution. Their difference is forced to have too many zeros unless it vanishes identically. ☐

Exercise 23.38 (Best constant formula)**.** Prove the best constant to continuous f is $(M + m)/2$.

Hint. Any constant has worst error at least half the range. ☐

Solution. $\max(M - c, c - m) \geq (M - m)/2$, with equality when the two terms are equal. □

Exercise 23.39 (Finite certificate)**.** Why is alternation a global certificate even though it checks finitely many points?

Hint. Root counting converts those sign constraints into an obstruction to every lower-degree improvement. □

Solution. The finite pattern forces any allegedly better polynomial difference to exceed its possible number of zeros. □

Counterexamples and hypothesis testing

Exercise 23.40 (Equal magnitude required)**.** Why do alternating signs at unequal submaximal errors not by themselves certify minimax optimality?

Hint. The theorem concerns points attaining the actual sup error. □

Solution. Nonactive points can alternate without constraining the worst-case error sufficiently. □

Exercise 23.41 (Wrong point order)**.** Why must the alternation points be ordered along the interval?

Hint. Root counting uses sign changes between adjacent points. □

Solution. Without order, one cannot infer distinct intermediate zeros. □

Exercise 23.42 (Too few points)**.** Give the conceptual failure of a degree-2 candidate with only three alternating extrema.

Hint. Degree two needs four. □

Solution. A quadratic perturbation can potentially reduce the maximum error because root counting no longer yields a contradiction. □

Applications and investigations

Exercise 23.43 (Certificate checking)**.** How can alternation be used to verify a numerical minimax approximation?

Hint. Locate extreme errors and compare magnitudes/signs. □

Solution. A near-equal alternating pattern at $n + 2$ points is the numerical signature of minimax structure. □

Exercise 23.44 (Remez update)**.** What does a Remez step do when it finds an error peak larger than the current level E?

Hint. Replace a weaker alternation point while preserving alternating order. □

Solution. It updates the finite support set so the next solve balances the true dominant errors more closely. □

Challenge problems

Exercise 23.45 (Perturbation intuition)**.** Explain why insufficient alternation suggests a sign-shaped perturbation can lower the maximum error.

Hint. Construct a low-degree function having selected signs on active regions. □

Solution. Without enough alternating constraints, polynomial degrees of freedom can shift the approximant toward high positive-error regions and away from high negative-error regions. □

Exercise 23.46 (Dual viewpoint)**.** Why is equioscillation analogous to balancing active constraints in optimization?

Hint. At optimum, several worst-case constraints are simultaneously tight with alternating signs. □

Solution. The balanced active set prevents any feasible low-degree perturbation from decreasing all maximal errors at once. □

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.

Chapter 24

Numerical Quadrature

Numerical quadrature replaces an integral by a weighted finite sum of function values. Interpolation gives a systematic derivation, while error formulas explain exactness and the role of node placement.

Learning goals

After this chapter, the reader should be able to:

- derive elementary quadrature rules by integrating an interpolating polynomial;
- compute degrees of exactness for given quadrature formulas;
- estimate quadrature errors from derivative bounds and remainder formulas;
- compare midpoint, trapezoidal, and Simpson-type rules on explicit integrands;
- construct composite rules and track how local error accumulates globally;
- choose a mesh size that achieves a prescribed error tolerance from an available error estimate;

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

24.1 Quadrature rules

Approximate $\int_a^b f$ by $\sum w_i f(x_i)$.

24.2 Interpolatory quadrature

Integrate an interpolation polynomial exactly to obtain weights.

24.3 Newton–Cotes rules

Equally spaced nodes yield midpoint, trapezoidal, Simpson, and higher rules.

24.4 Degree of exactness

A rule has degree m if it integrates all polynomials through degree m exactly.

Example 24.1 (Degree of exactness by testing monomials)**.** For the trapezoidal rule on $[0, 1]$,

$$Q(f) = \frac{1}{2}(f(0) + f(1)).$$

It is exact for 1 and x:

$$Q(1) = 1 = \int_0^1 1\,dx, \qquad Q(x) = \frac{1}{2} = \int_0^1 x\,dx.$$

But

$$Q(x^2) = \frac{1}{2} \neq \frac{1}{3}.$$

Therefore the degree of exactness is exactly 1. Testing the first failing monomial is what certifies the degree rather than merely a lower bound.

24.5 Error formulas

Remainders depend on higher derivatives and powers of step size.

24.6 Composite rules

Apply a low-order rule repeatedly on a fine partition.

Example 24.2 (Composite trapezoidal error scales quadratically in the mesh)**.** For $f \in C^2[a, b]$ and uniform step $h = (b - a)/n$, summing the one-panel trapezoidal errors gives

$$|E_T| \leq \frac{b-a}{12} h^2 \|f''\|_\infty.$$

Thus halving the mesh reduces the bound by a factor of four. To meet a target tolerance, solve this inequality for h or n. The global order comes from local cubic error accumulated over $O(1/h)$ panels.

24.7 Gaussian quadrature

Optimally chosen nodes increase polynomial exactness for a fixed number of samples.

Example 24.3 (Two Gaussian nodes integrate cubics exactly)**.** On $[-1, 1]$, the two-node Gauss–Legendre rule uses

$$x = \pm\frac{1}{\sqrt{3}}, \qquad w_1 = w_2 = 1.$$

Symmetry makes odd monomials integrate exactly. For x^2,

$$Q(x^2) = 2 \cdot \frac{1}{3} = \frac{2}{3} = \int_{-1}^1 x^2\,dx.$$

Constants are also exact, so every polynomial through degree 3 is integrated exactly. Two intelligently chosen samples therefore achieve higher exactness than two equally spaced endpoint samples.

24.8 Stability and adaptivity

Practical quadrature balances truncation error, smoothness, and sampling cost.

24.9 Core structural results

Theorem 24.4 (Trapezoidal error)**.** *For $f \in C^2[a, b]$, the one-panel trapezoidal error equals $-(b-a)^3 f''(\xi)/12$ for some ξ.*

Proof. Integrate the linear interpolation remainder or use a Peano-kernel argument. □

Theorem 24.5 (Simpson exactness)**.** *Simpson's rule is exact for polynomials of degree at most three.*

Proof. Check the rule on the basis $1, x, x^2, x^3$ after affine reduction to a symmetric interval. □

Theorem 24.6 (Gaussian exactness)**.** *An n-node Gaussian rule for a positive weight is exact for polynomials through degree $2n - 1$.*

Proof. Orthogonal-polynomial nodes make the error polynomial divisible by the degree-n orthogonal polynomial; division separates any degree-$2n - 1$ polynomial into a vanishing part plus a lower-degree remainder integrated exactly by the weights. □

24.10 Worked examples

Example 24.7 (Midpoint rule)**.** Approximate $\int_0^1 x^2 dx$ by one midpoint sample. The midpoint value is 1/4, so the one-panel approximation is 1/4 versus exact 1/3.

Example 24.8 (Trapezoidal rule)**.** Approximate the same integral. $(f(0) + f(1))/2 = 1/2$.

Example 24.9 (Simpson rule)**.** Apply Simpson to x^2 on $[0, 1]$. $(1/6)(f(0)+4f(1/2)+f(1)) = (1/6)(0 + 1 + 1) = 1/3$, exact.

Example 24.10 (Degree of exactness)**.** Why is trapezoidal exact for linear functions? It integrates the linear interpolant, which equals the function itself for degree at most one.

24.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 24.11 (Midpoint rule)**.** Approximate $\int_0^1 x^2 dx$ by one midpoint sample.

Solution. The midpoint value is 1/4, so the one-panel approximation is 1/4 versus exact 1/3. □

Problem 24.12 (Trapezoidal rule)**.** Approximate the same integral.

Solution. $(f(0) + f(1))/2 = 1/2$. □

Problem 24.13 (Simpson rule)**.** Apply Simpson to x^2 on $[0, 1]$.

Solution. $(1/6)(f(0) + 4f(1/2) + f(1)) = (1/6)(0 + 1 + 1) = 1/3$, exact. □

Problem 24.14 (Degree of exactness)**.** Why is trapezoidal exact for linear functions?

Solution. It integrates the linear interpolant, which equals the function itself for degree at most one. ☐

Problem 24.15 (Composite trapezoid)**.** What changes when the interval is subdivided?

Solution. Apply trapezoidal rule on each subinterval and sum; the global error decreases like h^2 under smoothness. ☐

Problem 24.16 (Composite Simpson)**.** What order is expected?

Solution. For sufficiently smooth functions the global error behaves like h^4. ☐

Problem 24.17 (Interpolatory weights)**.** How are weights obtained?

Solution. Integrate each Lagrange cardinal polynomial over the interval. ☐

Problem 24.18 (Positive weights)**.** Why are positive quadrature weights desirable?

Solution. They improve stability and preserve positivity for nonnegative sampled data. ☐

Problem 24.19 (Gaussian advantage)**.** Why can n Gaussian nodes beat n equally spaced interpolatory nodes?

Solution. The nodes are optimized jointly with weights, yielding degree of exactness $2n-1$. ☐

Problem 24.20 (Error and smoothness)**.** Why do high-order rules need higher derivatives?

Solution. Their error formulas involve higher derivatives; without regularity, formal high order may not translate into actual accuracy. ☐

Problem 24.21 (Adaptive idea)**.** How does adaptive quadrature choose where to sample more?

Solution. Estimate local error on subintervals and refine where the estimate is largest. ☐

Problem 24.22 (Quadrature synthesis)**.** How is quadrature connected to interpolation?

Solution. Interpolatory quadrature integrates a polynomial surrogate exactly; the quadrature error is therefore the integral of the interpolation error. ☐

24.12 Exercises with complete solutions

Exercise 24.23. Degree of trapezoidal exactness?

Hint. Linear. Derive the quadrature weights by integrating the interpolating basis polynomials, not by memorizing the rule. ☐

Solution. 1. ☐

Exercise 24.24. Degree of Simpson exactness?

Hint. Cubic. Test degree of exactness on monomials $1, x, x^2, \dots$ until the first failure. ☐

Solution. 3. ☐

Exercise 24.25. What does Gaussian n-point exactness reach?

Hint. $2n-1$. For an error estimate, insert the derivative bound into the interpolation or Peano-type remainder formula. ☐

Solution. Degree $2n - 1$. ☐

Exercise 24.26. Why composite rules?

Hint. Reduce step size. Exploit symmetry of the rule and interval to eliminate odd terms when possible. ☐

Solution. To lower error on long intervals. ☐

Exercise 24.27. What are quadrature weights?

Hint. Integrated basis functions. For a composite rule, apply the local error bound on each subinterval and then sum the contributions. ☐

Solution. Coefficients multiplying samples. ☐

Exercise 24.28. Is Simpson exact for x^4?

Hint. Express the mesh width h in terms of the number of panels before solving the tolerance inequality. ☐

Solution. Not generally. ☐

Exercise 24.29. What smoothness controls trapezoidal error?

Hint. Second derivative. Compare rules at the same step size using their order and the size of the relevant derivative. ☐

Solution. A bound on f''. ☐

Exercise 24.30. What is adaptive refinement based on?

Hint. Local error estimates. Check whether the formula's node/weight pattern requires an even number of subintervals, as in composite Simpson-type rules. ☐

Solution. Estimated local error. ☐

24.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 24.31 (Midpoint)**.** Approximate $\int_0^1 x^2 dx$ by one midpoint sample.

Hint. Use $f(1/2)$ times interval length. ☐

Solution. $1/4$. ☐

Exercise 24.32 (Trapezoid)**.** Approximate the same integral by one trapezoid.

Hint. Average endpoint values. ☐

Solution. $1/2$. ☐

Exercise 24.33 (Simpson)**.** Apply Simpson's rule to $\int_0^1 x^3 dx$.

Hint. Use weights $1, 4, 1$. ☐

Solution. $\frac{1}{6}(0 + 4(1/8) + 1) = 1/4$, exact. □

Exercise 24.34 (Degree test)**.** What is the degree of exactness of the midpoint rule on a symmetric interval?

Hint. Test $1, x, x^2$. □

Solution. It is 1: constants and linear functions are exact, quadratics are not. □

Exercise 24.35 (Gauss two-node)**.** Evaluate the two-node Gauss–Legendre rule on x^2.

Hint. Use nodes $\pm 1/\sqrt{3}$. □

Solution. The result is 2/3, exact. □

Proofs

Exercise 24.36 (Trapezoid exactness)**.** Prove trapezoidal quadrature is exact for all linear functions.

Hint. The linear interpolant equals the function. □

Solution. The rule integrates the endpoint interpolating line exactly. □

Exercise 24.37 (Simpson exactness)**.** Verify Simpson's rule on the basis $1, x, x^2, x^3$ after scaling to $[-1, 1]$.

Hint. Use symmetry for odd powers. □

Solution. Direct evaluation matches the exact moments for all four monomials. □

Exercise 24.38 (Composite error scaling)**.** Explain why a local $O(h^3)$ panel error leads to global $O(h^2)$ error over a fixed interval.

Hint. There are $O(1/h)$ panels. □

Solution. $(1/h) \cdot h^3 = O(h^2)$. □

Exercise 24.39 (Gaussian degree)**.** Explain why n Gaussian nodes can reach degree $2n - 1$.

Hint. Divide a polynomial by the degree-n orthogonal polynomial whose zeros are the nodes. □

Solution. The node-vanishing part integrates to zero by orthogonality, and the lower-degree remainder is matched by the weights. □

Counterexamples and hypothesis testing

Exercise 24.40 (More nodes not automatically stable)**.** Why can high-order equally spaced Newton–Cotes rules become unattractive?

Hint. Weights can grow and alternate, amplifying noise. □

Solution. Higher algebraic order can come with poor numerical stability. □

Exercise 24.41 (Exactness not universal accuracy)**.** Why does high polynomial degree of exactness not guarantee small error for every nonsmooth integrand?

Hint. Error formulas require smoothness/derivative control. □

Solution. A rule can be exact on many polynomials yet perform poorly on singular or highly oscillatory functions. □

Exercise 24.42 (Single-panel limitation)**.** Why can one Simpson panel be inaccurate on a long interval even though it is cubic-exact?

Hint. Noncubic remainder grows with interval length and fourth derivatives. □

Solution. Composite subdivision reduces the local scale and hence the remainder. □

Applications and investigations

Exercise 24.43 (Mesh selection)**.** How do you choose n from a composite trapezoid error bound?

Hint. Set the bound below tolerance and solve for $h = (b-a)/n$. □

Solution. The derivative bound and tolerance produce an explicit lower bound on n. □

Exercise 24.44 (Adaptive quadrature)**.** Why compare coarse and refined quadrature estimates locally?

Hint. The discrepancy is an empirical proxy for local truncation error. □

Solution. Subdivide where the estimates disagree most, concentrating samples where the integrand is hardest. □

Challenge problems

Exercise 24.45 (Moment equations)**.** Derive symmetric two-node weights/nodes on $[-1,1]$ by requiring exactness for 1 and x^2.

Hint. Let nodes be $\pm a$ with equal weight w. □

Solution. Constant exactness gives $2w = 2 \Rightarrow w = 1$; quadratic exactness gives $2a^2 = 2/3 \Rightarrow a = 1/\sqrt{3}$. □

Exercise 24.46 (Richardson idea)**.** If an error behaves like Ch^p, explain how two resolutions can eliminate the leading error term.

Hint. Combine Q_h and $Q_{h/2}$ using the factor 2^p. □

Solution. $(2^p Q_{h/2} - Q_h)/(2^p - 1)$ cancels the leading Ch^p term under the asymptotic model. □

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.

Chapter 25

Continued Fractions and Approximation Topics

Continued fractions provide a discrete approximation theory for real numbers. Their convergents are exceptionally good rational approximants and connect Euclidean algorithms, Diophantine approximation, and recursive computation.

Learning goals

After this chapter, the reader should be able to:

- compute finite and infinite continued-fraction convergents from recurrence relations;
- recover a continued-fraction expansion from the Euclidean algorithm in rational examples;
- prove basic determinant identities for consecutive convergents;
- use convergents to obtain rational approximations and quantify their error in elementary cases;
- compare continued-fraction approximants with simpler decimal or truncation approximations;
- identify which approximation statements require irrationality, infinite expansion, or additional hypotheses;

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

25.1 Simple continued fractions

Write a real number as $[a_0; a_1, a_2, \ldots]$ with positive integer partial quotients after the first.

25.2 Euclidean algorithm

Rational continued fractions terminate and encode the Euclidean algorithm.

25.3 Convergents

Truncating the expansion gives rational approximants p_n/q_n.

Example 25.1 (Convergents computed from the recurrence)**.** For

$$x = [1; 2, 2, 2, \ldots],$$

start with

$$p_{-2} = 0,\ p_{-1} = 1, \qquad q_{-2} = 1,\ q_{-1} = 0.$$

The first partial quotients $1, 2, 2$ give

$$\frac{p_0}{q_0} = 1, \qquad \frac{p_1}{q_1} = \frac{3}{2}, \qquad \frac{p_2}{q_2} = \frac{7}{5}.$$

The recurrence avoids repeatedly evaluating nested fractions and is the natural computational representation of continued fractions.

25.4 Recurrences

Numerators and denominators satisfy second-order recurrences.

25.5 Determinant identity

Adjacent convergents satisfy $p_n q_{n-1} - p_{n-1} q_n = (-1)^{n-1}$.

Example 25.2 (Adjacent convergents are forced to be very close)**.** From

$$p_n q_{n-1} - p_{n-1} q_n = (-1)^{n-1},$$

we get

$$\left| \frac{p_n}{q_n} - \frac{p_{n-1}}{q_{n-1}} \right| = \frac{1}{q_n q_{n-1}}.$$

Thus consecutive convergents are automatically close once denominators grow. The exact determinant identity is stronger than a vague convergence statement: it gives the separation between neighboring rational approximants explicitly.

25.6 Approximation quality

Convergents approximate unusually well relative to their denominators.

Example 25.3 (Continued fractions beat denominator-scale intuition)**.** If p_n/q_n is a convergent of an irrational x, its error is controlled on the scale $1/q_n^2$, substantially smaller than the naive grid spacing $1/q_n$ among rationals with denominator q_n. For the golden ratio, Fibonacci convergents

$$1,\ \frac{2}{1},\ \frac{3}{2},\ \frac{5}{3},\ \frac{8}{5}, \ldots$$

alternate around the limit and rapidly improve.

Continued fractions therefore select exceptional rationals rather than merely rounding to the nearest point on a denominator grid.

25.7 Quadratic irrationals

Eventually periodic continued fractions characterize quadratic irrationals.

25.8 Approximation perspective

Continued fractions complement polynomial approximation by optimizing arithmetic rather than functional error.

25.9 Core structural results

Theorem 25.4 (Convergent recurrences)**.** *With standard initial data,* $p_n = a_n p_{n-1} + p_{n-2}$ *and* $q_n = a_n q_{n-1} + q_{n-2}$*.*

Proof. Induct by rewriting the finite continued fraction after adjoining its last partial quotient. □

Theorem 25.5 (Adjacent determinant identity)**.** $p_n q_{n-1} - p_{n-1} q_n = (-1)^{n-1}$.

Proof. Insert the recurrences and observe the determinant changes sign at each step. □

Theorem 25.6 (Basic approximation bound)**.** *For an irrational simple continued fraction,* $|x - p_n/q_n| < 1/q_n^2$ *for infinitely many convergents, and standard sharper neighboring-denominator bounds hold for every convergent.*

Proof. Express x using the complete quotient after stage n and combine with the determinant identity; the denominator in the exact error formula exceeds q_n^2 at the appropriate stages. □

25.10 Worked examples

Example 25.7 (Rational expansion)**.** Find a continued fraction for 17/7. Euclidean algorithm: $17 = 2 \cdot 7 + 3$, $7 = 2 \cdot 3 + 1$, $3 = 3 \cdot 1$, so $17/7 = [2; 2, 3]$.

Example 25.8 (Convergents of $[1; 1, 1, \ldots]$)**.** Identify the convergents. They are ratios of consecutive Fibonacci numbers.

Example 25.9 (Golden ratio)**.** Show $\varphi = [1; 1, 1, 1, \ldots]$. The tail equals the whole number, so $x = 1 + 1/x$, giving $x^2 - x - 1 = 0$ and the positive root φ.

Example 25.10 (Recurrence computation)**.** Given partial quotients a_0, a_1, a_2, compute p_2, q_2 recursively. Use $p_{-2} = 0, p_{-1} = 1$, $q_{-2} = 1, q_{-1} = 0$, then apply the stated recurrences.

25.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 25.11 (Rational expansion)**.** Find a continued fraction for 17/7.

Solution. Euclidean algorithm: $17 = 2 \cdot 7 + 3$, $7 = 2 \cdot 3 + 1$, $3 = 3 \cdot 1$, so $17/7 = [2; 2, 3]$. □

Problem 25.12 (Convergents of $[1; 1, 1, \ldots]$)**.** Identify the convergents.

Solution. They are ratios of consecutive Fibonacci numbers. □

Problem 25.13 (Golden ratio)**.** Show $\varphi = [1; 1, 1, 1, \ldots]$.

Solution. The tail equals the whole number, so $x = 1 + 1/x$, giving $x^2 - x - 1 = 0$ and the positive root φ. □

Problem 25.14 (Recurrence computation)**.** Given partial quotients a_0, a_1, a_2, compute p_2, q_2 recursively.

Solution. Use $p_{-2} = 0, p_{-1} = 1$, $q_{-2} = 1, q_{-1} = 0$, then apply the stated recurrences. □

Problem 25.15 (Neighbor difference)**.** Use the determinant identity to compute $|p_n/q_n - p_{n-1}/q_{n-1}|$.

Solution. It equals $1/(q_n q_{n-1})$. □

Problem 25.16 (Why denominators grow)**.** Show $q_n > q_{n-1}$ for simple continued fractions beyond the first stage.

Solution. Since $a_n \geq 1$ and $q_{n-2} \geq 0$, $q_n = a_n q_{n-1} + q_{n-2} \geq q_{n-1}$, with strict growth once positive preceding terms occur. □

Problem 25.17 (Best-approximation intuition)**.** Why are convergents hard to beat?

Solution. The determinant identity forces nearby rational competitors with small denominators to remain separated, while the continued-fraction error is already on the order of inverse denominator squared. □

Problem 25.18 (Finite expansion ambiguity)**.** Why can a rational have two finite simple continued fraction expansions?

Solution. A final partial quotient $a_n > 1$ can be replaced by $a_n - 1, 1$, producing the same value. □

Problem 25.19 (Square root periodicity)**.** What is special about continued fractions of quadratic irrationals?

Solution. Their partial quotients are eventually periodic, and conversely eventual periodicity characterizes quadratic irrationals. □

Problem 25.20 (Euclidean algorithm link)**.** Why do rationals terminate?

Solution. Repeated reciprocal-and-floor steps reproduce Euclidean remainders, which eventually reach zero. □

Problem 25.21 (Approximation scale)**.** What does an error of order $1/q^2$ mean?

Solution. Doubling denominator can reduce error quadratically in the ideal scale, much better than a generic grid spacing of order $1/q$. □

Problem 25.22 (Continued-fraction synthesis)**.** How does this chapter fit the approximation part of the volume?

Solution. Earlier chapters approximate functions globally; continued fractions approximate numbers arithmetically, using recursive discrete structure to obtain near-optimal rational approximants. □

25.12 Exercises with complete solutions

Exercise 25.23. Find CF of 3/2.

Hint. Euclidean algorithm. Generate convergents recursively from numerator and denominator recurrences rather than expanding nested fractions from scratch. □

Solution. $[1; 2]$. □

Exercise 25.24. What are convergents?

Hint. Finite truncations. For a rational number, apply the Euclidean algorithm; the quotients are the finite continued-fraction coefficients. □

Solution. Rational truncations of the continued fraction. □

Exercise 25.25. What algorithm computes rational CFs?

Hint. Euclidean algorithm. Use the determinant identity for consecutive convergents to control their difference and relative primeness. □

Solution. Euclidean algorithm. □

Exercise 25.26. What famous CF has all ones?

Hint. Golden ratio. Write the error between the irrational number and a convergent in terms of the next tail of the continued fraction. □

Solution. φ. □

Exercise 25.27. How do q_n behave?

Hint. They grow. Consecutive convergents alternate around the target in the standard simple continued-fraction setting; check parity. □

Solution. They increase after the initial stages. □

Exercise 25.28. What identity relates adjacent convergents?

Hint. Determinant identity. Compare denominators before comparing approximation quality: continued fractions optimize error relative to denominator size. □

Solution. $p_nq_{n-1} - p_{n-1}q_n = \pm 1$. □

Exercise 25.29. What numbers have eventually periodic CFs?

Hint. Quadratic irrationals. A rational continued fraction terminates; an irrational simple continued fraction does not. □

Solution. Exactly quadratic irrationals. □

Exercise 25.30. What error scale do convergents achieve?

Hint. About inverse-square denominator. When proving a best-approximation statement, state the denominator restriction explicitly before comparing competitors. □

Solution. Order $1/q_n^2$. □

25.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 25.31 (Euclidean expansion)**.** Find a simple continued fraction for 13/5.

Hint. Run the Euclidean algorithm. □

Solution. $13/5 = [2; 1, 1, 2]$, with equivalent terminal variants possible. □

Exercise 25.32 (First convergents)**.** Find the first three convergents of $[1; 1, 1, \ldots]$.

Hint. Truncate after each partial quotient. □

Solution. $1, 2, 3/2$ under the standard indexing/truncation convention. □

Exercise 25.33 (Recurrence)**.** Given $a_0 = 2, a_1 = 3$, compute p_1, q_1.

Hint. Use standard initial data. □

Solution. $p_0 = 2, q_0 = 1$; $p_1 = 3 \cdot 2 + 1 = 7,\ q_1 = 3$. □

Exercise 25.34 (Golden ratio equation)**.** If $x = [1; 1, 1, \ldots]$, derive its quadratic equation.

Hint. The tail equals x. □

Solution. $x = 1 + 1/x$, so $x^2 - x - 1 = 0$. □

Exercise 25.35 (Neighbor distance)**.** Given consecutive convergent denominators $q_{n-1} = 5, q_n = 12$, what is the distance between the two convergents?

Hint. Use the determinant identity. □

Solution. $1/(5 \cdot 12) = 1/60$. □

Proofs

Exercise 25.36 (Recurrence proof)**.** Derive the convergent recurrence when appending one partial quotient.

Hint. Represent the finite continued fraction by a fractional-linear transformation or induct on truncations. □

Solution. The standard algebra yields $p_n = a_n p_{n-1} + p_{n-2}$ and the same recurrence for q_n. □

Exercise 25.37 (Determinant induction)**.** Prove the adjacent determinant identity by induction.

Hint. Insert the two recurrences and factor. □

Solution. The determinant at stage n is the negative of the determinant at stage $n - 1$, producing alternating sign. □

Exercise 25.38 (Coprime convergents)**.** Prove p_n and q_n are coprime.

Hint. Any common divisor would divide the adjacent determinant combination. □

Solution. Since $p_n q_{n-1} - p_{n-1} q_n = \pm 1$, the gcd must be 1. $\square$

Exercise 25.39 (Rational termination)**.** Explain why the continued fraction of a rational terminates.

Hint. The Euclidean algorithm has strictly decreasing positive remainders. $\square$

Solution. It reaches remainder zero after finitely many steps. $\square$

Counterexamples and hypothesis testing

Exercise 25.40 (Decimal truncation not optimal)**.** Why need truncating a decimal expansion not give the best rational approximant for a comparable denominator?

Hint. Continued fractions optimize arithmetic structure, not base-10 representation. $\square$

Solution. A convergent can be substantially closer with the same or smaller denominator. $\square$

Exercise 25.41 (Finite expansion ambiguity)**.** Why can a rational have two finite simple continued-fraction representations?

Hint. Use $[\dots, a_n] = [\dots, a_n - 1, 1]$ when $a_n > 1$. $\square$

Solution. The terminal 1 identity gives an equivalent expansion; a convention removes ambiguity. $\square$

Exercise 25.42 (Irrationality hypothesis)**.** Why does an infinite simple continued fraction represent an irrational number?

Hint. A rational Euclidean algorithm would terminate. $\square$

Solution. If it were rational, its continued fraction would have a finite expansion, contradicting genuine infinitude. $\square$

Applications and investigations

Exercise 25.43 (Rational approximation)**.** Why are convergents natural when a numerical system needs a small-denominator rational surrogate?

Hint. They provide unusually small error relative to denominator size. $\square$

Solution. They balance arithmetic complexity q against approximation error on roughly the $1/q^2$ scale. $\square$

Exercise 25.44 (Period detection)**.** What does eventual periodicity of a simple continued fraction suggest?

Hint. Recall Lagrange's theorem. $\square$

Solution. It characterizes quadratic irrationals, linking recursive arithmetic pattern to an algebraic quadratic equation. $\square$

Challenge problems

Exercise 25.45 (Pell connection)**.** Explain why convergents of $\sqrt{D}$ are related to good solutions of $p^2 - Dq^2 \approx 0$.

Hint. Rewrite the approximation error of p/q to $\sqrt{D}$. □

Solution. If p/q is very close to $\sqrt{D}$, then $p^2 - Dq^2 = (p - q\sqrt{D})(p + q\sqrt{D})$ can be a small integer, leading to Pell-type equations. □

Exercise 25.46 (Best-approximation intuition)**.** Use the determinant identity to explain why a fraction with a smaller denominator cannot easily lie between two neighboring convergents.

Hint. Cross-multiplication gives a minimum spacing between distinct rationals in terms of denominators. □

Solution. Neighboring convergents are separated by exactly $1/(q_n q_{n-1})$; a competing low-denominator rational would violate determinant/integer-spacing constraints if it were too close, underpinning best-approximation properties. □

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.